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Date: December 2025

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A review of the progress and challenges in ultrafast laser micro-welding of brittle optical materials

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Abstract

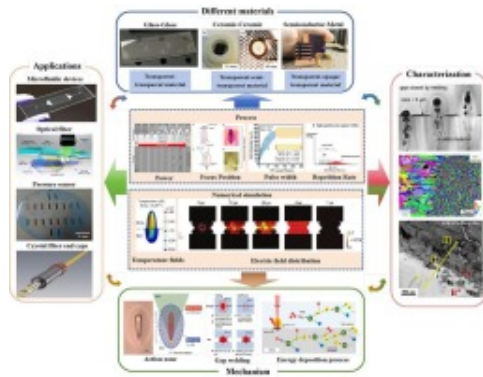
Ultrafast laser micro-welding (UFLMW), an innovative technique for joining brittle transparent optical materials such as glass, ceramics, sapphire, and silicon, has garnered significant attention in recent years. Firstly, this review comprehensively outlines the applications, phenomena and mechanism in ultrafast laser micro-welding of brittle optical materials, including the transparent optical material, transparent-to-semi-transparent optical material, and transparent optical material-to-opaque material. It delves into the latest advancements in the field, emphasizing phenomena and mechanism. The process characteristics and formation laws of ultrafast laser microwelding of brittle optical material are comprehensively analyzed and discussed, providing insights into the forming mechanisms. This review highlights the challenge of

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large contact gaps between brittle materials during welding, especially with complex-shaped materials, posing a critical bottleneck in advancing the technology. Despite progress, the physical understanding of micro-welding mechanisms between ultrafast lasers and brittle optical materials remains immature. Secondly, this review summarizes the recent breakthroughs in the numerical simulations of ultrafast laser micro-welded brittle optical material. Finally, this review delineates existing challenges and future prospects, emphasizing the importance of in-situ monitoring techniques such as pump-probe microscopy, plasma imaging, and acoustic emission for revealing micro-welding mechanisms and precise control of the ultrafast laser micro-welding process.

Graphical abstract



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Keywords

Ultrafast laser; Micro-welding; Brittle optical materials; Welding mechanisms; Numerical simulation

[Declaration of competing interest](#)

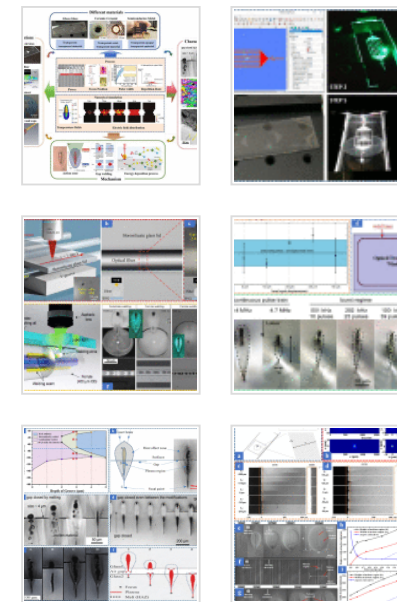
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1. Introduction

Brittle materials are defined as functional structural materials with a Vickers hardness greater than 10 GPa and a fracture toughness below $5\text{MPa}\cdot\text{m}^{1/2}$. These materials exhibit minimal plastic deformation and tend to fracture abruptly under external forces such as tension or impact. Representative examples include single-crystal sapphire, engineering ceramics, chemical vapor deposition (CVD) diamond, and specialty glasses. Brittle optical materials, such as glass, ceramics, sapphire, and silicon, are indispensable across various industries, including precision mechanics, biomedicine, optoelectronic sensing, photonic integration, and aerospace. These materials are highly valued as substrates, windows, and protective layers due to their superior optical properties, chemical stability, and high-temperature resistance [[1], [2], [3], [4]]. However, the utilization of brittle materials inevitably presents the challenge of connecting same or dissimilar materials. Conventional joining techniques, such as adhesive bonding [5], anodic bonding [6,7], brazing [8], and laser melting welding [9,10], face numerous limitations, including slow processing speeds, inadequate stability, the necessity for filler materials, stringent surface quality requirements. Additionally, these methods require the thermal expansion coefficients of the materials to be matched. These issues limit the applicability of traditional methods in modern industrial environments that demand high efficiency, precision, and cleanliness. Thus, there is an urgent need for innovative and efficient techniques to achieve precise bonding of brittle materials.

Ultrafast lasers, with their extremely short pulse widths and ultra-high peak power densities, can achieve optical power densities exceeding $10^{13}\text{W}/\text{cm}^2$ when focused. This induces nonlinear absorption in transparent materials, creating localized high temperatures at the laser focus, which is crucial for applications demanding robust structural integrity, such as in aerospace. Ultrafast laser micro-welding (UFLMW) leverages this nonlinear absorption to generate localized material melting at the focal

point, enabling precise micro-welding of transparent optical materials. This technique offers distinct advantages, including stable weld seams, high-strength connections, low thermal impact, high precision, minimal heat-affected zones, and localized processing [20,21]. In UFLMW, materials are typically categorized based on their transparency: welding between transparent optical materials, between transparent and semi-transparent optical materials, and between transparent optical materials and opaque materials. Transparent materials such as glass, crystals, and organic polymers primarily rely on nonlinear absorption and high peak power density for energy deposition. Welding of transparent and semi-transparent materials, such as doped crystals and ceramics, uses both nonlinear and linear absorption, while welding transparent to opaque materials (e.g., metals or semiconductors) requires careful control to ensure strong bonding. Compared to traditional bonding techniques, ultrafast laser welding eliminates the need for fillers or intermediate layers, resulting in stronger connections and enhanced precision. As ultrafast laser technology advances, it is increasingly recognized as a leading method for joining brittle materials, with broad applications in precision processing and high-strength bonding.

Pioneering work in this field was conducted by Tamaki et al. at Osaka University, who utilized femtosecond lasers with a repetition rate of 1 kHz, a pulse width of 85 fs, and a wavelength of 800 nm to achieve direct welding of quartz glass substrates [9]. Since then, numerous research groups have investigated the ultrafast laser micro-welding of both similar and dissimilar transparent materials. In this context, similar transparent materials refer to materials with comparable chemical composition and physical properties, such as fused silica-to-fused silica or sapphire-to-sapphire, while dissimilar transparent materials refer to the welding of materials with significantly different thermal, optical, or mechanical characteristics, such as glass-to-sapphire or sapphire-to-metal. This review aims to meet the growing demand for precision micro-welding of transparent brittle materials by providing a comprehensive analysis of ultrafast laser micro-welding techniques. This goal will be achieved through an exposition of the application-material-characterization-mechanism paradigm, encompassing a comprehensive review of the

current status of precision micro-welding techniques based on ultrafast lasers. Firstly, this review summarizes the applications, phenomena and mechanism of ultrafast laser micro-welding of brittle optical materials (see [Section 2](#)), including the transparent optical material (see [Section 2.1](#)), transparent optical material-to-semi-transparent optical material (see [Section 2.2](#)), and transparent optical material-to-opaque material (see [Section 2.3](#)). This paper analyzes the process characteristics and forming laws of ultrafast laser microwelding of brittle optical material in detail. Secondly, this review summarizes the recent breakthroughs in the numerical simulations of ultrafast laser micro-welded brittle optical material (see [Section 3](#)). Lastly, this review comes out with the existing challenges and future future prospects for this new welding technology (see [Section 4](#)). Through comprehensive summarization and synthesis, this paper aspires to provide researchers in the field of ultrafast laser micro-welding with a comprehensive compendium of information, fostering innovation and development within this domain.

2. Research progress of ultrafast laser micro-welding of brittle optical materials

In industrial applications, ultrafast laser welding offers significant advantages, particularly its ability to join materials with disparate thermal expansion coefficients. The weld pool is confined to just a few tens of micrometers, resulting in a minimal heat-affected zone and maintaining the parent material at nearly ambient temperature during welding. This effectively mitigates macroscopic thermal stresses caused by varying thermal expansion coefficients, making femtosecond laser welding ideal for encapsulating microfluidic and microelectronic devices. The non-linear absorption threshold further expands its applicability, allowing ultrafast laser welding to be used not only in microdevice and sensor encapsulation but also in the assembly of consumer electronics and aerospace window devices. Additionally, this technique enables the direct selective welding of multi-layer materials. The ability of ultrafast lasers to generate extremely high temperatures in the target region makes them especially suited for welding high-melting-point materials.

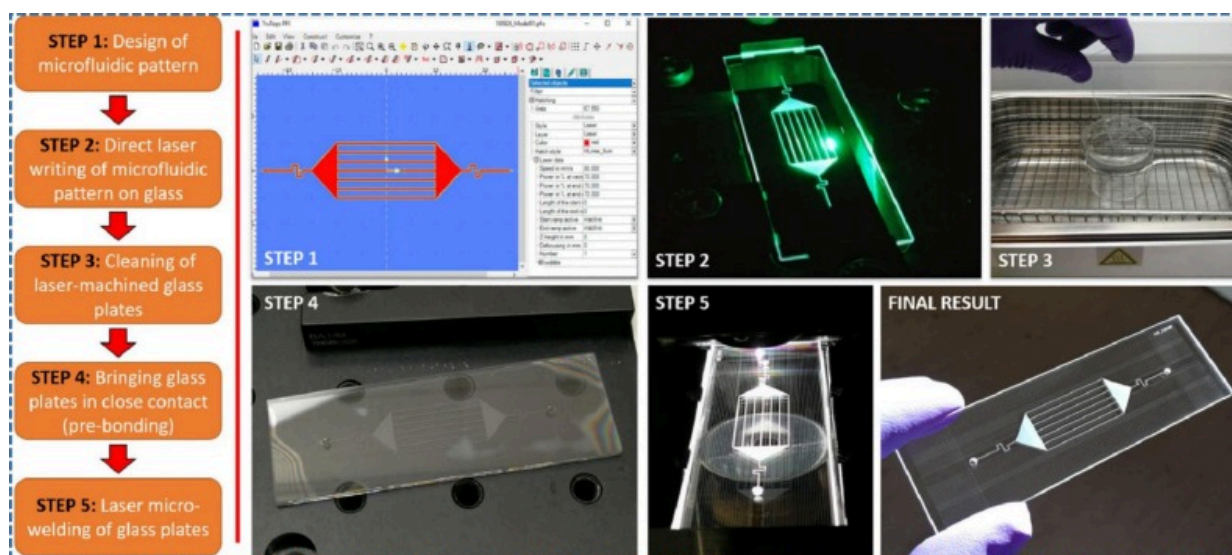
As a new research hotspot, ultrafast laser micro-welding technology has received extensive attention from research teams in various countries in the past decade. At present, ultrafast laser micro-welding technology can be successfully applied to the welding of different brittle materials such as glass-glass [23], [25], [26], [27], [28], [47], [48], [49], [50], [51], [52]], ceramic-ceramic [53,54], ceramic-metal [55,56], glass-metal [33], [34], [35], [36], [37], [40], [43], [55], [57], [58], [59], [60]] and glass-semiconductor [61], [62], [63]]. Next, this part offers a detailed examination of recent breakthroughs, encompassing diverse material combinations such as transparent optical material-to-transparent optical material, transparent optical material-to-semi-transparent optical material, and transparent optical material-to-opaque material within the domain of ultrafast laser micro-welding.

2.1. Micro-welding between transparent optical material

2.1.1. Application of micro-welding to transparent optical materials

A manufacturing technique where a femtosecond pulse laser system is the sole tool required for creating transparent glass substrate microfluidic devices [64]. The laser system is employed to directly generate microfluidic patterns on glass, drill holes for inlet/outlet ports on the glass cover, and weld two glass plates together to encase the laser-generated pattern from the top (as shown in Fig. 1). This technique allows for the rapid fabrication of fully functional microfluidic devices within a few hours, eliminating the need for photomasks, hazardous chemicals, or additional expensive tools. It enables the production of diverse microfluidic devices, including Hele-Shaw cells and microchannels, making it applicable for research in carbon dioxide storage, water recovery, and hydrocarbon recycling processes. Moreover, this method provides a means to embed small 3D objects within these devices. Researchers at Italy's Nanoecology Institute [65] successfully used near-infrared femtosecond lasers to weld transparent polymer materials (PMMA), encapsulating polymer-based microchannels-a technique promising for biomedical instruments. Building on this, Aschaffenburg University [66] extended the method to complex microfluidic chip encapsulation, while Hanbat National

University [67] applied it to blood pressure sensor encapsulation. Lab-on-a-Chip (LOC) devices are gaining traction in biomedicine, but mainstream methods like soft lithography have prototype limitations. Butkute et al. [68] proposed femtosecond laser micromachining for liver-on-chip (OOC) devices using selective laser etching (SLE) and welding on glass substrates. Perovskite solar cells (PSCs) face industrialization challenges due to humidity sensitivity. Chen et al. [12] addressed this with ultrafast pulse laser glass welding, enhancing perovskite thin film stability. Similarly, Young et al. [69] used femtosecond lasers for glass-to-glass welds in polymer-free solar modules, achieving promising results for hermetically sealed modules with reduced degradation and easier recycling.



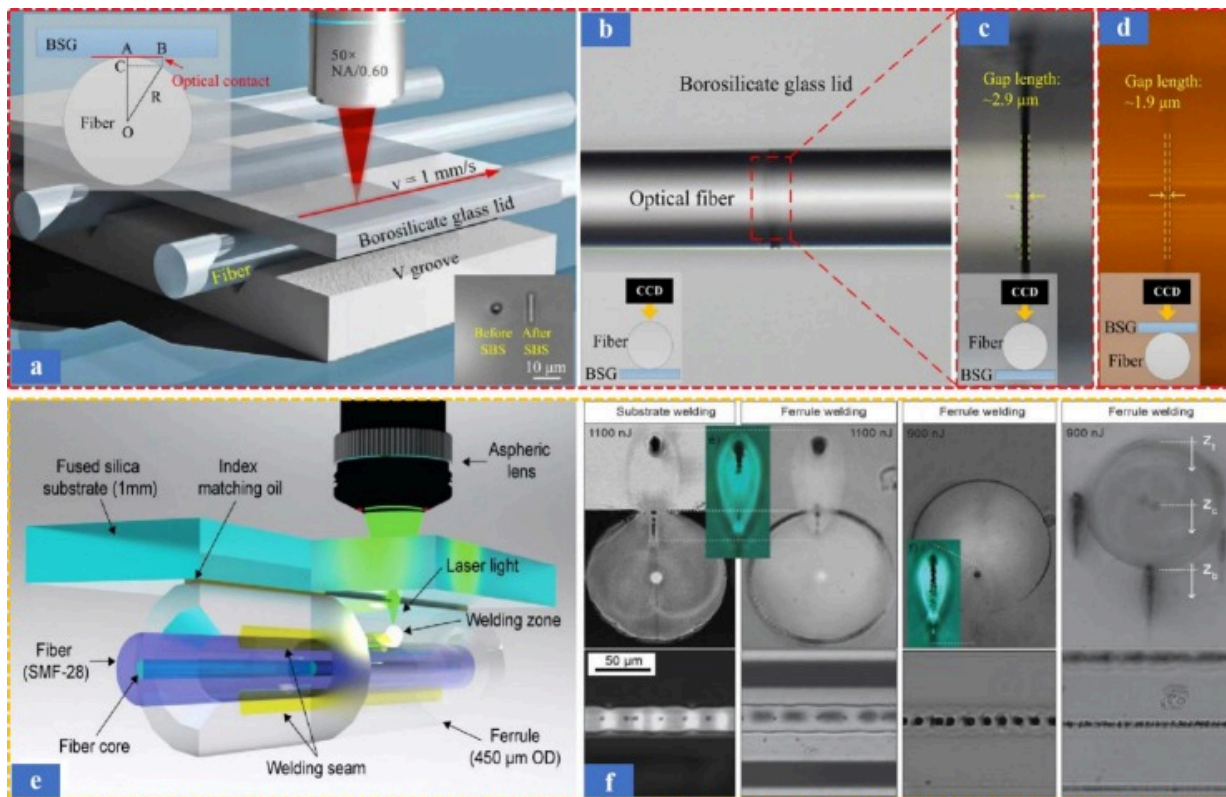
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Fig. 1. Rapid fabrication and encapsulation of glass microfluidic devices using ultrafast lasers [64].

The growing demand for faster high-performance computing (HPC) networks and increased data center transmission bandwidth-crucial for machine learning, data

processing, and computing-has led to innovations such as co-packaged optics (CPO) and near-packaged optics (NPO) to overcome the limitations of Moore's law. Photonic integrated circuits (PICs), which often feature very small mode field sizes, face significant losses when directly coupled with commercial optical fibers like single-mode fiber (SMF). Additionally, the high temperatures generated by next-generation high-speed chips, such as switch ASICs, CPUs, and GPUs, render traditional epoxy-based fiber arrays unsuitable for coupling fibers to PICs. To address these challenges, an epoxy-free bonding method utilizing femtosecond laser welding of borosilicate glass (BSG) 3.3 and optical fibers has been proposed and successfully demonstrated ([Fig. 2\(a-d\)](#)) [70]. This method achieved low-loss, polarization-independent fiber-to-fiber coupling, highlighting the reliability and effectiveness of the bonding technique. The technology is scalable and can potentially be automated for high-density multi-fiber edge coupling with PICs. Femtosecond laser welding was further applied to optical silica fiber (SMF-28) by focusing through fused silica substrates and ferrules to create all-glass weld seams ([Fig. 2\(e-f\)](#)) [71]. Thus, femtosecond laser writing and welding offer a versatile approach to photonics fabrication and packaging, particularly for fiber Bragg gratings (FBGs), enabling reliable high-frequency vibration sensing in high-temperature and strain environments.



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Fig. 2. Femtosecond laser welding of optical fiber (a) Schematic setup for the femtosecond laser welding between the BSG lid and optical fiber [70]; (b-d) Microscopy image from fiber top of the two aligned SMFs which were welded on a BSG lid [70]; (e) Experiment arrangement for femtosecond laser welding of a single mode optical fiber (125 μm diameter) to a fused silica ferrule with pseudo-continuous welding seams [71]; Microscope images recorded in backlighting show cross sectional (top row) and side (bottom row) views of welding seams formed between SMF-28 fiber (125 μm diameter) and fused silica [71].

In summary, welding of brittle transparent materials is crucial in fields such as microfluidic devices, lab-on-a-chip systems, solar cell modules, and optical fiber packaging. Ultrafast lasers, with their non-linear energy deposition characteristics, enable the welding of transparent brittle materials without the need for interlayers, making them particularly suitable for non-bonded optical devices. These lasers play a pivotal role in the micro-welding of transparent materials, significantly advancing electronic packaging. Companies like PolarOnyx employ femtosecond fiber lasers for glass welding, microfluidic device sealing, and bonding dissimilar materials. Major industry players such as Huawei and Samsung utilize ultrafast lasers for OLED foldable smartphone packaging, adhering to stringent environmental standards. Beyond packaging for sensors and optoelectronic devices, this technology is also applicable in micro-mechanical systems, aircraft components, and microelectronic circuit manufacturing. Offering precise welding with minimal heat impact, ultrafast laser welding ensures stable, low-damage connections for a wide range of microdevices, presenting significant potential for diverse applications. In particular, ultrafast laser welding schemes for transparent glass materials have shown great promise [72,73]. Extensive research has demonstrated the feasibility of using ultrafast lasers to achieve stable bonding in fused silica, borosilicate glass, and low thermal expansion glass, with the fracture strength of welded joints comparable to that of bulk materials [74,75,17]. This section summarizes recent domestic and international research on ultrafast laser micro-welding of homogeneous and heterogeneous glass materials.

2.1.2. Optical contact between glasses

Research on ultrafast laser welding of glass emphasizes the critical need for pre-existing optical contact or minimal gaps between components to ensure successful bonding. Studies have shown that maintaining gaps within the range of 100–500nm is essential to prevent interface cracking or ablation [48]. To achieve this, researchers frequently employ optical contact to minimize gaps, thereby reducing residual stress at the weld seam and enhancing weld quality (see [Table 1](#)). Tamaki et al. [11] were among the first to weld quartz pieces using a titanium sapphire femtosecond laser. However, their hardware

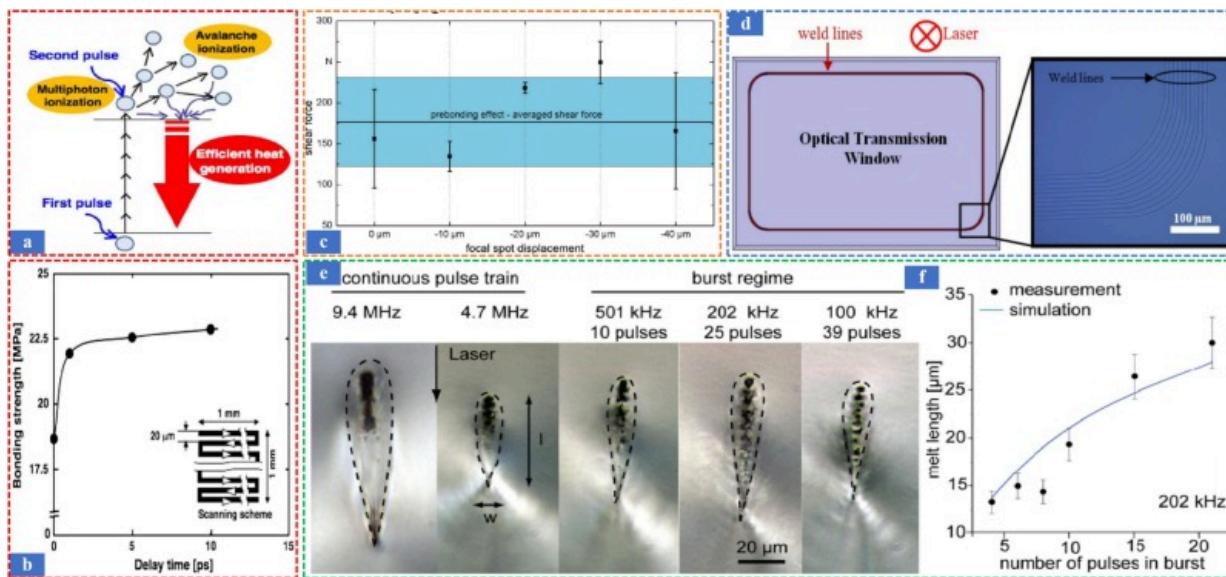
limitations led to a low pulse repetition rate, restricting welding to gaps smaller than a quarter wavelength. This pioneering work marked the introduction of ultrafast laser welding in glass, garnering significant attention despite the absence of detailed analysis on weld quality and strength. Further advancements were made by Watanabe et al. [20], who utilized a femtosecond laser with a 1 kHz repetition rate to achieve direct selective welding between dissimilar transparent glasses. To enhance weld strength and focus tolerance, Sugioka et al. [13] introduced dual-pulse femtosecond laser welding (Fig. 3(a)). This technique allowed precise control over electron excitation processes such as multi-photon and avalanche ionization, leading to more efficient free electron generation. The dual-pulse method produced larger heat-affected zones compared to single-pulse welding. Experimental results showed that dual-pulse femtosecond laser welding achieved a welding strength of 22.9MPa (Fig. 3(b)), approximately 22% higher than conventional single-pulse methods. Consequently, using a dual-pulse train of ultrafast laser pulses significantly enhances the quality and strength of glass welds. Cvecek et al. [14] investigated the effect of focus offset on welding strength using two 1 mm-thick borosilicate D263T glass samples and a focal length of 6mm. Their experiments (Fig. 3(c)) revealed that a defocusing amount of 20–30 μ m achieved a significant shear strength of 50MPa, while 0 and 30–40 μ m defocusing halved the seam strength, indicating a 20 μ m defocusing tolerance. Richter et al. successfully micro-welded various glass types, including fused silica, borosilicate, Zerodur, BK7, and ULE, using a high repetition rate femtosecond laser with 450fs pulse width and 9.4MHz repetition rate (Fig. 3(e-f)) [17,76,16]. They utilized a burst-mode laser operation, where femtosecond pulses were grouped into bursts with an intra-burst pulse repetition rate of approximately 9.4MHz (corresponding to $\sim$ 100ns pulse spacing), and bursts themselves were separated by intervals of about 10ms. This scheme produced larger melting structures and reduced laser-induced stress. In their study on borosilicate glass welding, a second-order polynomial model was developed using response surface methodology to describe the relationship between welding quality and process parameters [18]. This model, reflecting the influence on lap shear strength and welding width, predicted results with an error margin of less than 6%, proving its significance for optimizing picosecond laser welding

parameters. Elemental diffusion of Si and O during welding improved the weld zone's micro-hardness, especially with more adherends at the seam. Pan et al. [77] used a femtosecond laser with a 0.25 numerical aperture microscope objective to weld optically contacted quartz glass specimens. They found that maintaining a gap width below 100nm significantly affected splatter during welding. He et al. [78] introduced a transparent material welding method using ultrafast lasers for precision sensor glass mother tubes. This enhanced the sensor's compressive resistance from 30MPa with adhesive sealing to 50–60MPa, thereby extending its lifespan.

Table 1. Recent works on ultrafast laser microwelding of glass with optical contact.

Weleing Materials	Laser Parameters			Contact State	Shear or Tensile Strength/MPa	Reference
	Pulse Width	Wave-Length	Repetition Frequency			
Silica Glass	130fs	800nm	1 kHz	Optical Contact	—	[11]
Borosilicate Glass /Fused Silica	85fs	800nm	1 kHz	Optical Contact	15.3MPa	[12]
Photostructurable Glass	360fs	1045nm	200kHz	Optical Contact	22.9MPa	[13]
Borosilicate Glass /D236T	10ps	1064nm	1 MHz	Optical Contact	50 MPa	[14]
Silica Glass/BK7	55fs	790nm	250kHz	Optical Contact	15MPa	[15]
Zerodur/ULE ULE/Borosilicate Borosilicate/Fused	450fs	1030nm	9.4MHz	Optical Contact	75MPa	[16]

Weleing Materials	Laser Parameters			Contact State	Shear or Tensile Strength/MPa	Reference
	Pulse Width	Wave-Length	Repetition Frequency			
Silica						
BK7/Fused Silica						
Fused Silica	450fs	515nm	9.4MHz	Optical Contact	54MPa	[17]
Borosilicate glass	10ps	1064nm	0.8–1.6MHz	Optical Contact	3.25–6.92MPa	[18]
PG /ZMAS	180fs	1030nm	1MHz	Optical Contact	---	[19]



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Fig. 3. Welding of different glass combinations under optical contact (a-b) Double-pulse femtosecond laser welding of glass [13]; (c) Relationship between defocusing amount and shear strength [14]; (d) Ultrafast laser welding of optically contacted BK7 and quartz glass [15]; (e-f) High-repetition-rate welding of optically contacted quartz glass [17].

Despite over a decade of laboratory research, ultrafast laser micro-welding of glass has achieved weld strength nearly comparable to the original materials. However, practical engineering application faces challenges. Ultrafast lasers offer instantaneous welding but are constrained by focus space, requiring precise positioning. Welding also demands optical contact between surfaces, necessitating prior polishing and cleaning for plasma overflow prevention. This complexity limits effectiveness on large or irregular surfaces. Further advancements are crucial to improve applicability and efficiency in real-world scenarios.

2.1.3. Non-optical contact between glasses

Most research on ultrafast laser welding of glass has been conducted under conditions of optical contact, requiring gaps less than a quarter wavelength, typically around 100nm, to minimize melted volume and control thermal pressure and plasma dissipation. Due to the uneven pressure induced by the fixture on the glass samples, the strength between the samples may decrease due to residual stress after welding. Additionally, the design and fabrication of versatile clamping devices are costly and inefficient for real-world applications. However, achieving such precision is challenging, especially for large-area welding, hindering widespread engineering applications. To address this, researchers are exploring non-optical contact welding methods, potentially opening new practical and scalable applications (**see Table 2**). Chen et al. [22] demonstrated picosecond laser welding of micron-scale gaps, successfully welding glasses with 3 μ m gaps, as shown in **Fig. 4(a)**. Richter et al. [23] achieved fused silica glass welding using femtosecond laser with single-pulse energy. When the femtosecond laser was focused on the interface between two materials, the resulting weld seam was narrow, and the gap between the two glass interfaces was not completely filled, as shown in **Fig. 4(c)**. When the

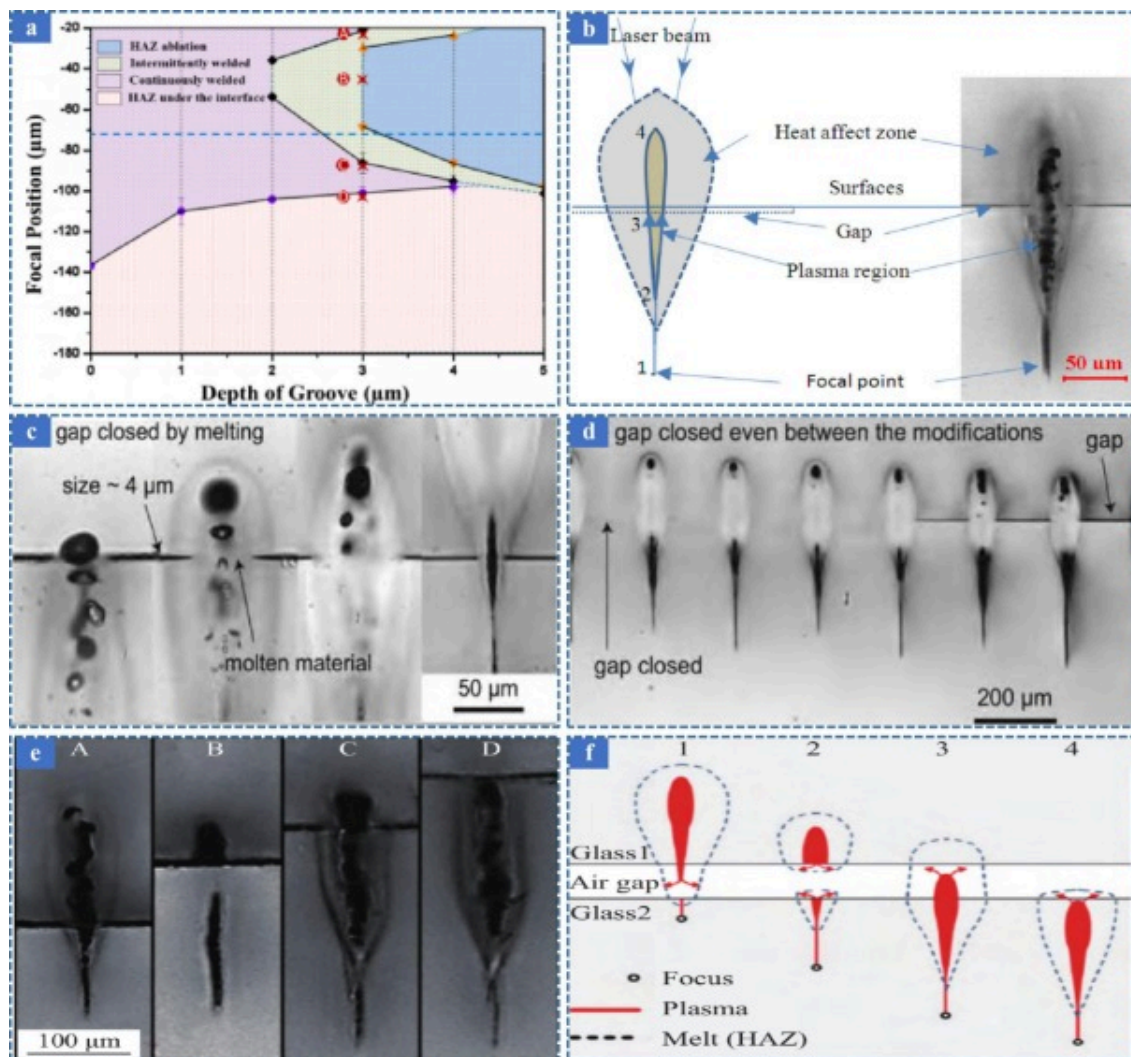
femtosecond laser focal spot is focused below the interface gap and on the upper surface of the glass substrate, the weld zone formed by the cooling of the molten zone fills the gap with a width of about $4\mu\text{m}$, as shown in Fig. 4(d). When larger gaps were present in the welded samples, laser-induced surface thermal ablation and spattering were more pronounced. The ejected, minute molten particles would fall, mix, and fill the gaps within the focal and adjacent regions of the sample, creating a bridging effect between upper and lower interface surfaces, as illustrated in Fig. 4(e-f) [22]. When focused below the gap, the molten zone filled the gap, with larger gaps experiencing more pronounced thermal ablation and spattering. Laser focus in the lower glass achieved welding, progressing through plasma ablation, laser action zone ablation, intermittent and continuous welding stages. Richter et al. [48] utilized four laser pulses with $10\mu\text{J}$ pulse energy at 20ns intervals to weld quartz glass with a $3\mu\text{m}$ contact gap, achieving a maximum welding strength of 73MPa through molten material expulsion and solidification-induced contraction. Notably, sample bending feasibility may be limited to thin samples due to material rigidity. Non-optical contact approaches offer potential solutions for industrial applications by addressing challenges associated with optical contact.

Table 2. Recent works on ultrafast laser microwelding of glass with non-optical contact.

Weleing Materials	Laser Parameters			Contact State	Shear or Tensile Strength/MPa	Remarks	Reference
	Pulse Width	Wave- Length	Repetition Frequency				
Fused Silica Borosilicate glass /Fused silica	5.9ps	1030nm	400kHz	Non- optical contact	—	Two glasses with $3\mu\text{m}$ gap were successfully welded.	[22]

Weleing Materials	Laser Parameters			Contact State	Shear or Tensile Strength/MPa	Remarks	Reference
	Pulse Width	Wave- Length	Repetition Frequency				
Fused Silica	50fs	1030nm	200kHz	Non- optical contact	75MPa	Two glasses with 4μm gap were successfully welded.	[23]
Soda-lime glasses	10ps	1030nm	1MHz	Non- optical contact	64MPa	The welding of glasses with natural stacking contact.	[24]
EAGLE XG ® alkaline earth sodium pyroborate- aluminosilicate glass	800fs	515nm	600kHz	Non- optical contact	20MPa	Two glasses with 3μm gap were successfully welded.	[25]
Corning third generation gorilla glass	2ps	1064nm	—	Non- optical contact	30MPa	The gap is 2μm, a high strength weld with smooth microstructure can be obtained.	[26]

Weleing Materials	Laser Parameters			Contact State	Shear or Tensile Strength/MPa	Remarks	Reference
	Pulse Width	Wave- Length	Repetition Frequency				
Fused silica substrates (JGS2)	1000fs	1030nm	600kHz	Non- optical contact	80.26MPa	Two-layer vertical welding of fused silica glass;	[27]
Soda lime glass	13ps	1064nm	50kHz	Non- optical contact	6.5MPa	The non- optical soda lime glass that has a gap up to ~5μm.	[28]



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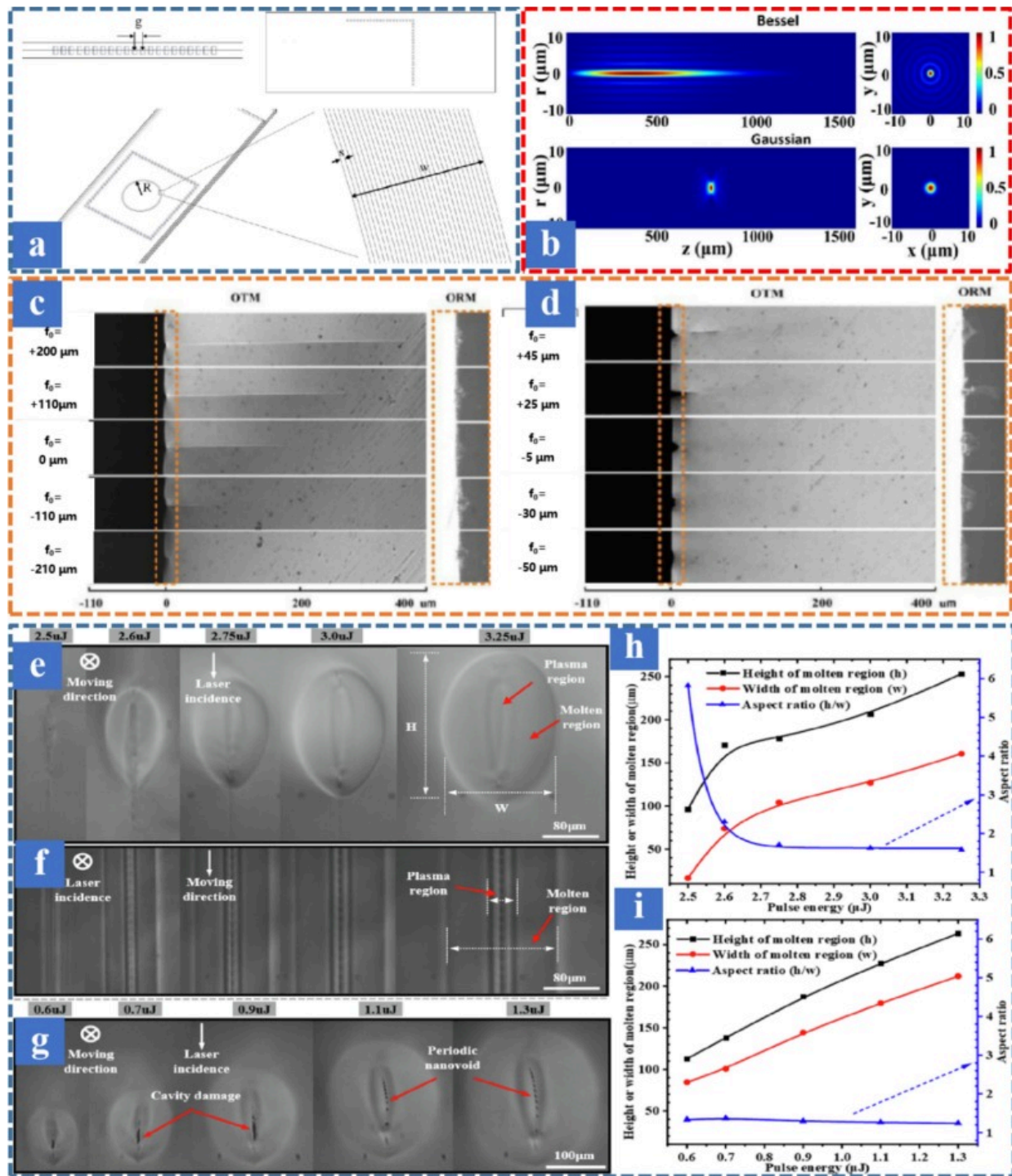
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Fig. 4. Welding effect diagram and interface diagram of different combinations of glass under non-optical contact (a) Welding effect diagram under different focus position and gap width [22]; (b) Welding cross section [22]; (c) The focus is located at the interface [23]; (d) The focus is located below the interface [23]; (e) Welding cross-sections of samples corresponding to different focal positions under non-optical contact conditions

[22]; (f) Welding schematic diagram corresponding to different laser focusing positions under non-optical contact conditions [22].

To address the limitations of optical contact in glass welding, Duan et al. [49,45] introduced a high repetition rate picosecond laser sealing welding method (Fig. 5(a)). This method begins with oscillation scanning to locally weld overlapping glasses with large gaps, reducing the gap to about 1.5 μm . This is followed by plasma-free sealing welding in the closed area. The rapid oscillating scan expands the interaction range between the laser and glass, enhancing the connection of glass layers. Zhang et al. [50,51] developed the Bessel ultrafast laser micro-welding method, significantly increasing the focus position control tolerance (Fig. 5(b)). This technique expands the laser focus position window by four times compared to Gaussian laser welding, promoting industrial application. Fig. 5(c) shows a side-view micrograph of samples welded with a zero-order Bessel beam, achieving a focal depth of 410 μm , much greater than the Gaussian beam's depth in Fig. 5(d). When the femtosecond laser Bessel beam is focused on the sample interface, rapid heating occurs within the focal region, leading to material melting and flow across the interface, driven by thermal gradients and subsequent laser pulses. This process eliminates the gap between silicon and borosilicate glass, resulting in complete material mixing within the melting pool, as confirmed by optical transmission microscopy images (Fig. 5(e-i)). The elliptical molten region formed outside the laser focus is attributed to diffused heat and the accumulation process. A narrow, smooth substructure within the molten region is likely due to electron plasma relaxation at the optical border, acting as a linear heat source for the outer molten region. The elongated geometry and nonlinear stability of the Bessel beam play crucial roles in this process. Consequently, using Bessel beams significantly improves focus position control tolerance by approximately 5.5 times. Compared to Gaussian laser welding, the Bessel ultrashort pulse laser welding process increases the axial focus position window by four times, significantly advancing industrial applications of ultrashort pulse laser micro-welding technology. Ultrafast laser glass welding typically involves focusing the beam with a high numerical aperture microscope objective, limiting the welding to a tiny focal volume,

with short working distances, low speeds, and narrow weld widths, making it less suitable for industrial use. Huo et al. [25] demonstrated strong and stable bonding between rough copper foil and quartz glass using a green femtosecond laser (520nm, 300fs, 250kHz). Under optimized conditions, the joint achieved a maximum shear strength of 17.19MPa. A two-temperature model was used to simulate electron and lattice temperatures during welding. Elemental diffusion, interfacial reactions, and the formation of a modified zone in glass due to excess energy were investigated. Cu–O–Si bonds were identified at the interface. The joint maintained ~12MPa shear strength after thermal cycling (0–100°C) and heating at 150°C. This study demonstrates that effective and robust welding of quartz glass to rough copper foil can be achieved using green femtosecond laser irradiation.



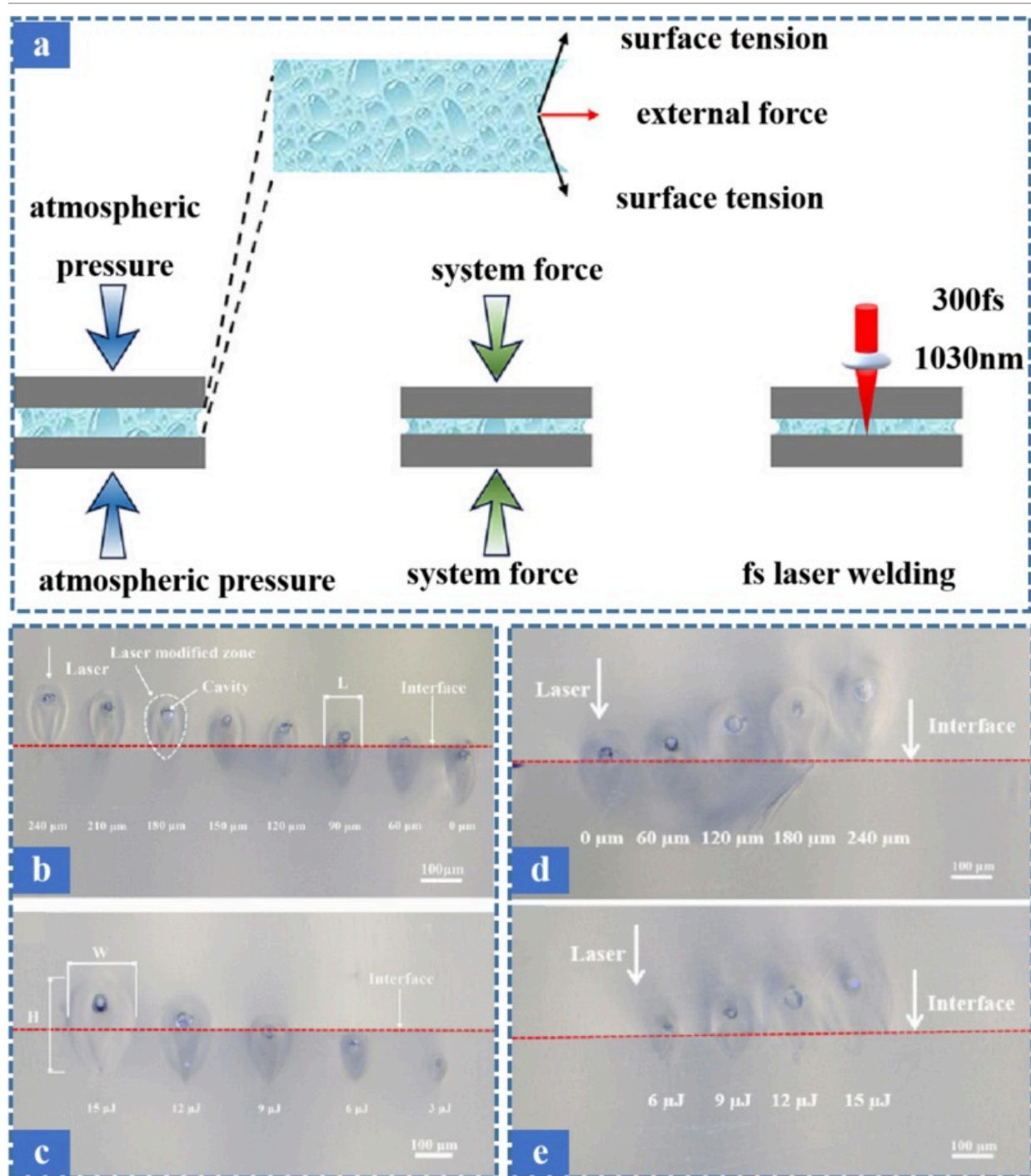
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Fig. 5. Welding effect diagram and interface diagram of glass under non-optical contact (a) A schematic diagram of large gap glass welding method [49]; (b) Gaussian beams converging through a focusing objective and Bessel beams with the same transverse confinement strength [50]; (c) Specimen diagram of femtosecond laser Bessel beam welding at different focusing positions [50]; (d) Specimen images of femtosecond laser Gaussian beam welding at different focusing positions [50]; (e-i) Modification lines inside the aluminosilicate glass written by ultrafast Bessel beam and Gaussian beam at pulse duration of 1.2 ps, repetition rate of 1 MHz, and scanning speed of 1 mm/s [51].

The development of three-dimensional glass structures and flexible glass in electronics introduces challenges in controlling gaps between glass surfaces. To address this, Zhang et al. [26] developed a real-time gap detection method based on optical thin-film interference, enabling precise control of picosecond laser welding. By optimizing welding parameters according to gap size and laser focal point positions, high-strength welds with smooth microstructures were achieved, which are essential for electronic packaging. Tan et al. [27] explored vertical femtosecond laser welding of double-layer glass, resulting in finer structures and higher bonding strengths than horizontal welding. However, industrial application is limited by slow processing speeds and tight gap requirements. Yoshitake et al. [52] advanced ultrafast micro-welding of glass substrates, achieving faster processing and increased allowable gaps while maintaining welding strength comparable to traditional methods. Void formation during welding did not significantly impact strength, offering an alternative to conventional ultrashort pulsed laser welding. Jia et al. [28] introduced a multi-scan picosecond laser welding technique for non-optical contact transparent materials, achieving significant welding strength through a three-step process. This method is simple and offers a new solution for practical ultrafast laser welding applications. Chen et al. [79] enhanced the effects of nonlinear absorption and heat accumulation by employing ultrafast lasers with ultrahigh repetition rates for non-optical contact glass welding. They shaped the laser beam to suppress microcrack

formation, ensuring weld quality and facilitating engineering applications. These advancements open new avenues for practical ultrafast laser welding. Therefore, it is more practical to study how to improve the welding strength of glass in a simpler method under non-optical contact conditions. To address these issues, Gao et al. [80] propose a jig-free method to weld silica glass by femtosecond laser double-pulse under a non-optical contact condition. By adjusting the optical path of two pulsed laser beams, the time difference in reaching the sample to be processed can be controlled, allowing for double-pulse processing. Chen et al. [81] investigated and compared two methods for realizing connections between nonoptical contact glasses via fs laser welding (Fig. 6(a)). In method 1 (Fig. 6(b-c)), a homemade fixture is used to reduce the gap between the glass samples. In method 2 (Fig. 6(d-e)), a water layer is introduced in the gap to achieve high-quality connections via femtosecond laser welding. Through a comparison of shear strength using two different methods, it was found that a maximum shear strength of 18.3MPa was achieved using water intermediate layer-assisted femtosecond laser welding, which is twice as high as that achieved through the use of fixtures to apply mechanical pressure.



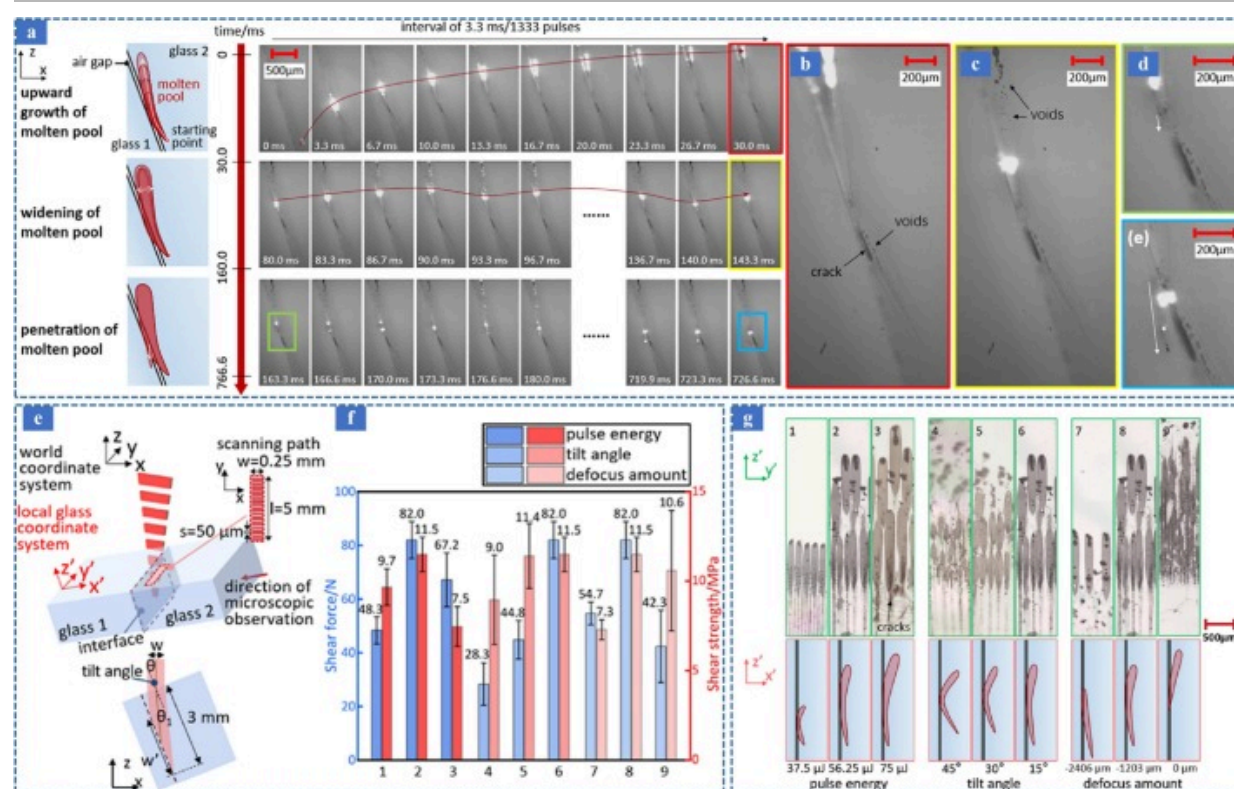
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Fig. 6. (a) Improved Femtosecond Laser Welding of Nonoptical Contact Glass by Pressure and a Water Intermediate Layer; (b) Cross-sectional morphology of the modified zone at different focus positions at a pulse energy of 9 μJ , repetition rate of 1087 kHz, and scanning speed of 1 mm/s; (c) Cross-sectional morphology of the modified zone in glass under different pulse energies; (d-e) Morphologies of the zones modified by the water intermediate layer-assisted femtosecond laser welding in glass. [81].

Ultrafast laser-based glass welding is effective for glass packaging but faces challenges such as laser energy loss and microcracks. Previous research focused on enhancing weld strength, suppressing microcracks, and developing non-optical contact welding methods. However, due to plasma shielding and thermal diffusion, the laser forms a double structure laser action zone, wasting energy and increasing microcrack risks. Multifocal shaping offers a solution, but traditional algorithms like Gerchberg-Saxton (GS) lack uniformity, affecting weld quality. Chen et al. [82] proposed a multi-focus picosecond laser welding method based on an improved GS algorithm, enhancing uniformity to 99.8%. This method regulates molten pool formation, preventing excessive longitudinal expansion, increasing weld width by 75%, and inhibiting microcracks by scattering stress distribution. This advancement holds significant implications for engineering applications of ultrafast laser glass welding. In current literature on ultrafast laser glass welding, the glass pieces are typically arranged in an overlapping configuration, with the laser directed vertically through the top surface of the upper glass to the welding interface. However, this approach has limitations, particularly when the opposite surface is not laser-permeable due to surface treatments or the presence of electronic components. Additionally, if the upper glass is thicker than the laser's focal length-especially when using an objective lens with a short focal length-the laser cannot effectively reach the welding interface. Although systems employing galvanometer setups with longer focal lengths or Bessel beams with extended focal depths can mitigate some thickness-related issues, they are inadequate for thicker glass. An alternative method, glass butt welding,

has been successfully demonstrated using oblique laser incidence [83]. This technique addresses the challenge of beam propagation within tilted glass, offering a theoretical foundation for focusing inside the material. High-speed camera observations reveal three stages of molten pool evolution during this process (Fig. 7(a-d)): the pool grows opposite to the laser propagation direction, intersects with the welding interface, and eventually forms a connection that expands or penetrates further under subsequent pulses. This elucidates the mechanism of oblique-incidence ultrafast laser glass butt welding. The study further investigates the impact of laser pulse energy, glass tilt angle, and defocus on welding strength (Fig. 7(e-g)), achieving a maximum shear strength of 11.5 MPa.



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Fig. 7. Oblique incidence of ultrafast laser for glass butt welding (a-d) Evolution of the molten pool in the glass based on high-speed photography[83]; (e) Scan trajectory diagram; (f) Welding shear force and shear strength; (g) Schematic diagram of molten pool in glass and microscope observation diagram [83].

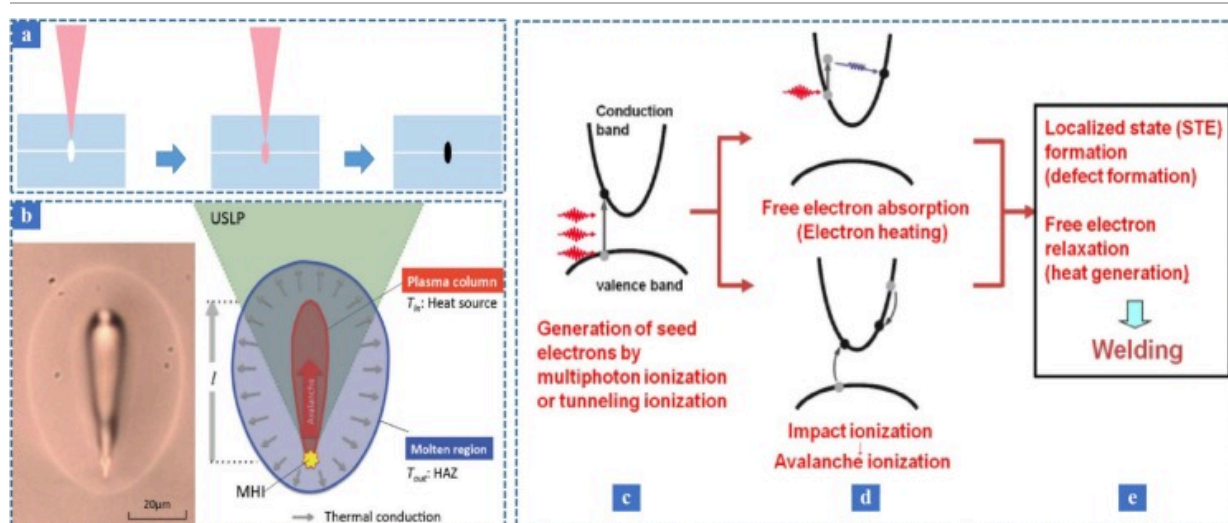
The extensive research on ultrafast laser micro-welding technology has primarily focused on glass contact gaps not exceeding 10 μ m. However, in practical scenarios, variations in contact gap due to material properties and surface roughness pose challenges. External force application is often necessary to reduce the contact gap, especially for large-area or complex-shaped glass welding. This introduces complexity and hampers the practical engineering application of ultrafast laser glass welding technology. Addressing this challenge is crucial for advancing the broader adoption and effectiveness of ultrafast laser micro-welding in real-world applications.

2.1.4. Mechanism of ultrafast laser micro-welding between transparent optical material

Ultrafast lasers, unlike continuous and long-pulse lasers, exhibit nonlinear photon absorption at the focal point in transparent materials, causing localized melting and forming a weld pool. This enables welding between materials that are relatively or partially transparent to the laser. The high peak power density of ultrafast lasers selectively induces nonlinear effects such as multi-photon absorption, avalanche ionization, and plasma evolution.

Maintaining optical contact between glass substrates is essential during the welding process. Optical contact is defined as having a gap less than one-fourth of the wavelength, reducing air within the contact area and minimizing optical reflection while temporarily bonding the surfaces through van der Waals forces. Therefore, it is necessary to analyze ultrafast laser welding of transparent materials at different spacings. Analyzing ultrafast laser welding at different spacings reveals the molten pool formation mechanism. Fig. 8(a) and Fig. 8(c-e) illustrates the process: a focused ultrafast laser at the interface of two

transparent samples induces multiphoton ionization. Ionized electrons gain energy, leading to avalanche ionization and increased ion temperature, causing localized melting and solidification upon cooling, thus welding the glass. The electronic excitation and relaxation processes in glass induced by ultrashort pulse laser irradiation are shown in Fig. 8(c-e). Relaxation process is shown in Fig. 8(c-e). From Fig. 8(c-e), it can be seen that the electronic excitation and relaxation processes in glass can be divided into the following three parts [84]: 1) When the laser electromagnetic field is very strong, electrons first pass through the ultrashort multi-photon absorption (multi-photon ionisation) or tunneling ionisation of pulsed laser light excitation from the valence band to the conduction band; 2) The excited electrons can sequentially absorb several laser photons and move themselves to a higher energy state, achieving electron heating. Once electrons are excited into the conduction band, they can absorb energy from the laser field via inverse bremsstrahlung absorption (IBA), also known as free-carrier absorption (FCA), which contributes to electron heating. Simultaneously, these energetic electrons may undergo impact ionization by colliding with lattice atoms, generating secondary electrons. This combination of IBA/FCA and impact ionization constitutes avalanche ionization, a critical process for electron density buildup during femtosecond laser-matter interaction; 3) Some free electrons relax in electron-cavity pairs, localising the stored energy and thus producing localised states such as self-trapped excitons (STE). STE relaxes within a few hundred picoseconds to form permanent defects in the trap. Other free electrons convert their energy into lattice vibrations, which generate heat and then relax to the ground state. Glass heating occurs tens of picoseconds after laser irradiation and the illuminated region returns to room temperature after a few tens of milliseconds [85,86].



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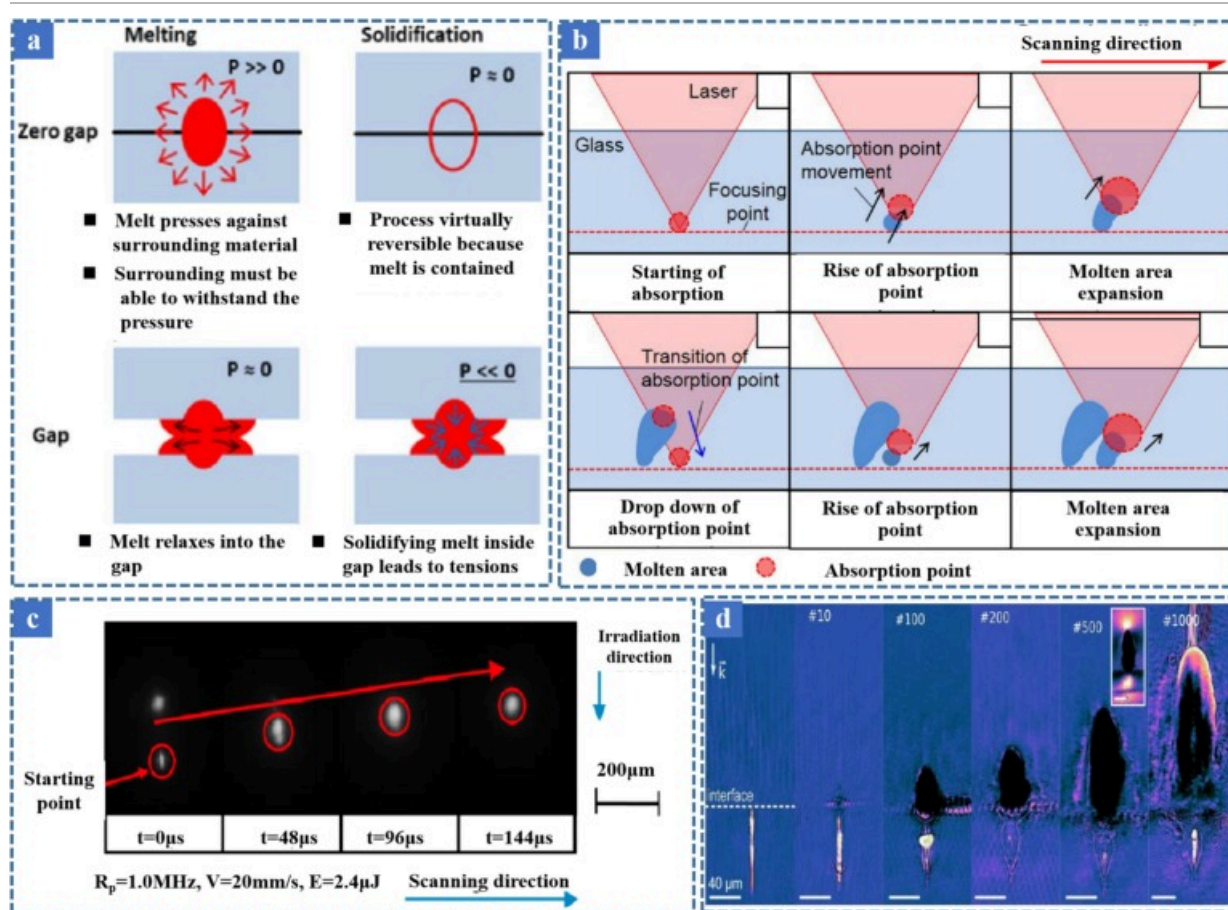
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Fig. 8. Ultrafast laser welding mechanism of transparent material-transparent material (a) Ultrafast laser micro-welding process of brittle transparent material; (b) The water droplet double structure action zone produced by ultrashort pulse laser acting on glass material [47]; (c) Generation of free electrons by multiphoton ionisation or tunneling ionisation; (d) Avalanche ionisation after electron heating and collisional ionisation; (e) Formation of localised states and free electron relaxation [84].

When a femtosecond laser beam with medium pulse energy is focused on the interface of the stacked glass substrate, by this heating, only localised melting near the interface is induced and rapid re-solidification, which leads to glass joining. Reducing the peak power density below the ionization threshold, often by defocusing, returns the laser action zone to the focal point. The internal plasma transfers energy to surrounding materials via thermal diffusion, forming an external melting modification zone. This double structure, comprising the internal plasma action zone and the external melting modification zone, constitutes the laser action zone [47], as shown in Fig. 8(b). Unlike metal welding joints, which consist of weld, fusion, and heat-affected zones due to linear absorption and mild

energy deposition, ultrafast laser welding of glass generates a strong central plasma action zone. This heats surrounding material through thermal diffusion, changing the refractive index and forming a melting modification zone.

Fig. 9(a) shows molten pool variations with and without gaps [87]. After the laser pulse, the melt pool cools and solidifies. Under zero-gap conditions, cooling contraction returns the material to its original volume and stress. With larger gaps, some melt diffuses into the gap and solidifies under tensile stress. Welding large-gap glass involves high-energy multi-pulse sequences to optimize the affected region. This strategy melts the lower-layer material, creating an expanded molten pool. Molten material bridges the gap through reversible and irreversible expansions. Reference [88] discusses using four laser pulses for quartz welding at 10mm/s and 1 mm/s, with focal depths from 650µm to 5µm intervals. When the laser-induced structure approaches the surface, softened material protrudes and deforms due to thermal expansion and plasma pressure, leading to reversible expansion [74,89]. If close enough, van der Waals forces and temperatures above the softening point cause the material to extend or diffuse towards the interface, forming connections upon cooling and reducing the gap through compressive stress.



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Fig. 9. Mechanisms of plasma formation during ultrafast laser micro-welding of transparent and transparent materials (a) The schematic diagram of glass welding with and without gap [87]; (b) The schematic diagram of the formation mechanism of the molten pool during the ultra-fast laser scanning process [11]; (c) The process of plasma motion [90]; (d) Pump detection diagram of ultra-short pulse laser welding glass action zone [91].

The schematic diagram of the formation mechanism of the molten pool during the ultrafast laser scanning process is shown in Fig. 9(b) [11]. Plasma first produces a nonlinear absorption point at the laser focal point, developing along the optical axis due to energy accumulation, and finally returning to the focus. Due to laser scanning, the absorption point deviates, forming a melting area. High peak power density at the laser focus leads to multi-photon ionization and tunnel ionization, generating plasma at the focus [91]. Fig. 9(c) illustrates plasma formation near the focus, expanding upward and disappearing, with periodic changes in structure height, size, and brightness [90]. Plasma shielding prevents subsequent laser penetration, causing the action point to rise with each pulse, creating a plasma column, shown in Fig. 9(d). This column has a high free electron density, around $10^{19}/\text{cm}^3$. For welding glass with larger gaps, post-processing reduces residual tensile stress, ensuring quality. Optical contact minimizes gaps and residual stress, enhancing welding quality. Irreversible expansion decreases material spacing, facilitating connection of the convex interface.

In addition to the above welding mechanism, there are also methods to implement welding by using the self-focusing effect of ultrashort pulse laser [20,29,60,92,93]. This mechanism does not require a high pulse repetition frequency, only the laser needs to have sufficient peak power conditions. The ultrashort pulse laser propagates in the glass material, and the filament is formed due to the self-focusing effect, and its length is mainly determined by the pulse energy. When the filament passes through the optical contact interface between two transparent materials, the plasma generated in the filament will transfer the absorbed laser energy to the surrounding glass materials. At the same time, the accumulation of pressure in the focusing area will cause microscopic deformation of the interface, prompting the molten material of the upper and lower materials in the filament area to mix together to form a weld [94]. Since there is no thermal accumulation effect, there is almost no thermal diffusion. The rapid cooling of the laser-affected zone results in a weld width of only a few μm .

When only single pulses or low-repetition-rate pulses (e.g., 1 kHz) are used, the deposited energy quickly dissipates through thermal diffusion, allowing the material to cool down

rapidly. As a result, only minor permanent structural modifications or localized damage typically occur at the focal point, rather than actual melting. This phenomenon is commonly referred to as “cold processing” in the context of ultrafast lasers. The key to successful ultrafast laser micro-welding lies in the thermal accumulation effect, which requires a very high laser repetition rate—typically exceeding 100 kHz and often reaching the MHz range. At such high repetition rates, the next laser pulse arrives before the heat generated by the previous pulse has fully dissipated, leaving the lattice in a “preheated” state. Consequently, the series of pulses acts like a continuous heat source, gradually increasing the temperature until it surpasses the melting point of the material, such as $\sim 1700^{\circ}\text{C}$ for fused silica, thus forming a localized molten pool. In this case, the pulse interval ($\sim 1\mu\text{s}$) is shorter than the thermal diffusion time ($\sim 2\mu\text{s}$), allowing effective heat accumulation.

In contrast to low repetition rate regimes, when the repetition rate of an ultrafast laser exceeds approximately 100 kHz, the time interval between consecutive pulses becomes shorter than the characteristic thermal diffusion time from the focal region. As a result, energy delivered by successive pulses accumulates locally, leading to a progressive temperature rise in the focal volume before reaching thermal equilibrium. This pronounced heat accumulation effect becomes especially evident at repetition rates around 250 kHz, where the continuous injection of energy causes not only local melting but also extended thermal diffusion into the surrounding material over longer time scales[95]. This high-repetition-rate regime, often referred to as thermal ultrafast laser direct writing (ULDW), results in a modification zone that exceeds the focal spot size due to significant melting. While nonthermal ULDW is widely used for precise microstructuring with minimal collateral damage, the thermal regime opens up additional opportunities. Thermal accumulation has been shown to play a critical role in inducing complex structural transformations and enhancing device performance in transparent materials. For instance, it can promote more symmetric waveguide geometries, reduce optical propagation losses, and significantly improve processing

efficiency, making it highly relevant for applications in integrated photonic device fabrication.

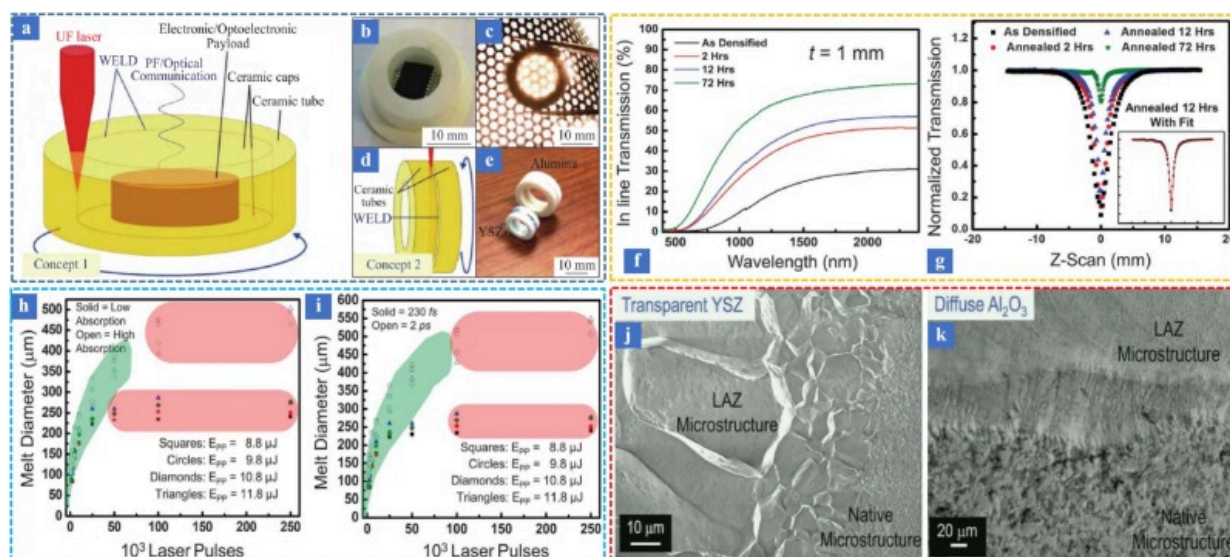
These studies collectively illustrate that thermal accumulation can be harnessed not only to mitigate stress and prevent crack formation in brittle optical materials, but also to unlock new phase-transformation pathways for nanoscale patterning [96]. Such engineered heating regimes substantially broaden the capabilities of ultrafast laser micro-welding and direct-write processes across diverse material classes.

2.2. Micro-welding between transparent and semi-transparent optical material

In recent years, ultrafast pulse (UF) lasers have emerged as advanced tools for ceramic welding. Studies have indicated that glass, particularly borosilicate, processed by ultrafast lasers exhibits lower fracture strength and thermal shock resistance compared to typical engineering ceramics such as zirconia and alumina. Consequently, the thermal-mechanical properties of ceramics, which are often the focus in laser welding, do not pose a significant challenge. However, a stark difference between ceramics and glass lies in their optical properties, with ceramics often being translucent or opaque to lasers. This distinction constitutes the primary divergence in the laser welding processes for ceramics and glass.

When an ultrafast laser is precisely focused on the interface micro-area for welding, it initiates a nonlinear multi-photon absorption process, leading to material melting at the interface. The success of ultrafast laser welding of ceramics hinges on adjusting the ceramics' optical transparency, allowing precise control of laser energy and material interaction. Researchers, including Garay from the University of California, have explored ultrafast pulsed lasers for welding yttria-stabilized transparent zirconia (YSZ) and polycrystalline alumina ceramics (Fig. 10(a-e)) [53]. Together, the data in Fig. 10(f-g), show that both the linear and NLA (linear and nonlinear absorption) of YSZ can be tuned by simple annealing treatments. The high linear transparency makes it possible to focus deep in the material, similar to glasses, and the large NLA enables efficient laser-material

coupling in the beam focus, allowing precise placement of energy at part interfaces that lead to localized melting useful for welding. The LAZ for both samples is similar at a low number of pulses (2500 and 5000) but is larger for the high-absorption sample as the number of pulses increases (Fig. 10(h)). For ceramics with high melting points, high-repetition-rate femtosecond lasers generate thermal accumulation, forming a molten pool at the interface that solidifies into a welded region. The laser-affected zone (LAZ) (Fig. 10(j)) shows larger, elongated grains, indicative of melting, compared to the native ceramic microstructure, which resembles the transparent portion. An example weld using this strategy was made on diffuse alumina (Fig. 10(k)). Adjusting thermal processing parameters can produce YSZ ceramics transparent to near-infrared lasers with high nonlinear absorption, enabling ytterbium-doped fiber laser welding of ceramics. This method creates ceramic tubes with high gas tightness for electronic encapsulation and enhances joint strength. The technique demonstrates ultrafast pulsed lasers' potential for welding high-melting-point materials and can be applied to other ceramics like alumina. It holds significant promise for industries such as ceramic microelectromechanical systems, photonic packaging, and more.

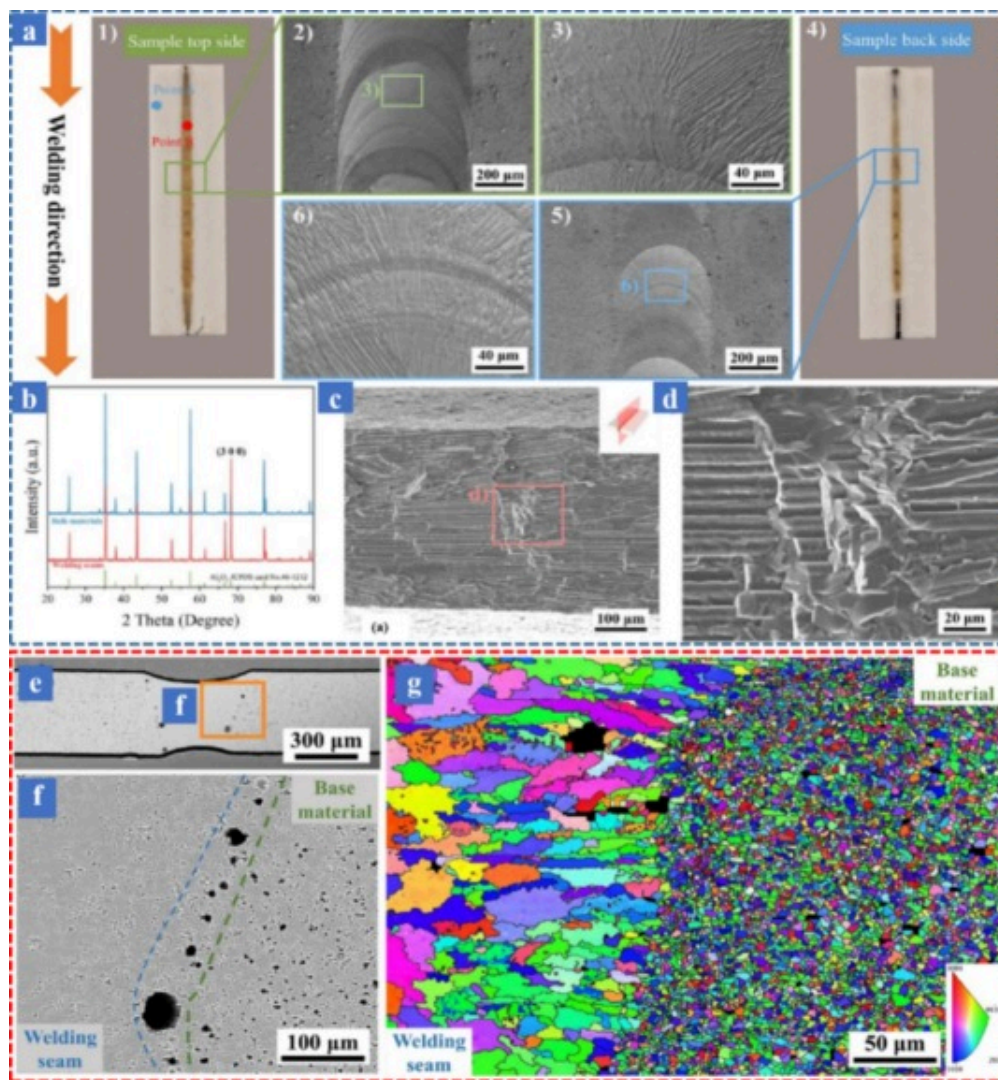


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Fig. 10. Ultrafast laser welding of yttria-stabilized zirconia (YSZ) ceramics [53]; (a-e) Two concepts for ultrafast laser welding of ceramics [53]; (f) Linear transmission measurements of YSZ samples with varying heat treatments (annealing time in air) [53]; (g) Nonlinear transmission (Z-scan) measurements of YSZ with varying annealing time [53]; (h) Effect of number of laser pulses on resulting melt diameter for samples with low and high absorption [53]; (i) Effect of number of laser pulses on resulting melt diameter for 2-ps and 230-fs pulse widths [53]; (j) SEM micrographs of cross section of YSZ near LAZ; (k) SEM micrographs of cross section of alumina near LAZ [53].

Penilla et al. [53] demonstrated room temperature ultra-short pulse laser (USPL) welding for ZrO_2 (Al_2O_3) ceramics, while Yuan et al. [54] achieved USPL welding of alumina ceramics, analyzing microstructures and fracture morphology. Optical and SEM images (Fig. 11(a)) showed defect-free joints with ripple-shaped seams and columnar crystals, indicating melting and resolidification. XRD analysis (Fig. 11(b)) confirmed no phase transformation, identifying both phases as $\alpha\text{-Al}_2\text{O}_3$. The fracture surface along the welding direction (Fig. 11(c-d)) revealed columnar crystals with a defined growth direction. The alumina in the welding seam formed a preferred orientation, influencing the XRD pattern intensity (Fig. 10(e-f)). The welding seam exhibited lower porosity than the base material, but large pores were observed at the fusion line, likely due to trapped gas during solidification. EBSD imaging (Fig. 10(g)) showed grain coarsening in the welding seam without a heat-affected zone, attributed to high welding speed and ceramic grain stability. Optimizing welding parameters achieved a maximum bending strength of 266MPa, 60% of the base metal strength. This study proposes a welding mechanism based on nonlinear absorption of ultra-short pulse lasers, offering valuable insights for further research in this innovative technology.



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Fig. 11. Direct welding of diffuse alumina ceramics by ultrashort pulse laser (USPL) [54] a) Typical morphology of the sample bonded by USPL; 1) Optical image of the sample top side, 2), 3) High-magnification SEM images of 1), 4) Optical image of the sample back side, 5), 6) High-magnification SEM images of 4); b) XRD patterns at points A and B in Fig.

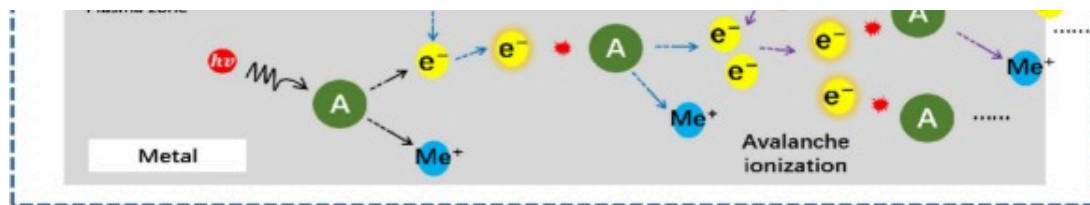
11a; c) Morphology of the typical fracture surface of the welding seam; d) High-magnification image of c); e) Typical morphology of the cross-section surface of the sample; f) High-magnification image of e); g) EBSD image of the region between the welding seam and base material [54].

2.3. Micro-welding between transparent optical material and opaque material

2.3.1. Mechanism of ultrafast laser micro-welding between transparent optical materials-opaque materials

Ultrafast laser pulses, when focused at the interface between transparent and opaque materials, induce nonlinear absorption in the transparent material and linear absorption in the opaque material. Nonlinear absorption processes, like multiphoton absorption and tunnel ionization, coexist with linear absorption in the opaque material, leading to plasma generation within the focusing volume. Electrons rapidly transfer their energy to the lattice, establishing thermal equilibrium, and subsequently causing localized melting and re-solidification at the interface, effectively connecting the two materials [38], as illustrated in Fig. 12(a-b). A schematic of ultrafast laser welding of metal and glass via thermal conduction is shown in Fig. 12(c). Here, the high peak power density irradiates the interface, causing nonlinear absorption in the upper glass layer through multiphoton absorption (MPA), leading to ionization. The conduction band free electrons formed absorb additional photons, resulting in collision ionization and avalanche ionization, gradually forming plasma. This high-temperature plasma transfers energy to the lattice via electron-phonon coupling, causing the material to melt or vaporize around the focal point. The lower metal layer undergoes linear absorption, primarily through inverse bremsstrahlung absorption, leading to a high-density plasma state that amplifies laser absorption. This results in electron-electron and electron-phonon scattering, and lattice heat conduction, culminating in a phase explosion that generates high-temperature and high-pressure gas and liquid, which mix dynamically [97,39].





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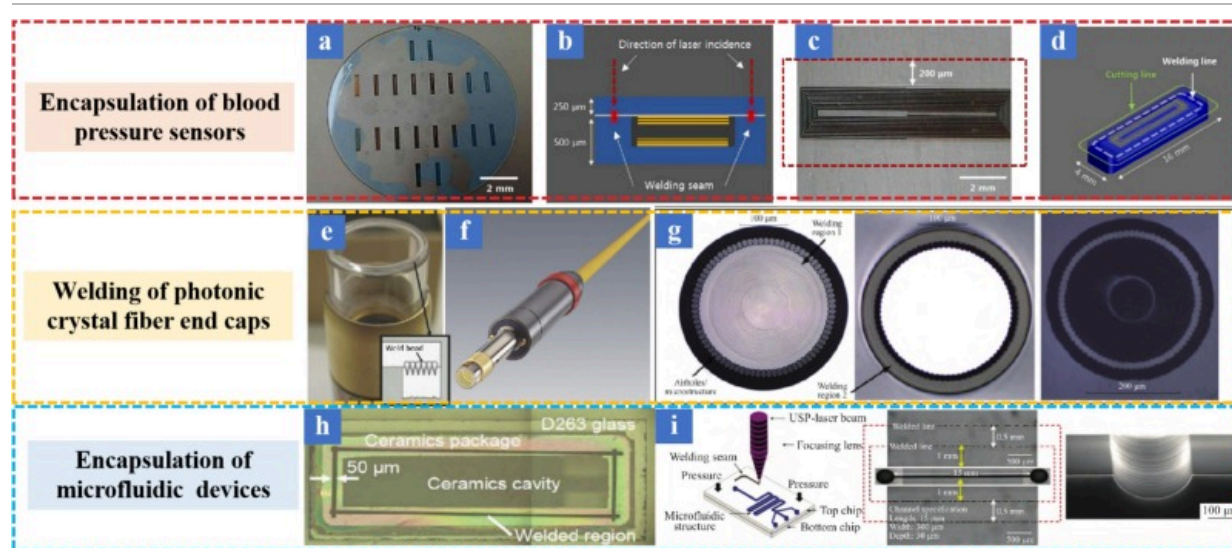
Fig. 12. Ultrafast laser welding phenomena of transparent-opaque materials (a) Ultrafast laser welding of transparent-opaque materials; (b) Ultrafast laser micro-welding process of transparent material-opaque material [38]; (c) Ultrafast laser microwelding heat transfer between transparent and opaque materials [39]; (d) Teardrop structure; (e) Nonlinear absorption by ultrafast laser pulses; (f) Optical-contact welding at different focal positions [98]; (g) Schematic diagram of energy deposition process of Ultrafast laser welding transparent materials and metals [99].

When joining transparent materials and metals, the metal substrate undergoes thermalization via linear absorption, differing from the transparent material's nonlinear absorption (Fig. 12(g)) [99]. This leads to electron–electron and electron–phonon scattering, forming a highly constrained plasma and melts. As the temperature reaches the thermodynamic critical point, a phase explosion occurs, generating high-temperature and high-pressure metal vapor and melts, which mix and fill the gap between the substrates, ensuring a reliable weld joint. The mixed melts attach to the opposite transparent material, transmitting heat and bridging gaps, forming a solid weld upon cooling. The ultra-fast laser focus on the interface of two materials, the high-energy laser of the focus induces multiphoton ionization in glass, the speed of the ionizing free electrons increases by absorbing laser photon energy, and the electrons collide with other atoms, producing avalanche ionization. The coupling of electron and phonon makes the temperature of ion and unionized part rise to the melting point of the material and the welding is successfully realized [98], as shown in Fig. 12(e). The welding area presents a teardrop structure, as shown in Fig. 12(d). The figure is a schematic representation of

internal processing, and the tear-drop structure formed when the laser is focused within the material. The position of the tear-drop structure varies according to the position of the focus relative to the interface between the glass and the silicon. Therefore, it is speculated that the change of the focal position will affect the width of the weld and the length and density of the filamentous structure, which will, in turn, affect the strength of weld.

2.3.2. Application of micro-welding to transparent optical material and opaque material

With the threshold effect of non-linear absorption, ultrafast laser welding can be applied not only for the encapsulation of microdevices and sensors, but also for the joining of consumer electronics and aerospace window devices (as depicted in [Fig. 13\(a-d\)](#) [67] and [Fig. 13\(i\)](#)) [66,100]. Additionally, this technique can be employed for direct selective welding of multi-layer materials. Given the ability of ultrafast lasers to create extremely high temperatures in the target region, they are particularly well-suited for welding high-melting-point materials. By modulating the laser-induced thermal effects, the minimum feature size of the ultrafast laser welding area can be reduced to the micron scale, enabling ultra-precise selective welding of small structures like photonic crystal fiber end caps [101], as illustrated in [Fig. 13\(e-g\)](#) [101]. Itoh et al. [102] have applied femtosecond laser welding to the sealing of ceramic samples. [Fig. 13\(h\)](#) [102] displays a photograph of a glass substrate sealing a ceramic chamber. The traditional production of glass microfluidic devices entails a complex multistep process that necessitates a combination of various manufacturing techniques, including photolithography, chemical/dry etching, and thermal/anodic bonding. All of these methods necessitate additional equipment such as furnaces, wafer bonding machines, and cleanrooms, significantly increasing the manufacturing costs of glass microfluidic devices. In summary, with a wide range of applications in areas such as the packaging of microdevices and sensors, and the joining of consumer electronics and aerospace window devices will inevitably involve the welding of transparent and opaque materials.



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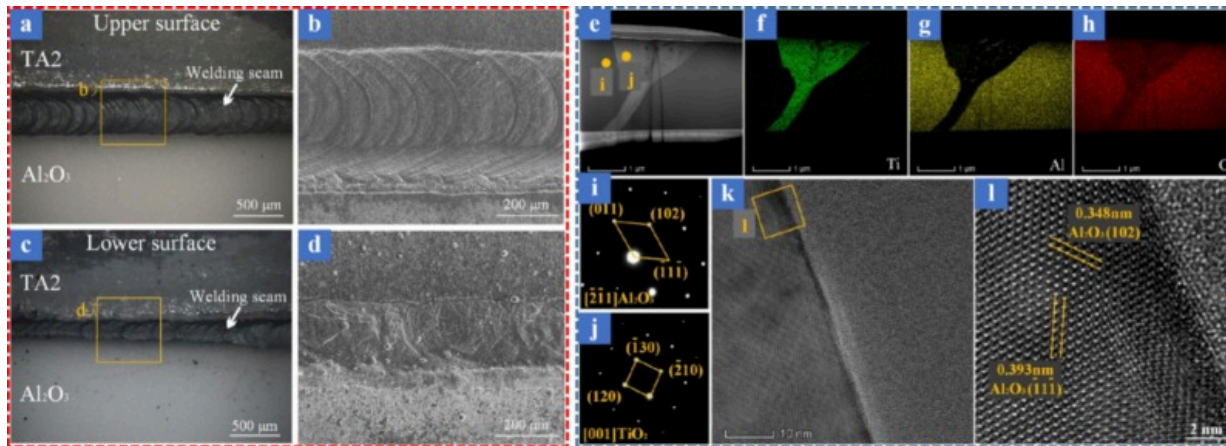
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Fig. 13. Application of Ultrafast laser microwelding in assembling of brittle material-based components (a-d) Ultrafast laser selective encapsulation of blood pressure sensors [67]; (e-g) Laser selective welding of photonic crystal fiber end caps [101]; (h) Sealing of ceramic chambers with glass substrate [102]; (i) Ultrafast laser selective encapsulation of microfluidic devices [66,100].

2.3.3. Micro-welding between ceramic and metal materials

The significant differences in thermal expansion coefficients between ceramics and metals, coupled with the brittle nature of ceramics, pose challenges for ultrashort pulse laser welding at room temperature without preheating. Such differences can lead to residual stress and joint cracking. Moreover, the contrasting metallurgical properties of ceramics and metals raise concerns about their compatibility under ultrashort pulse laser welding. However, ultrashort pulse laser (USPL) welding presents a viable method for joining ceramics and metals without preheating. Yuan et al. [56] successfully welded alumina ceramic and titanium using USPL, achieving defect-free bonding at room

temperature. Fig. 14(a, c) illustrates the surface morphology of the Al_2O_3 -TA2 sample welded with a laser power of 19.52 W and a welding speed of 0.8 mm/s, with magnified images of the welding seam shown in Fig. 14(b, d). Transmission electron microscopy (TEM) analysis identified TiO_2 and Al_2O_3 phases at the interface, confirming strong bonding. The HADDF image (Fig. 14(e)), combined with EDS mapping (Fig. 14(f-h)), indicates that the white phase contains Ti and O, while the dark phase contains Al and O. Selected Area Electron Diffraction (SAED) patterns (Fig. 14(i, j)) confirm that the dark phase is Al_2O_3 and the white phase is anatase TiO_2 . Additionally, brookite TiO_2 was observed (Fig. 14(k, l)), with high-resolution TEM images demonstrating excellent bonding at the $\text{Al}_2\text{O}_3/\text{TiO}_2$ interface. The joint achieved a maximum bending strength of 134.9 MPa, exceeding traditional methods. Current techniques face challenges due to thick transition layers that impede heat flow, reducing device longevity. To address this, Song et al. [55] proposed a laser-assisted bonding method for direct ceramic-metal joining without intermediate materials. By applying a laser to β - Si_3N_4 ceramic, a Si microcrystalline layer with α - Si_3N_4 nanocrystals was created. The face-centered cubic (fcc) Si and hexagonal close-packed (hcp) β - Si_3N_4 substrates exhibited coherent orientation relations. The resulting defect-free $\text{Si}_3\text{N}_4/\text{Cu}$ interface, formed by reacting Si and Cu at 805–900 °C for 30 min, demonstrated strong and stable bonding. This novel laser-assisted approach is promising for ceramic-metal components in high-power device applications.



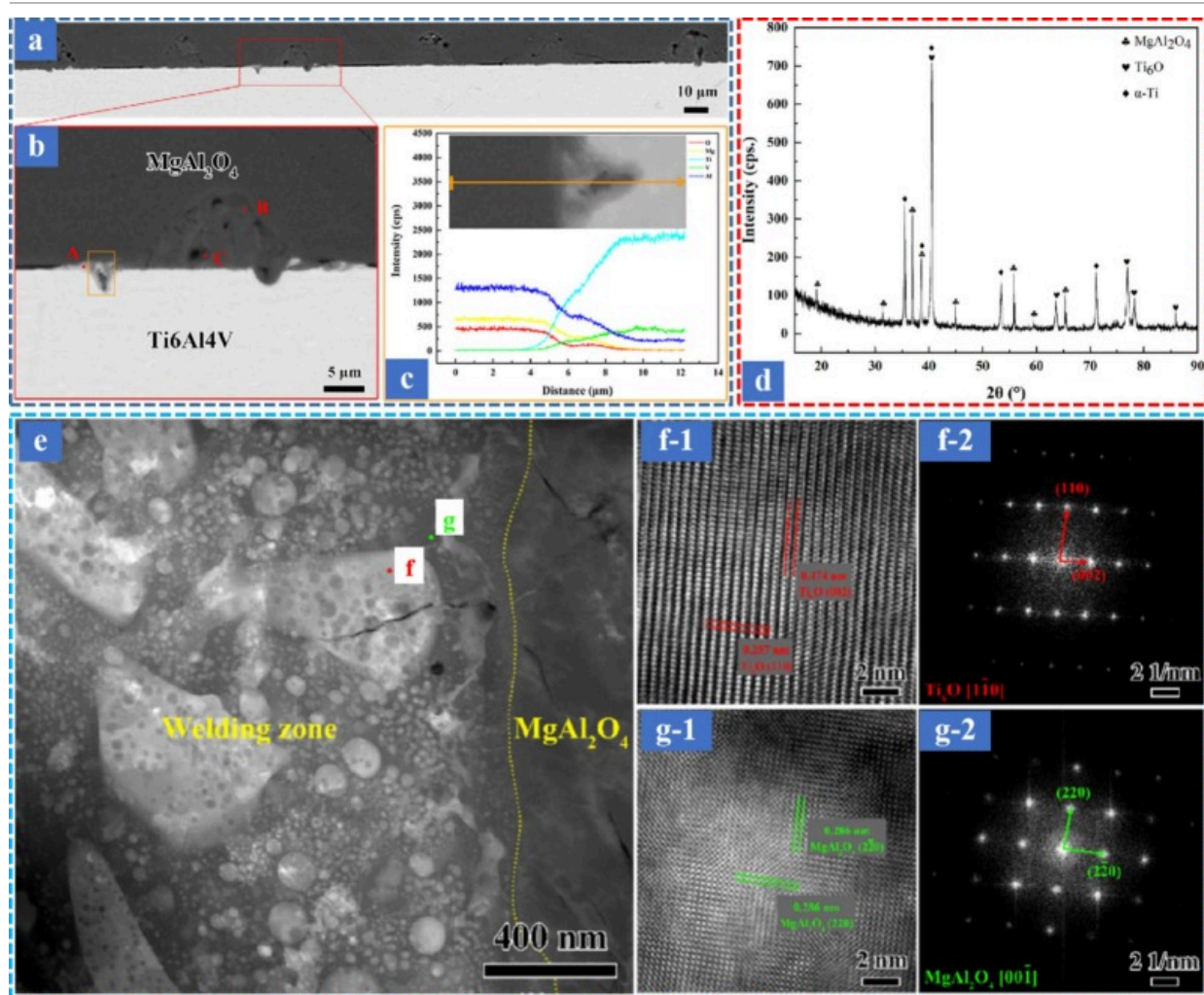
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Fig. 14. Ultrashort pulse laser welding of alumina ceramic and titanium[56] The typical optical morphology of the $\text{Al}_2\text{O}_3/\text{TA2}$ sample top (a) and back (c) sides bonded under laser power of 19.52W and welding speed of 0.8mm/s; (b), (d). high-magnification images of a), c); (e-h) TEM image of the Al_2O_3 /welding seam interface: e) the HAADF image, f)-h) element distributions of Ti, Al and O, i) and j) SAED patterns indicated in (e, k)HRTEM image of the interface in (e, l) high-magnification of (k) [56].

Transparent magnesia alumina spinel (MgAl_2O_4) is a key optical material known for its high strength, corrosion resistance, and transmittance, making it valuable for various industrial applications. Ti6Al4V, a widely used titanium alloy, offers high specific strength, corrosion and oxidation resistance, as well as good toughness at room temperature and creep resistance at elevated temperatures. Effective joining of MgAl_2O_4 to Ti6Al4V could enhance their combined potential. However, the disparate physical and chemical properties of ceramics and metals present significant challenges for effective bonding. Traditional joining methods often fall short of meeting modern industrial demands for high efficiency, minimal thermal damage, and precision. Recent advancements suggest that ultrashort pulse laser (USPL) welding could overcome these

challenges. Penilla et al. [53] demonstrated the feasibility of femtosecond laser welding for translucent ceramics, broadening the scope of USPL applications. Pan et al. [43] achieved a high-strength bond of 108.35 MPa between sapphire and Fe-36Ni alloy using femtosecond lasers, significantly improving upon previous results. Despite these advancements, there has been limited research on joining MgAl_2O_4 with Ti6Al4V. A recent study explored the innovative use of ultrashort pulse lasers with ultra-high peak power to bond transparent MgAl_2O_4 and Ti6Al4V [103]. The high energy near the laser focus facilitates the mixing and reaction of MgAl_2O_4 and Ti6Al4V, creating a robust joint. The resulting bond showed homogeneous mixing of phases, with the formation of a Ti_6O phase at the interface (Fig. 15). The study also examined the effects of laser power and scanning speed on the joint's microstructure and mechanical properties, highlighting the superior efficiency and precision of femtosecond laser welding compared to traditional methods.



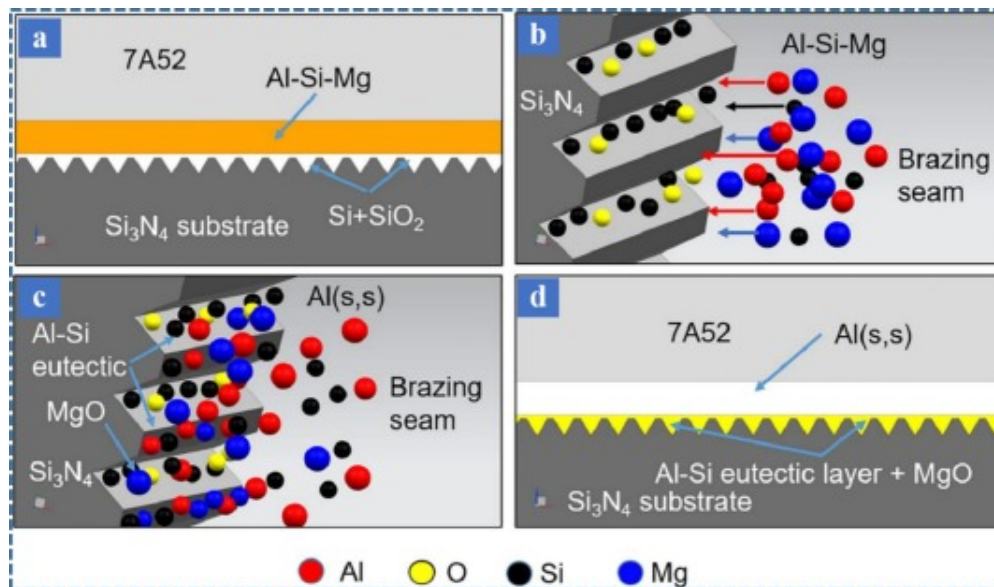
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Fig. 15. Reliable joining between MgAl_2O_4 and Ti6Al4V by ultrashort pulse laser [103] (a) The SEM image of the MgAl_2O_4 /Ti6Al4V brazing joint welded by femtosecond laser; (b) Magnification of the region in (a); (c) Element distribution linescanning of the welding zone; (d) The phases of the joint tested by XRD; (e-g) TEM image of the MgAl_2O_4 /welding zone interface: (e) the HADDF image, (f-1) the HRTEM image of Ti_6O , (f-2) the diffraction

pattern of Ti_6O , (g-1) the HRTEM image of MgAl_2O_4 , (g-2) the diffraction pattern of MgAl_2O_4 [103].

Fabricating ceramic–metal composites is challenging due to poor wetting between dissimilar materials. Aluminum-based brazing materials, commonly used in vacuum brazing of aluminum alloys, exhibit poor wetting on Si_3N_4 ceramics below aluminum's melting point (660.3°C). Techniques like active brazing and ceramic surface metallization can improve wetting but risk damaging substrates or forming brittle intermetallic compounds. Additionally, metallization of ceramic surfaces with aluminum alloys can generate brittle intermetallic compounds, reducing the joint's mechanical properties. Zhang et al. [104] proposed a femtosecond laser surface modification method for joining Si_3N_4 and 7A52 aluminum alloy. Their study demonstrated that laser processing created periodic grooves on the ceramic surface, optimizing the depth-to-width ratio to increase bonding area and enhance wetting. Fig. 16 illustrates the femtosecond laser-assisted brazing mechanism. Laser-induced thermal decomposition of Si_3N_4 produced a recrystallized Si layer, improving the diffusion of Al and Mg at the interface. This structure also altered the joint's fracture path, resulting in a 52.4% increase in strength compared to unmodified joints, all without requiring surface metallization. In conclusion, femtosecond laser processing significantly improved the ceramic–metal joint strength through surface modifications, yielding a more robust bond.



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Fig. 16. Connection diagram of Si_3N_4 /7A52 under femtosecond laser intensification [104].

The current application of ultrafast pulsed laser welding predominantly focuses on small ceramic parts, typically not exceeding 2mm in size. However, there is a promising trajectory towards optimizing this welding technology to facilitate its extension to large-scale welding tasks. Efforts are also underway to adapt the technology for welding components made of different materials and varying shapes. This evolving welding technology holds the potential to revolutionize the utilization of ceramic components, making them applicable in harsh environments. Furthermore, it opens new avenues for integration into optoelectronic devices or electronic packaging scenarios, where visible light wave penetration is essential. The advancements in ultrafast pulsed laser welding are poised to enhance the versatility and utility of ceramic components across various industries and applications.

2.3.4. Micro-welding between glass and metal materials

Heterogeneous material connections, particularly involving metals and glass, are crucial in various industrial applications. Ultrafast lasers excel in creating precise, localized heating zones, enabling simultaneous nonlinear absorption in transparent materials and linear absorption in opaque ones [75,105]. This technique facilitates direct welding of brittle materials to metals without bonding layers, streamlining the process and enhancing efficiency. Notably, ultrafast laser pulses reduce uneven heating, preventing high stress gradients that could lead to micro-cracks in weld joints. This technology preserves transparency and optical transmission efficiency of brittle transparent materials during welding, crucial for industrial applications. Recent advancements include achieving direct welding between aluminum silicate glass and metals using femtosecond laser pulses, yielding high shear joint strength. Various studies have demonstrated successful welding of optical materials like quartz, borosilicate glass, and sapphire to metals such as aluminum, titanium, and stainless steel, withstanding extreme conditions [29,33,34,[37], [57], [58], [59]]. **Ultrafast laser micro welding technology has enabled high-strength welding of glass materials with single-crystal silicon, alloys, and other heterogeneous materials (see Table 3).** Mechanical properties of weld joints remain stable even after extensive temperature cycling, meeting relevant standards requirements. Ongoing research explores welding diverse materials such as aluminum, copper, stainless steel, silicon, and silicon carbide to glass substrates, showing promising results for industrial applications.

Table 3. Recent works on ultrafsst laser microwelding of glass-metal.

Welding Materials	Laser Parameters			Remarks	Reference
	Pulse Width	Wave-Length	Repetition Frequency		
Cu/non-alkali glass	130fs	800nm	1 kHz	1.Optical polishing treatment and mechanical pressing;	[29]

Welding Materials	Laser Parameters			Remarks	Reference
	Pulse Width	Wave-Length	Repetition Frequency		
	600ns	527nm	1 kHz	2.Joint strength by fs laser: >16MPa.	
Cu/borosilicate	360fs	130fs	1 kHz	1.Polishing treatment and pressure assistance; 2.Joint strength: 20MPa.	[30]
Cu/fused silica W/fused silica	70fs	790nm	250kHz	1.Polishing treatment and clamping assistance; 2.Cu/fused silica joint strength: 9MPa.	[31]
NiTi/glass	35 fs	800nm	1 kHz	1.Bonding micrometric NiTi particle to glass; 2. Free of cracks.	[32]
Al/SiO ₂ Cu/SiO ₂ SS/borosilicate SS/Sapphire	10ps	1030nm	400kHz	1.Polishing treatment and clamping assistance; 2.In all cases the welds fractured around the modified region within the glass.	[33]
Cu/Alumina-silicate Al/Alumina-silicate Steel/Alumina-silicate	160fs	800nm	1 kHz	1.Polishing treatment and no pressure assistance; 2. Cu/glass joint shear strength: 2.34MPa.	[34]
Al/fused silica glass	5.9ps	1030nm	400kHz	1.Polishing treatment and clamping assistance; 2.Joint formation process:	[35]

Welding Materials	Laser Parameters			Remarks	Reference
	Pulse Width	Wave-Length	Repetition Frequency		
				nonlinear absorption, creation of plasma cavity and secondary keyholes; 3. Formation of nanocrystalline Si and, transitional alumina phases γ - and δ -Al ₂ O ₃ .	
Al-Bk7, Al-SiO ₂ and Nd:YAG-AlSi	5.9ps	1030nm	400kHz	1. Polishing treatment and pressure assistance; 2. Bonding of two materials with close thermal properties.	[36]
Al 6082/SiO ₂ Al 6082/BK7	5.9ps	1030nm	400kHz	1. Pressure assistance and no optically polishing treatment; 2. Optimized Al 6082/BK7 bonding strength: 13 MPa.	[37]
Cu/glass	550fs	1064nm	1 MHz	1. No $\nearrow$ polishing treatment on metal and no pressure assistance; 2. Joint shear strength: 1 MPa.	[38]
304 SS/soda lime glass	270fs	1040nm	400kHz	1. Surface grinding treatment but non-optical contact condition; 2. Joining mechanism: mechanical mixing, wetting and adsorption;	[39]

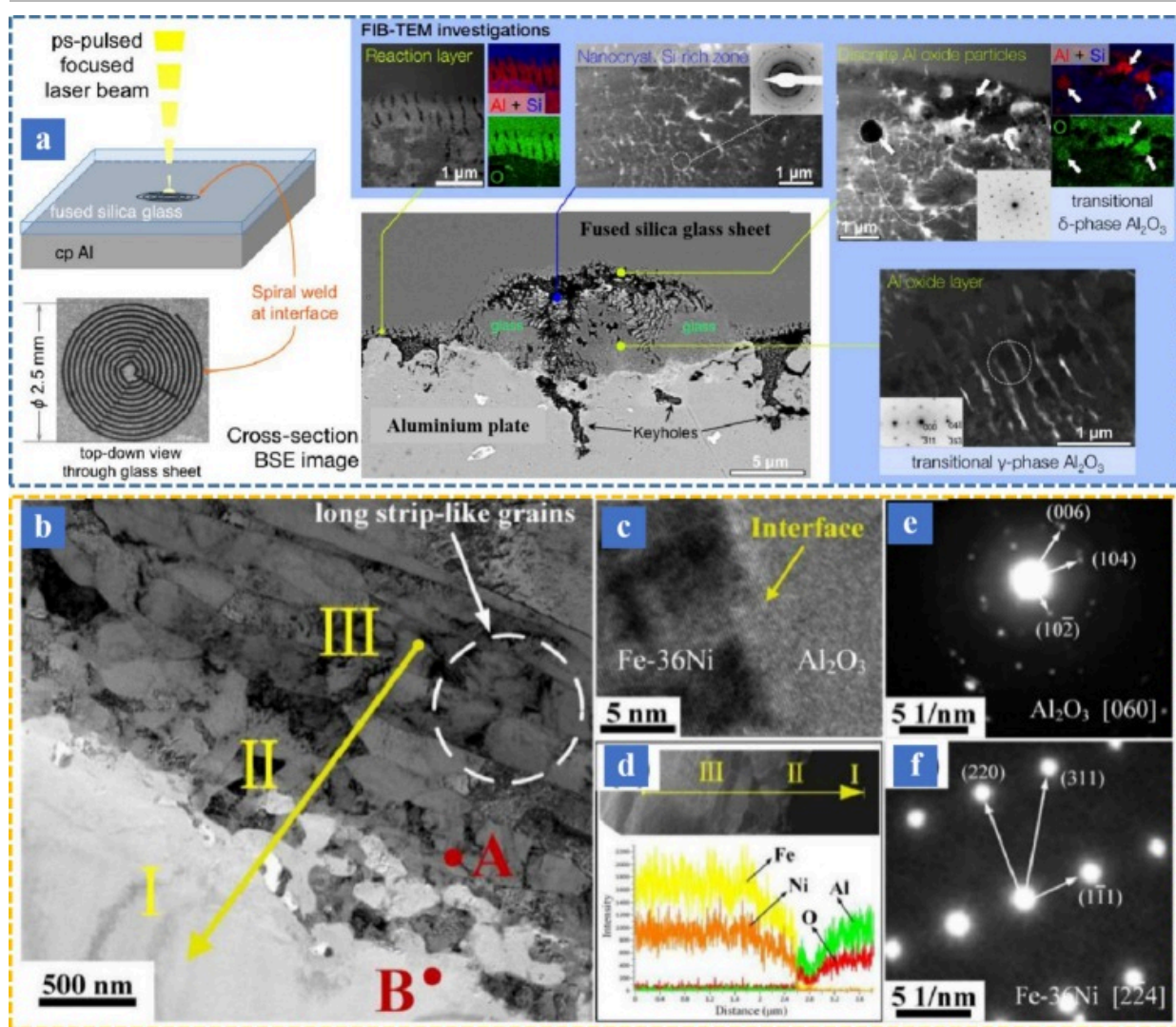
Welding Materials	Laser Parameters			Remarks	Reference
	Pulse Width	Wave-Length	Repetition Frequency		
				3.Optimized joint strength: 8.79MPa.	
Al 6082/SiO ₂	35 fs	800nm	1 kHz	1.Pressure assistance and no polishing treatment; 2.Forming a mechanical pin structure consisting of adhesive regions and a concave-convex structure between two substrates; 3.Shear strength: >100MPa.	[40]
Kover/soda lime glass	300fs	1030nm	1 kHz	1.No ↗ polishing treatment and no pressure assistance; 2.Material mixing and interdiffusion occurred; 3.Shear strength: 2MPa;	[41]
Tin powder/borosilicate glass Nickel powder/borosilicate glass Copper powder/borosilicate glass 304 stainless steel	350fs	1030nm	1.5MHz	1.Successfully and reproducibly deposited continuous tracks of powder materials on glass; 2.An ↗ inhomogeneous material composition of the interface region.	[42]

Welding Materials	Laser Parameters			Remarks	Reference
	Pulse Width	Wave- Length	Repetition Frequency		
powder/borosilicate glass					
Invar 36 alloy/sapphire	800fs	1030nm	200kHz	1.Polishing treatment and pressure assistance; 2.Welding interface consisting of a<1 μm mixed phase region with Fe-36Ni particles and sapphire; 3.Bonding mechanism: chemical bonding and mechanical interlocking; 4.Optimized shear strength: 108.35MPa.	[43]
K9 glass/304 stainless steel	----	1070nm	----	1.With a large gap of hundreds of microns; 2.The texturing pattern on the metal surface is designed to control the solder distribution, and the welding strength is improved by 3.5 times compared to that without patterns.	[44]
	8ps	1030nm	400kHz		
Fused silica glass/304 stainless steel	10ps	1030nm	200kHz	The welding of glass and metal with a gap;	[45]

Welding Materials	Laser Parameters			Remarks	Reference
	Pulse Width	Wave-Length	Repetition Frequency		
				Create an ultra-large molten pool.	
Fused silica/Al6082	35 fs	800nm	1 kHz	A mJ-level femtosecond laser system for aluminum and fused silica welding; Silicon nanoparticles and transitional γ-Al₂O₃ precipitate near the center of the weld; A maximum shear strength of 8.94MPa.	[46]

Connecting brittle materials to metals no longer requires polishing and pressure assistance to achieve optical contact. Matsuyoshi et al. [38] demonstrated the advantages of high repetition rate bonding substrates in a non-optical contact situation. In their experiment, copper-glass micro-welding was performed using a femtosecond laser with a repetition rate of 1 MHz, without pressure assistance or surface pretreatment of the copper substrate. The joint strength of the copper-glass sample and the post-welded sample under this condition was 1 MPa, equivalent to the bond strength between glass and polished copper.

In recent years, the research focus has gradually shifted from parameter mapping studies to the revelation of connection mechanisms. More and more related research projects are employing advanced imaging/characterization instruments to analyze the microstructure and composition at the interface, aiming to understand the behavior of weld formation. Ciuca et al. [35] achieved precise micro-welding joints between fused silica glass and aluminum using picosecond pulse laser technology. Electron microscopy methods analyzed the microstructure and interface reaction behavior of the weld (Fig. 17(a)). The weld involved forming a constrained plasma cavity, rapid heating by non-linear absorption in the glass, and secondary keyholes in the metal. The energy absorbed by the silica glass melted the aluminum substrate over a wider area. Extensive reactions in the weld zone formed nanocrystalline silicon and transitional alumina phases. Amorphous alumina with ultrafine γ - Al_2O_3 crystallites and a thin Si layer were observed outside the main interaction volume. Only γ - and δ - Al_2O_3 phases were confirmed in the weld zone, attributed to their favorable nucleation kinetics at high supercoolings. Pan et al. [43] used femtosecond laser micro-welding to connect sapphire and Fe-36Ni alloy, achieving a complete joint without voids or microcracks and an interface width of less than $1\text{ }\mu\text{m}$. TEM analysis of the joint domain (Fig. 17(b-f)) revealed that zone I is sapphire, zone II is the interface domain, and zone III is Fe-36Ni alloy. Zone II comprises Fe-36Ni and sapphire particles, verified by SAED patterns (Fig. 17(e) and (f)). HRTEM images showed chemical bonding between Fe-36Ni and sapphire without other phases forming at the interface (Fig. 17(c)). The lack of element diffusion or metallurgical reaction is attributed to the metallurgical inactivity between Fe-36Ni and sapphire and the high localization and small thermal effect of femtosecond laser irradiation. Chemical bonding and mechanical interlocking occurred during the “cold” femtosecond laser ablation process, resulting in serrated interface features.



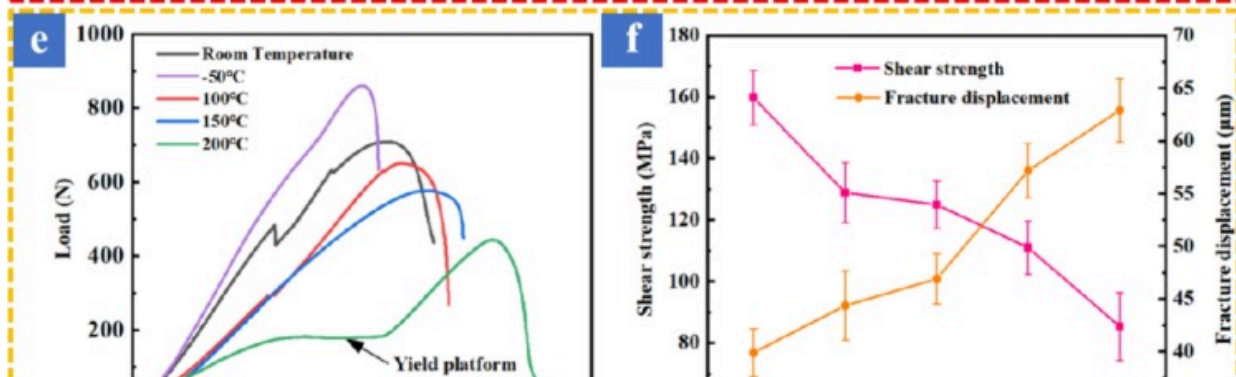
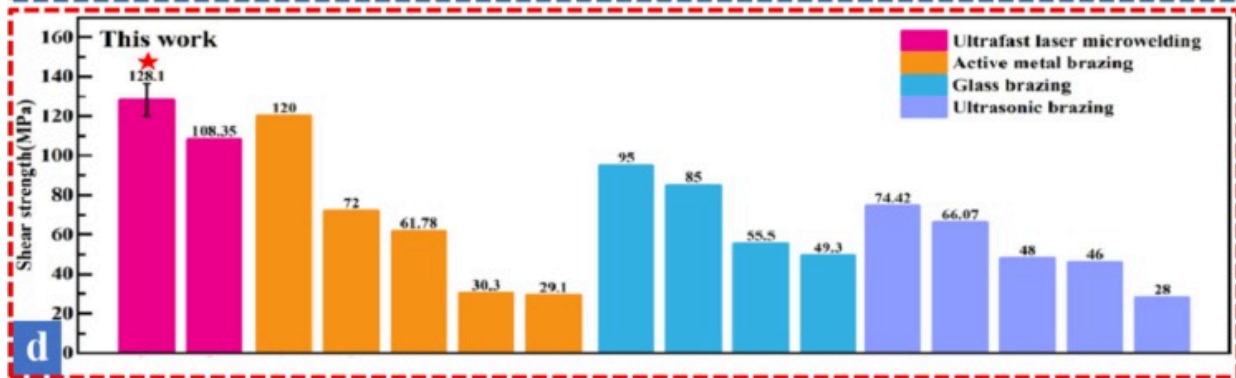
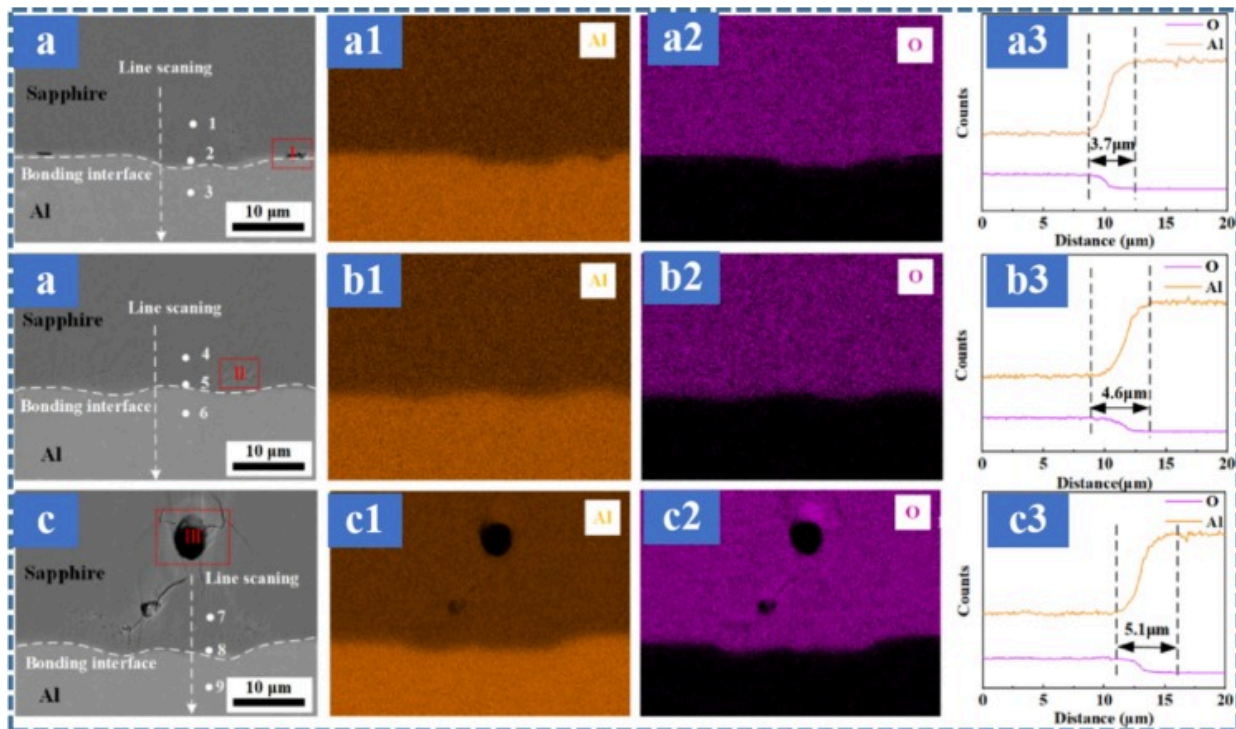
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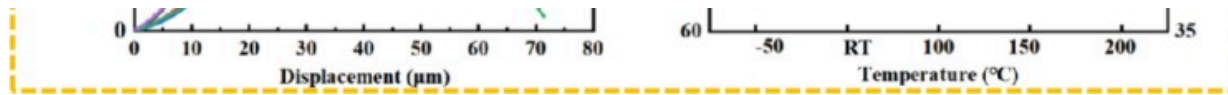
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Fig. 17. Ultrafast laser welding of glass-Metal connections (a)SEM images and element distribution of welded seam between glass and aluminum [35];(b-f)SEM images and element distribution of welded seam between sapphire glass and metal [43].

Ultrafast laser micro-welding has the advantages of high spatial selectivity and small thermal effect, and is considered to be a promising alternative method for transparent material connection. With the absorption of nonlinear photons, high energy will be deposited in the laser focus area, where melting/expansion/solidification processes and strong material interactions can occur, thereby promoting joint forming. On this basis, people try to use ultrafast laser to connect different combinations of materials, including oxide /metal [35,43], oxide/semiconductor [[61], [62], [63],93] and oxide/oxide [53]. Although the bonding strength has been improved by optimizing the structural design, improving the processing flow and modulating the laser energy, it is still necessary to understand the internal bonding mechanism and further improve the comprehensive performance to meet the requirements of practical applications. Robust joints of sapphire and aluminum were successfully achieved by direct ultrafast laser transmission microwelding [106]. Effects of laser parameters on the microstructures and mechanical properties were comprehensively studied. The welding process highly depended on the laser ablated metal nanoparticles within the heterojunction, as well as the thermal accumulation under multiple laser pulses irradiation. As shown in Fig. 18(a-c), SEM and EDS results of the cross sections further exhibited the clear and tortuous interfaces between sapphire and aluminum. There was no macroscopic mixing or blending between two base materials, indicating a typical characteristic of nonblending joints of brittle materials. Fig. 18(d) summarizes the bonding strength comparison of sapphire/sapphire and sapphire / metal under different methods. The results show that the sapphire/aluminum joint welded by ultrafast laser transmission hybrid welding has the highest shear strength at room temperature. Among them, the shear strength of ultrafast laser welded sapphire/aluminum joint is 72.1% higher than that of brazed sapphire/5A06 aluminum alloy joint (74.42 MPa). Compared with traditional material bonding methods such as brazing and diffusion bonding, ultrafast laser micro-welding has higher local heating capacity and smaller heat-affected zone, which can significantly reduce the thermal residual stress at the joint. To evaluate the performance of sapphire/aluminum composite structures under extreme operating conditions as heat dissipation substrates, shear tests were conducted on laser-welded joints at various temperatures under a laser

fluence of $0.56\text{J}/\text{cm}^2$. The load–displacement curve is shown in Fig. 18(e), and the corresponding shear strength and maximum fracture displacement statistics are shown in Fig. 18(f). Interestingly, when the test temperature increases, the shear strength of the ultra-fast laser welded joint gradually decreases, and the fracture displacement gradually increases. Especially at -50°C , the shear strength can reach $159.8\pm 8.9\text{MPa}$ due to the increase of deformation impedance, showing excellent low temperature strength. This work demonstrates the feasibility of using ultrafast laser to achieve high-quality sapphire/aluminum joint, which is promising in heat-dissipation substrate manufacturing in power semiconductor devices.





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Fig. 18. High-strength and impermeable sapphire/aluminum joints fabricated by ultrafast laser microwelding [106] (a-c) Cross-sectional morphologies and corresponding EDS results of sapphire/aluminum joints at different laser fluences: (a)-(a3) $0.48\text{J}/\text{cm}^2$; (b)-(b3) $0.56\text{J}/\text{cm}^2$; (c)-(c3) $0.64\text{J}/\text{cm}^2$; (d) Shear strength of sapphire/sapphire and sapphire/metal joints achieved by different methods; (e-f) Load-displacement curves of sapphire/aluminum joints at different testing temperatures [106].

Femtosecond laser welding of aluminum to fused silica has primarily focused on mechanical properties and microstructure, but the fusion process remains inadequately understood. Zhang et al. [46] investigated a millijoule(mJ)-level femtosecond pulse laser welding technique (Fig. 19(a)), identifying five distinct zones in the weld seam (Fig. 19(b)): base fused silica (B-FS), fused silica fusion zone (FS-FZ), aluminum-fused silica mixture zone (Al-FS-MZ), aluminum fusion zone (Al-FZ), and base aluminum alloy (B-Al). In the rapidly cooled aluminum pool, epitaxial grains formed along the temperature gradient, with high-angle grain boundaries (HAGBs). Supercooling and the presence of amorphous SiO_2 inhibited the γ -to- α phase transition in the Al-FZ. Fig. 19(c-e) and (f-k) show the presence of Al nanoparticles and amorphous Al_2O_3 near the Al-FZ and Al-FS-MZ boundaries, creating $\text{Al}_2\text{O}_3/\text{Al}$ composite microstructures. The weld exhibited interlaced amorphous Al_2O_3 and micron-scale alumina particles, with no $\gamma\text{-Al}_2\text{O}_3$ present. Rapid cooling suppressed nucleation, forming Al_2O_3 through the reaction $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$. Fig. 19(l) outlines the weld seam formation mechanism in four stages: 1) Laser pulse ionization and nonlinear absorption, 2) Plasma formation and heat transfer, 3) Material diffusion and solidification, and 4) Grain refinement and consolidation. This process produced a durable joint with a maximum shear strength of 8.94MPa. Further studies on

Figure 1 Schematic diagram of the femtosecond laser welding process and the resulting microstructure. (a) Schematic of the welding setup showing fused silica, aluminum alloy, and the femtosecond laser with force and welding direction. (b) SEM image of the weld cross-section showing layers: B-FS, FS-FZ, Al-FS-MZ, Al-FZ, and B-Al. (c) EDS spectrum showing peaks for Si, Al, and O. (d) EDS line scan showing concentration profiles of Al, Si, and O. (e) EDS elemental maps for Al, Si, and O. (f-k) High-magnification SEM images of the weld interface showing the lamellar structure. (l) Schematic of the welding process at different time scales: 0 fs (laser focus), fs~1 ps (plasma formation), and 1 ps~1 ns (molten SiO₂ and micro-droplets). (m) Schematic of the weld structure at different time scales: ns~μs (Al-FZ, Al-FS-MZ, FS-FZ, B-FS) and μs~ms (B-Al, Al-FZ, Al-FS-MZ, FS-FZ, B-FS).

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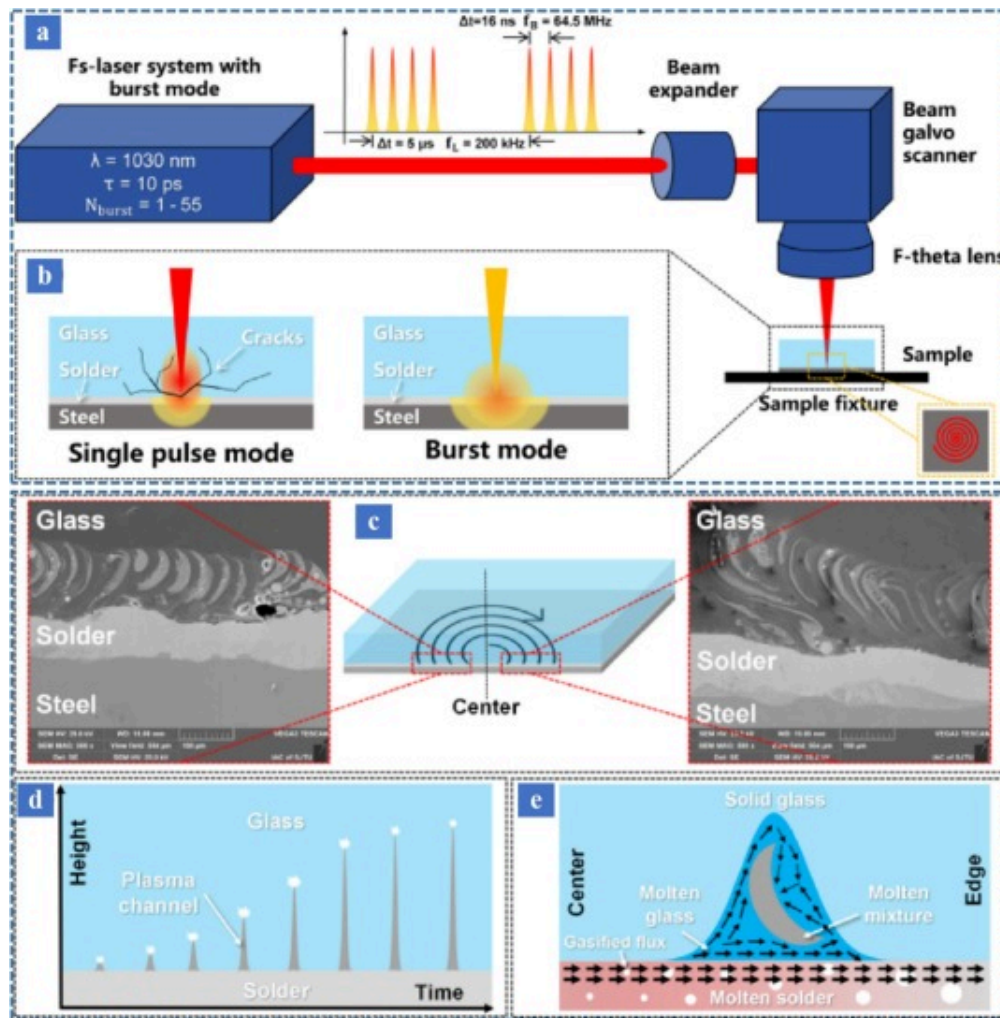
Fig. 19. Fused silica-aluminum alloy welding during mJ-level femtosecond laser. (a) A schematic configuration of Al-SiO₂ welding system; (b) The cross-section of welded joint with five characteristic zones; (c) HAADF-STEM image of TF1; (d) EDS-HAADF spectra image from (c); (e) EDS-HAADF distribution maps of Al, O, Si, Mg; (f-k) The analyses of the aluminum oxide phase in the Al-O rich region; (l) Schematic diagram of weld form processes [46].

Wang et al. [39] employed a high-repetition-rate femtosecond laser to micro-weld 304 stainless steel to glass. SEM characterization revealed an asymmetric molten zone, with a larger molten area on the glass side than the metal side. An indented interface formed due to the erosion of both molten materials, creating a mechanical interlock. However, EDS results indicated minimal element diffusion, suggesting the primary bonding mechanism is mechanical rather than chemical. Zhang et al. [40] studied the process window for femtosecond laser welding of Al6082 to SiO₂ glass. Pin structures caused by ablation were observed at the interface, crucial for bonding. The microstructure of the glass-aluminum micro-weld revealed four distinctive regions on the aluminum side under high magnification: (i) the original region, (ii) and (iii) the heat-affected zones, and (iv) the ablation zone. Regions (i) and (ii) had dense structures, while region (iii) displayed a gas structure formed from rapid solidification of overheated aluminum due to ultrashort laser ablation. Despite the enforced bonding between substrates through mechanical pin structures, the gas microstructure reduced the shear strength of the welding sample. In optical mechanical systems, adjusting the position and orientation of optical components is necessary before connection. The light source, fixed on a metal base, uses a glass lens to collimate emitted light. Machining deviations in the metal base can cause incorrect collimated light direction when the glass lens contacts the metal base, leading to a larger divergence angle. Adjusting the lens ensures the correct direction, creating a gap between the lens and the metal base. Therefore, a reliable method is needed to connect glass and metal with a gap (>100μm). Tian et al. [44] proposed a method combining laser brazing with ultrafast laser welding to connect glass and metal

with gaps of several hundred micrometers. This method involves three steps: 1) Applying solder paste and positioning the glass and metal; 2) Using continuous laser welding; 3) Using ultrafast laser welding to connect the solder and glass. Different texture patterns, solder distributions, and ultrafast laser scanning paths result in varying welding outcomes. Designing texture patterns on the metal surface controls solder distribution, increasing welding strength by 3.5 times compared to cases without texture patterns. This method offers a universally applicable approach for connecting glass and metal with large gaps.

Jiang et al. [107] employed a femtosecond laser-induced selective micro-welding technique to join sapphire and Invar alloy, examining the effects of laser power on the sealing performance, macro-morphology, and shear strength of the joints. They characterized welding defects, elemental distribution, and fracture behavior using scanning electron microscopy, energy-dispersive X-ray spectroscopy, and laser confocal microscopy. The study revealed that the sapphire and Invar alloy underwent melting, mixing, and diffusion due to nonlinear and linear absorption effects, resulting in interdigitated interfaces and indicating both metallurgical bonding and mechanical interlocking mechanisms. Shear strength increased monotonically with laser power, peaking at 145.3MPa with a laser power of 10.19W. Fractures primarily occurred through cleavage, with Fe, Ni, Al, and O present on both fracture surfaces, further illustrating the metallurgical reactions induced by ultrafast laser irradiation. Tian et al. [44] introduced a combined method of laser brazing and ultrafast laser welding to connect glass and metal with large gaps. Their approach involved applying solder paste, positioning the glass and metal, and using continuous and ultrafast laser welding to bond the solder and glass. Texture patterns on the metal surface enhanced solder distribution, increasing welding strength by 3.5 times compared to non-textured cases. However, the requirement for continuous wave lasers increased equipment costs and process complexity. Ren et al. [45] proposed using ultrafast laser bursts to weld glass and metal with large gaps filled with solder paste, allowing flexible glass positioning and reliable connections even at gaps of hundreds of micrometers. The ultrafast laser burst, focused on the glass's bottom surface

(Fig. 20(a)), melted the solder, which bonded to stainless steel through element diffusion (Fig. 20(b)). The molten pool volume increased significantly with ultrafast laser bursts, reaching approximately 350µm in height, much greater than conventional ultrafast lasers (10–20µm). Cross-sectional analysis revealed extensive material mixing and stable connections at multiple interfaces. Crescent-shaped periodic structures analyzed showed an axisymmetric distribution centered on the scan starting point, with intervals corresponding to the spiral scan line spacing of 30µm (Fig. 20(c)). The initial laser pulse generated strong plasma that enhanced the mixing of glass and solder components, increasing energy absorption from subsequent pulses. This process formed a plasma channel that grew until defocusing halted further plasma excitation (Fig. 20(d)). The transition to crescent-shaped plasma channels was driven by temperature gradients and viscous forces (Fig. 20(e)). During welding, molten solder and glass flowed from the center to the edge due to temperature differences and escaping gases, forming crescent shapes when obstructed by the solid glass wall. This behavior indicates intense fluid motion in the glass-metal molten pool during burst mode.



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Fig. 20. Ultrafast laser bursts welding glass and metal with solder paste to create an ultra-large molten pool (a) Experimental setup and laser pulse output information; (b) welding sample and process in different laser output modes; (c) Morphology of the scanned area at different locations; (d) plasma channel growth process; (e) mechanism of crescent-shaped structure formation [45].

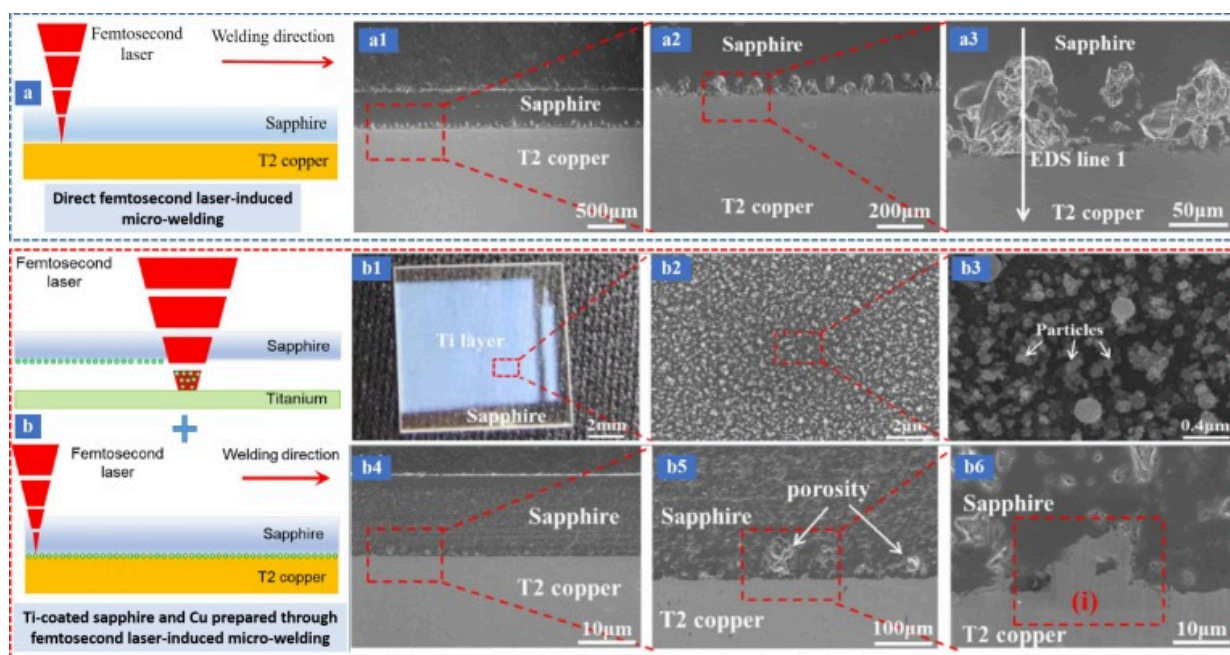
The ultrashort pulsed (USP) laser microwelding of sapphire/Invar36 alloy controlled by the surface roughness of metal was investigated for the first time [108]. The surface roughness (S_a) of Invar alloys gradually decreased from 0.944 to 0.029 μm from the prime surface to grounded and polished surface. However, the joint shear strength first increased and then decreased with the lowered S_a , the maximum shear strength reached 107.87 MPa at the $S_a \sim 0.131 \mu\text{m}$. Compared to other surfaces with low S_a , the relatively high surface roughness enhanced the interfacial thermal deposition both spatially and temporally which in turn promoted the diffusion of interface elements and the formation of jagged mechanical interlocking structures. Therefore, the appropriate rough metal surface was beneficial for the enhancement of sapphire/ metal dissimilar joints. This report is of great significance in simplifying the surface preparation process in the USP laser microwelding of transparent hard and brittle materials with metals, therefore promoting this technique from lab to industry. Welding of glass and metal materials by ultrafast laser has emerged as important research in advancing welding technology. Current research predominantly centers on single parameter experiments, yet efficiently correlating welding parameters between welding quality remains challenging. Welding of quartz glass to 304 stainless steel using a picosecond laser is conducted [109]. The welding thresholds are proposed and established through experimental analysis, and the impact of the effective quantity of incident pulses on this threshold is investigated. Three welding behaviors are confirmed by contrasting the welding threshold with alternative thresholds and demonstrated by observing the cross-section of weld. The efficient and intuitive study of welding thresholds has revealed the welding laws under the combined influence of multiple parameters. The proposal of three types of welding behaviors further elucidates the connection between material properties and welding processes, throwing some new light on parameters selection in ultrafast laser welding applications.

The welds defects and instability due to the low absorptivity (<5%) of copper at the fundamental wavelength (800–1080 nm) of ultrafast lasers remain a challenge. Processing quality is significantly improved using green laser radiation, due to its higher absorptivity of copper (~40%). Copper and other metals have been welded using green lasers to

achieve stable joints [\[\[110\], \[111\], \[112\]\]](#). Microporosity in the copper substrate was achieved using green lasers [\[113\]](#). Moreover, under ultrafast laser irradiation, the absorptivity of copper can be transiently increased due to rapid electron heating and modification of the material's dielectric function, enabling enhanced energy deposition beyond the typical linear absorption behavior. These transient changes can improve coupling efficiency at high peak intensities. The green femtosecond lasers are likely to provide higher quality welding of copper and silica glass. Femtosecond pulse and green light from the second harmonic generation were combined to increase the copper absorptivity [\[114\]](#). The silica glass and rough copper foil were effectively welded. Under the optimized parameters, a maximum shear strength of 17.19MPa was obtained. The electron and lattice temperatures during the welding process were simulated using two-temperature model. The micro-scopical mechanism, element diffusion, and chemical reaction were investigated. A modified region in the glass was formed due to excessive laser energy and scattered subsequent laser pulses. Cu–O–Si bonds were detected on the welds. Welding stability at various temperatures was characterized, with shear strength maintained at approximately 12MPa after thermal cycling from 0 to 100 °C and heating at 150 °C. This study demonstrated that effective and stable welding of silica glass and rough copper foil can be achieved using green femtosecond lasers.

To improve interface bonding strength, reduce residual stress, and optimize interface formation in the connection of transparent materials and metals using femtosecond lasers, a deeper understanding of the physical mechanisms under optical contact conditions is essential. Adding an intermediate transition layer has proven effective in mitigating residual stress and stress gradients in dissimilar joints. Yu et al. [\[115\]](#) investigated femtosecond laser welding of sapphire and copper, focusing on the bonding interface formation mechanism under non-optical contact conditions. They examined the impact of pre-fabricating a titanium (Ti) film on the sapphire surface using femtosecond laser-induced backward transfer (LIBT) on the bonding strength of the sapphire/copper joint. Direct femtosecond laser welding of sapphire and copper resulted in an interface with a mixture of sapphire and copper, achieving a shear strength of approximately

46.4MPa (**Fig. 21(a)**). This indicates that molten sapphire and laser-ablated copper mixed at the interface, filling gaps and forming an effective metallurgical bond (**Fig. 21(a1-a3)**). When a Ti film was pre-fabricated on the sapphire surface using LIBT, a Ti transition layer formed at the bonding interface, increasing the shear strength to approximately 115.5MPa (**Fig. 21(b)**). The Ti coating, composed of nano-sized spherical particles (20–100nm in diameter) with a density of about 98 NPs/ μm^2 , was uniformly distributed on the sapphire surface (**Fig. 21(b1-b3)**). The joint interface with Ti coating exhibited reduced gaps and damage zones in the sapphire, with copper protruding upwards at the interface (**Fig. 21(b4-6)**). The Ti layer effectively alleviated residual stresses and enhanced bonding between copper and sapphire, improving joint strength. These findings offer valuable insights for ultrafast laser bonding between other transparent brittle materials and metals.

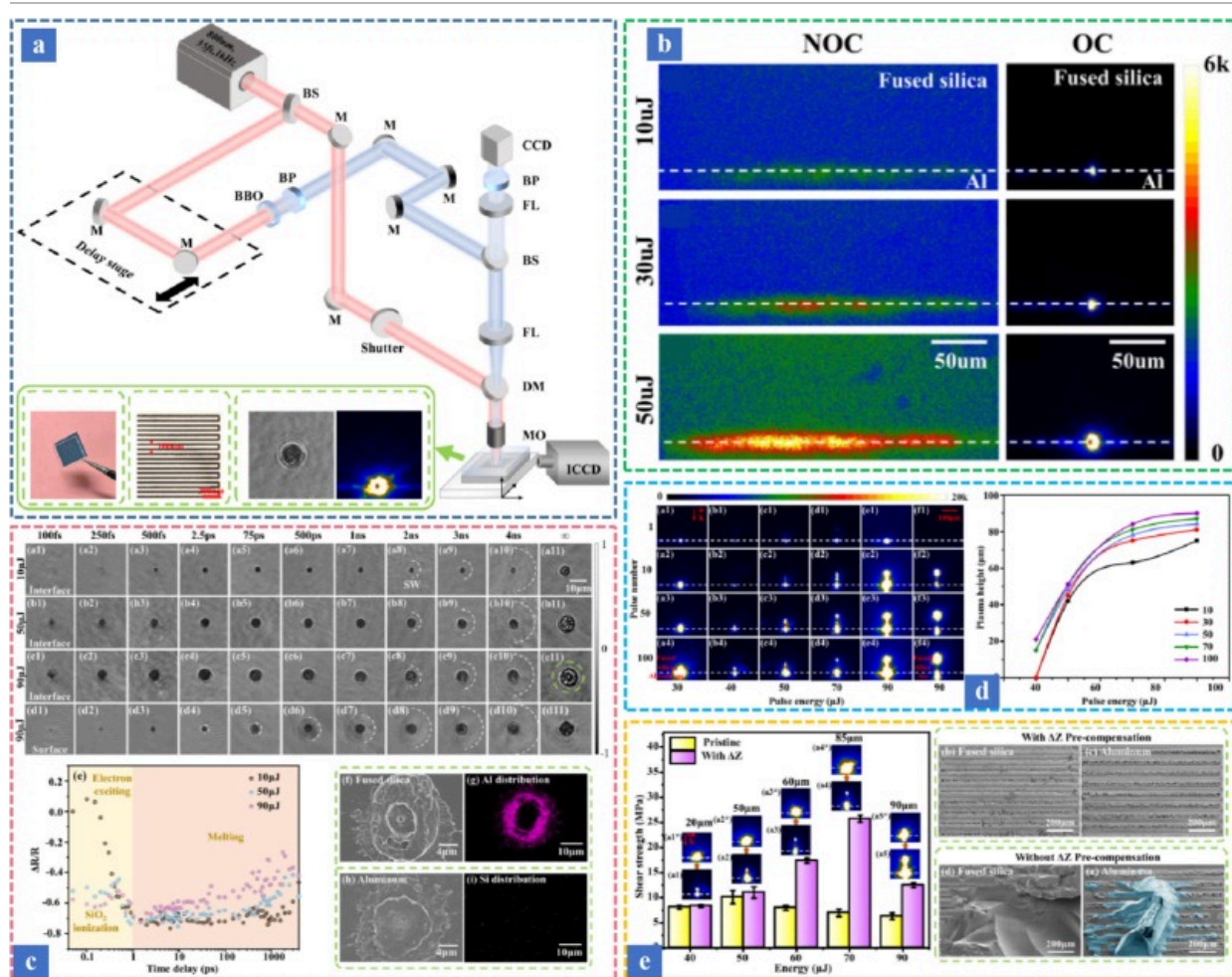


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Fig. 21. Femtosecond laser welding of sapphire-copper using a thin film titanium interlayer (a) Direct femtosecond laser-induced micro-welding, (a1-a3) Micro-morphologies in the bonding area by direct femtosecond laser-induced micro-welding; (b) Schematic diagrams for welding after femtosecond LIBT of coating, (b1-b3) Ti coating prepared on the sapphire surface through femtosecond LIBT, (b4-b6) Cross-sectional morphologies of the joint of Ti-coated sapphire and Cu prepared through femtosecond laser-induced micro-welding [115].

Current research mainly investigates how processing parameters influence welding strength, focusing on single variables rather than their combined effects. Despite extensive experimentation, an efficient rapid laser welding method has not been established, leading to low research efficiency. Yang et al. [116] introduced the concept of a welding threshold based on pulse energy density and the effective number of incident pulses for ultrafast laser welding of 304 stainless steel and quartz glass. Their study examines the incubation effect of this threshold, identifying three distinct types joining behaviors. By analyzing the welding threshold under varying pulse numbers, they characterized weld morphology across these behaviors, ultimately identifying suitable parameters for different applications. Femtosecond laser welding is a precise, clean technique known for high accuracy, minimal heat-affected zones, and low waste. However, the connection process is not yet fully understood. To clarify the welding mechanism at the contact interface, pump-probe microscopy and plasma imaging were utilized [117]. Under femtosecond laser pulses, aluminum melts and re-solidifies within picoseconds, effectively bonding different materials. Plasma confinement in the interaction zone after laser irradiation enables joints with higher shear strength due to concentrated energy absorption (Fig. 22). In fused silica, self-focusing and nonlinear absorption impact shear strength, while focal point precompensation reduces fractures, increasing shear strength to 25.75MPa. These findings underscore the potential of ultrashort laser welding for precise bonding of heterogeneous materials.

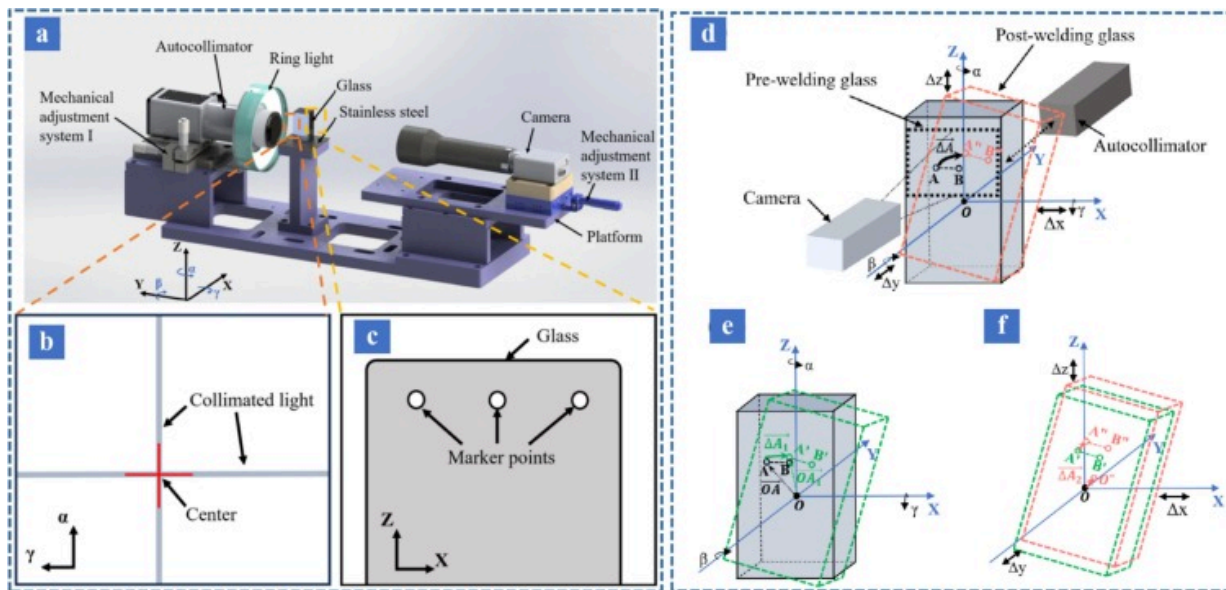


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Fig. 22. Mechanism and optimization of femtosecond laser welding fused silica and aluminum (a)Experimental scheme for weld and insitu observation; (b). Plasma behaviors at single pulse energy of 10, 30 and 50 μJ; (c)Time-resolved reflectivity at the OC interface irradiated with different pulse energies; (d)Plasma behaviors recorded at the gate width of 5 ns after 1–100 pulses interact with the OC interface at different pulse energies [117].

Ultrafast laser welding is a technique for connecting glass and metal materials, offering advantages such as precisely localized heating, rapid processing, and a minimal heat-affected zone compared to traditional welding methods. A significant challenge in this process is focusing the laser precisely at the interface between the two materials. Accurate connection of glass and metal without fixture clamping is crucial in optomechanical coupling scenarios. Therefore, a measurement system with micrometer-scale accuracy for monitoring glass positioning during ultrafast laser welding is essential. To address this, a new online measurement scheme (Fig. 23) has been proposed, utilizing a single camera and a single autocollimator to determine the relationship between the glass position and observed quantities [118]. This system measures changes in glass position across five degrees of freedom (DOF): rotational angles around the X, Y, and Z axes, and displacements along the X and Z axes. An experimental setup was developed to track glass pose during the welding process and to obtain a graph showing the variation in glass pose over ultrafast laser processing time. The results indicate that ultrafast laser welding induces periodic changes in the angular orientation (rotational pose) of the glass around the X and Z axes. These variations are attributed to the helix periodic scanning of the glass-metal contact interface and phenomena such as material heating, melting, vaporization, and splashing.



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Fig. 23. Measurement of glass position in ultrafast laser welding of glass and metal (a-c)Schematic diagram of the measurement system; (d-f)Schematic diagram of the glass position decoupling [118].

In summary, the process of forming heterogeneous welds using ultra-short, high-energy-density laser pulses is highly complex, primarily due to its transient nature and the mismatch of thermal properties between the two components. Challenges such as the elimination of residual stresses, control of interface units, and the revelation of resulting synergistic strengthening mechanisms remain to be addressed. In ultrafast laser welding of different materials, focusing the laser to the interface between the two materials is a major challenge. These issues significantly impact the joint quality and optical performance of specimens, thereby limiting the scope and progress of brittle material-metal ultrahigh-frequency laser micro-welding. Despite extensive research into the microstructure and composition of heterogeneous interface connections, these challenges

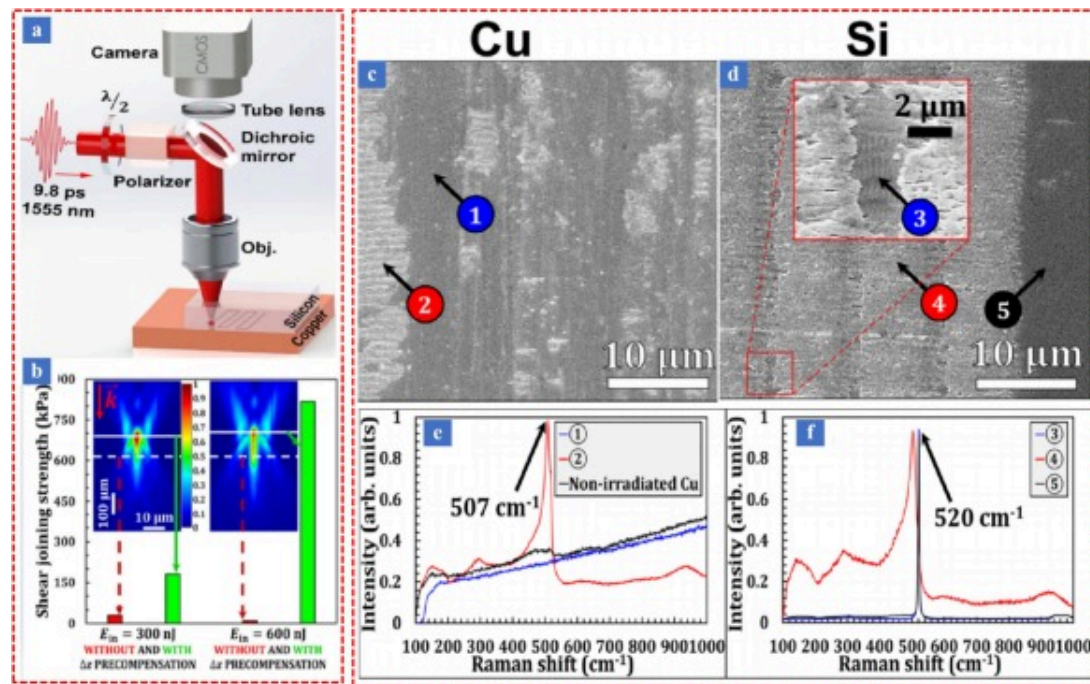
remain incompletely resolved. To further comprehend the connection mechanisms and achieve successful heterogeneous welding joints, new characterization techniques are required beyond SEM and TEM.

2.3.5. Micro-welding between semiconductor and metal materials

Ultrafast laser micro-welding has successfully connected glass to metal, semiconductors, and polymers, and more recently, ceramic-ceramic structures, achieving shear bonding strengths in the MPa range. However, its application to semiconductor–metal structures remains underexplored despite its potential in photovoltaics, sensors, and broader microelectronics. This limitation stems from the significant nonlinear propagation effect in semiconductors, which causes uneven energy deposition at the interface. Currently, the most effective strategy to counteract these nonlinear effects involves using nanosecond laser pulses. While effective for intra-volume functional silicon, this approach's long pulse width leads to thermal processes that hinder precise energy control during welding. Overcoming these challenges is essential for expanding ultrafast laser micro-welding's applicability to semiconductor–metal structures, thereby unlocking its potential in various microelectronic applications. Addressing these issues will enhance the precision and reliability of ultrafast laser micro-welding in this domain, broadening its utility in advanced technological applications.

Hence, the picosecond regime emerges as a promising compromise solution to mitigate nonlinear propagation and thermal–mechanical effects in laser welding applications. Although nonlinear propagation in silicon had not been previously characterized, a groundbreaking study by Chambonneau et al. [119] has elucidated the challenges of energy deposition at the semiconductor–metal interface during picosecond laser pulse propagation in silicon through nonlinear transmission investigation (Fig. 24). Historically, nonlinear propagation effects in semiconductors have posed limitations, but ultrafast laser welding of semiconductor–metal interfaces, such as silicon and copper, successfully circumvents these physical constraints. In their experiments, researchers observed filament formation induced by picosecond pulses in silicon and used nonlinear

propagation imaging to measure the nonlinear focus shift. They then compensated for this shift to maximize optical intensity at the silicon-copper interface, enhancing energy deposition. This pre-compensation strategy facilitated effective welding, achieving a shear bonding strength of up to 2.2 MPa, suitable for various applications involving semiconductors and metals in complex microelectronic devices. In summary, given the strong nonlinear propagation effects in narrow bandgap semiconductors, the ultrafast laser welding technology based on nonlinear focal shift pre-compensation proposed in this study can be applied to many semiconductor–metal structures. Additionally, this technique is applicable to some semiconductor–semiconductor configurations, provided the first material is transparent to the laser wavelength.



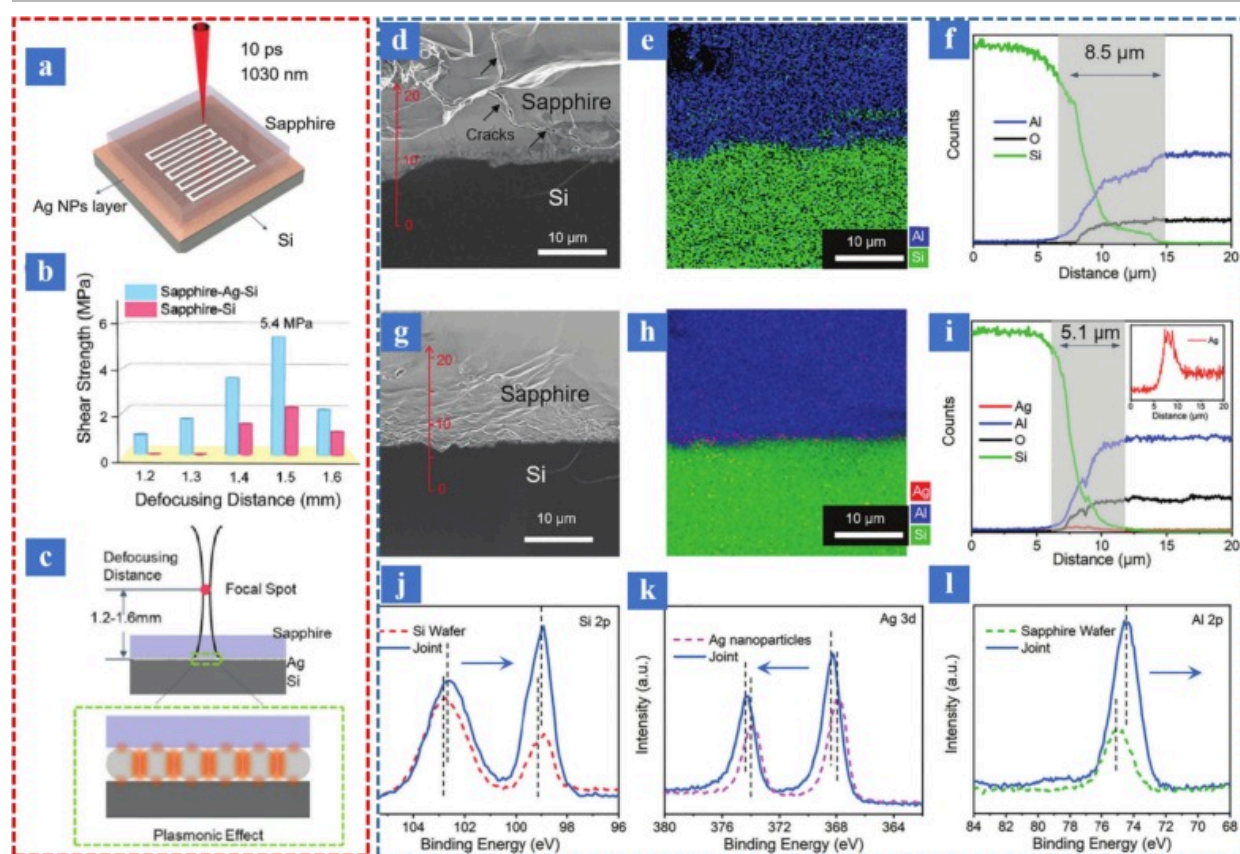
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Fig. 24. Taming ultrafast laser filaments for optimized semiconductor–metal welding (a) The schematic diagram of the ultrafast laser welding setup; (b) Under conditions without

(red) and with (green) precompensation for nonlinear focal shift, the laser-induced shear connection strength between silicon and copper is shown for input pulse energies of 300 nJ and 600 nJ; (c-f) Material analyses of laser welds ($E_{in}=950$ nJ) [119].

Recent work has demonstrated that ultrafast laser bonding of silicon–silicon interfaces can be enhanced by the incorporation of a nanometer-thick metallic film at the interface, which significantly improves interfacial energy absorption and bonding efficiency [120]. This interlayer strategy effectively improves energy confinement and facilitates localized thermal and chemical interactions at the bonding interface. In Fig. 25, a representative example illustrates the use of an Ag nanoparticle film between sapphire and silicon wafers [93], where plasmon-enhanced absorption and interfacial reactions (e.g., Ag-Si and Al-O-Ag bond formation) enable robust microwelding. While the previous work focused on Si-Cu bonding without an interlayer [119], these findings underscore the broader value of interfacial structure engineering in optimizing ultrafast laser micro-welding across diverse material systems. In summary, these studies highlight the effectiveness of interfacial nanolayers in enhancing energy absorption and mitigating stress, thus expanding the applicability of ultrafast laser micro-welding to a wide range of brittle and dissimilar materials.



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Fig. 25. Metal nanoparticles assisted ultrafast laser plasmonic microwelding of Sapphire–Semiconductor interconnects (a) Schematic of the experimental setup for ultrafast laser microwelding; (b) Shear strength of Sapphire–Si and Sapphire–Ag–Si joints after ultrafast laser microwelding with different defocusing distances; (c) Illustration of ultrafast laser plasmonic microwelding of heterogeneous Sapphire–Ag–Si structure under defocusing conditions; (d–l) Cross-sectional SEM images and XPS spectra of the interfacial microstructures of Sapphire–Si and Sapphire–Ag–Si [93].

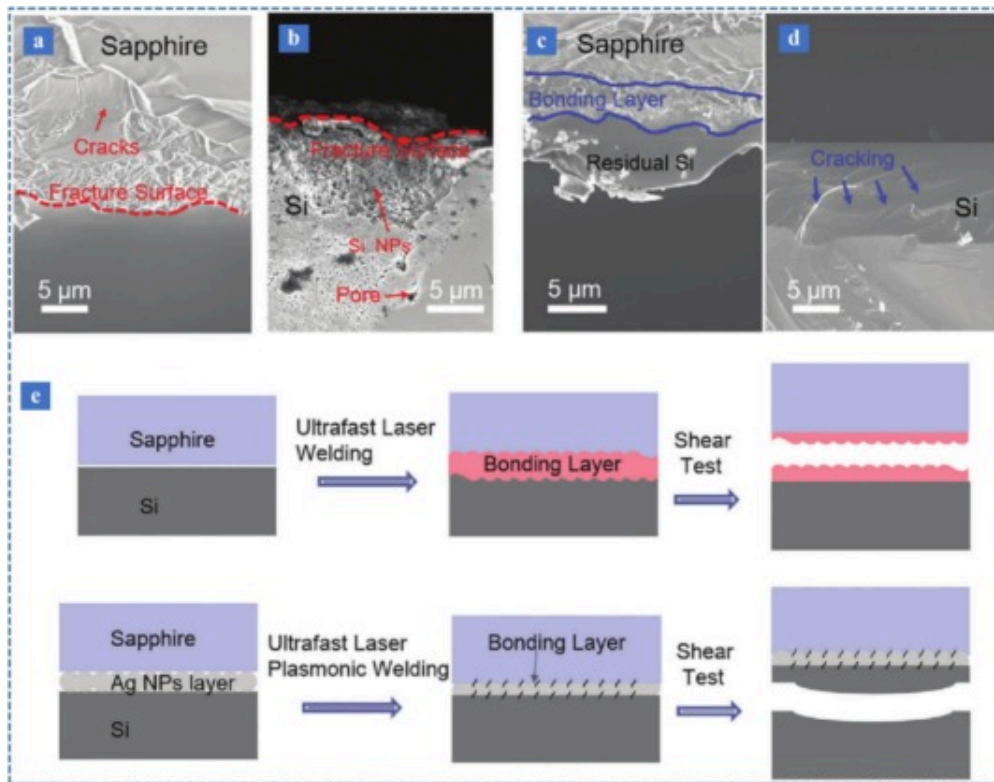
2.3.6. Micro-welding between oxides and semiconductors materials

Ultrafast lasers, with their high spatiotemporal precision, have been effectively employed in microwelding heterogeneous materials such as Cu-Si [119], Si-GaAs [94], and Alumina-YSZ [53]. The success of these welds is attributed to the precise energy deposition at the junction, enabling localized fusion or mixing of materials. However, welding transparent materials often requires a complex experimental setup for precise laser beam focus and positioning to minimize base material damage. Additionally, residual stress-induced cracking, a common issue in the fusion welding of brittle materials, can degrade joint performance. Thus, controlling spatial energy distribution and managing synergistic material evolution under ultrafast laser irradiation are critical for integrating high-performance brittle materials, especially differential oxides and semiconductors.

One approach to overcoming these challenges is the incorporation of metal layers, which can alleviate residual stress and improve material compatibility during ultrafast laser irradiation. For instance, as illustrated in Fig. 25(a), an Ag nanoparticle (NP) thin film is used as an energy confinement layer between oxide (sapphire) and semiconductor (Si) wafers [93]. The plasmonic effects within the dielectric-metal-dielectric structure (Sapphire-Ag-Si) enhance thermal diffusion and chemical interactions between Ag and the adjacent materials, enabling the microwelding of sapphire and Si wafers (Fig. 25(c)). During this process, Ag NPs sinter at the junction, bridging the oxide and semiconductor while forming Al-O-Ag and Ag-Si bonds at the respective interfaces (Fig. 25(d-l)). The resulting heterogeneous joint exhibits high shear strength, up to 5.4MPa, with fractures occurring within the Si wafer. Additionally, the metal nanolayer effectively mitigates residual stress-induced microcracking in brittle materials.

Cross-sectional images of the fractured joints revealed that in the Sapphire-Si bond, the fracture occurred within the bonding area (Fig. 26(a-b)). This region exhibited ablated Si nanoparticles, microcracks, and pores, attributed to the intense material evolution (e.g., ablation, redeposition) under ultrafast laser irradiation. In contrast, the fracture in the Sapphire-Ag-Si joint occurred within the Si wafer itself (Fig. 26(c-d)), indicating a dense and robust bond without a vulnerable zone. A portion of Si remained on the Sapphire wafer post-fracture, underscoring the strong bonding achieved. The shear modulus of

bulk Si exceeds 60 GPa, and the Sapphire-Ag-Si joint demonstrated a shear strength of up to 5.4MPa, confirming the formation of chemical reactions and products within the junction. A schematic of the fracture mechanisms in the Sapphire-Ag-Si and Sapphire-Si joints is presented in Fig. 26(e). Wafer-scale Sapphire and Si interconnects exhibited robust strength and impermeability even after 200 thermal cycles between -40°C and 120°C . This plasmonic microwelding technique offers a viable solution for overcoming material incompatibility and stress-induced cracking in small-scale brittle materials integration. Additionally, this metal nanoparticle-assisted plasmonic microwelding approach holds promise for broad applications in high-performance microdevice development, including MEMS, MOEMS, and microfluidics.



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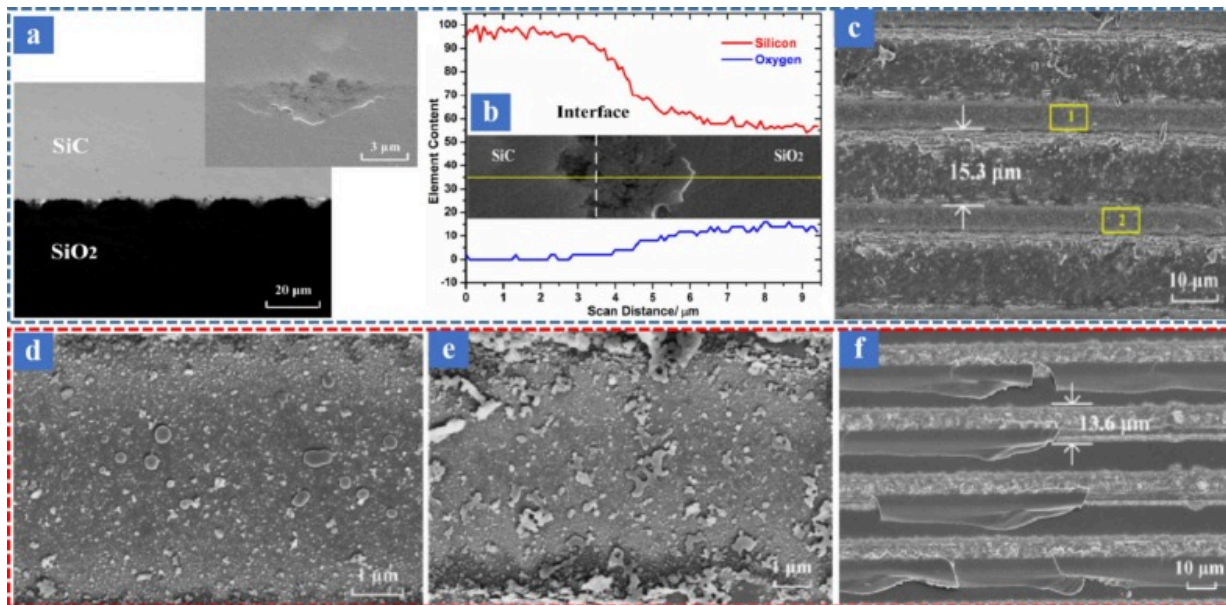
Fig. 26. Cross-sectional SEM images of the joints after fracture: (a) Si side and (b) Sapphire side from the Sapphire–Si joint; (c) Si side and (d) Sapphire side from the Sapphire–Ag–Si joint. Fracture surface profiles in (a) and (b) were sketched by red dashed lines; (e) Schematic illustrations of the fracture mechanism of ultrafast laser microwelded Sapphire–Si and Sapphire–Ag–Si joints. The fracture occurred inside Si wafer in Sapphire–Ag–Si and within the bonding layer in Sapphire–Si [93].

2.3.7. Micro-welding between glass and silicon materials

Silicon and glass are frequently used together in microelectromechanical systems (MEMS) packaging. Single-crystal silicon is favored for its outstanding electrical, thermal, and mechanical properties. While epoxy resin has been used for bonding silicon and glass, its performance can be affected by changes in humidity and temperature. Anodic bonding, however, matches glass with a coefficient of thermal expansion (CTE) close to that of silicon. This technique has become common in MEMS packaging due to its precise geometry and low bonding temperature. However, challenges such as longer processing times and the need for wide flat surfaces exist.

In the study conducted by Cheng et al. [121], femtosecond laser welding was employed to join quartz glass and monocrystalline silicon with a sample gap of less than 100nm. The joint strength ranged from 7 to 54MPa, depending on specific welding processes. Welding materials with significant differences in thermal expansion coefficient and melting point poses a challenge. Zhang et al. [61] successfully welded molten quartz (melting point 1720°C) and silicon carbide chips (melting point 3100°C) using an 800nm near-infrared femtosecond laser. The cross-section of welding lines with 20µm spacing is depicted in Fig. 27(a), with the magnified region shown in the upper right corner, revealing the elimination of the inner gap between materials after laser irradiation. Element distribution within the cross-section of the welding line, obtained by EDS line analysis (Fig. 27(b)), indicates mixing and interdiffusion of materials along the longitudinal direction within the irradiated region. The impact of laser ablation on welding quality was investigated by increasing the internal gap between silicon carbide chips and molten

quartz. Samples without optical contact showed a shear joint strength of about 5.7MPa, significantly lower than the 13.8MPa shear joint strength of samples with optical contact (Fig. 27(c)). No large residual molten quartz remnants were observed on the silicon carbide surface after shear separation, with an average weld seam width of around 15.3 μ m. In Fig. 27(d-e), numerous nanoscale droplets were found in the unirradiated area between adjacent weld lines for non-optically contacted samples. Conversely, for samples with optical contact, laser ablation in the focal region was effectively suppressed (Fig. 27(f)), with a measured average weld seam width of about 13.6 μ m, notably smaller than that of non-optically contacted samples.

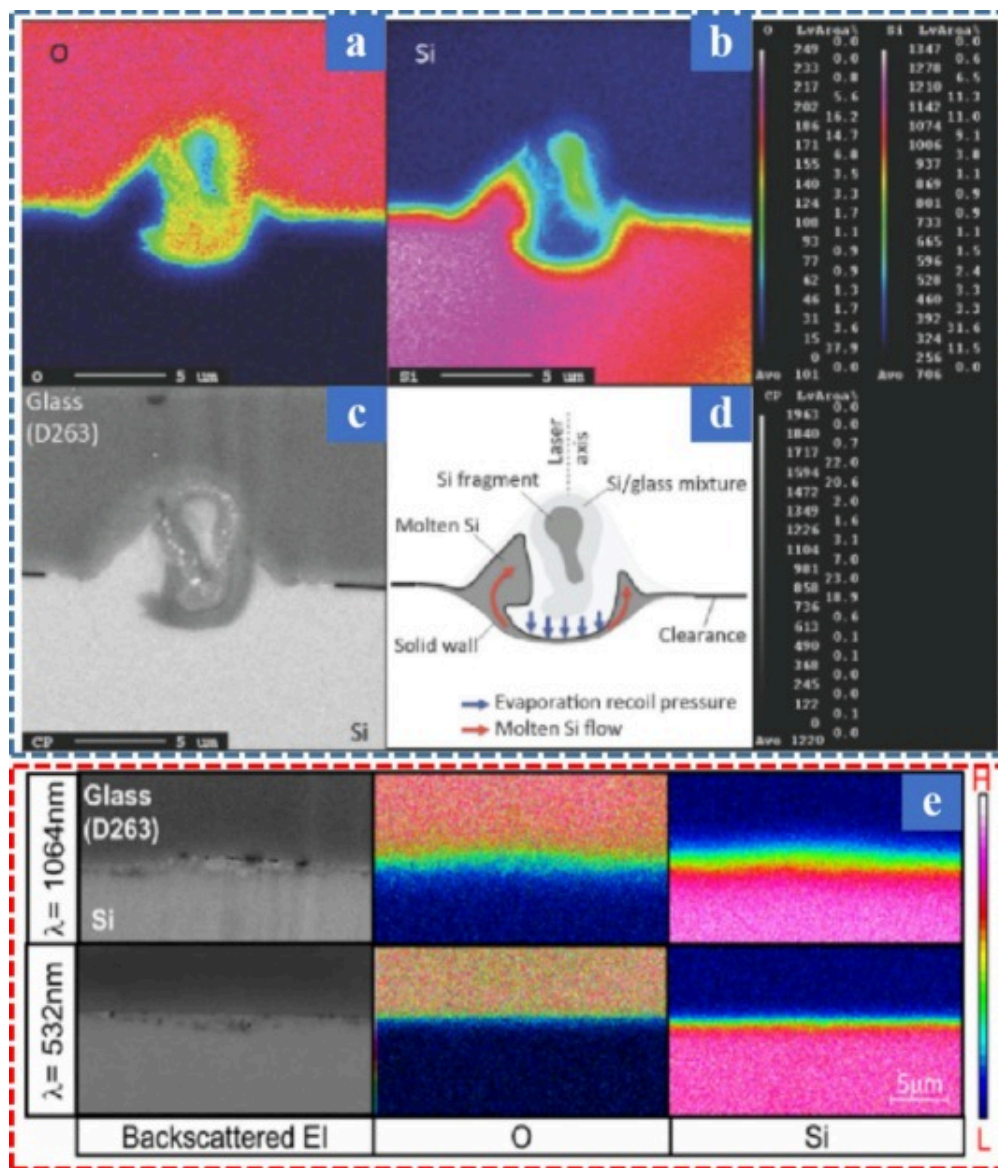


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Fig. 27. Ultrafast Laser Micro-Welding of Glass-Silicon (a) Morphology and (b) chemical elements analysis of the cross section of welding line irradiated at pulse energy of 1.0 μ J and speed of 0.5mm/s; (c-f) SEM images of the samples after post-joining separation: (c) SiC surface with non-optical contact, (d) larger version of region 1, (e) larger version of region 2, (f) SiC surface with optical contact [61].

Researchers have explored both nanosecond (ns) and femtosecond (fs) laser pulses for welding silicon and glass, aiming to achieve mechanical strength comparable to anodic bonding. Miyamoto et al. [62] introduced a novel method using 20ps high-repetition-rate laser pulses, with a spatial resolution of 15–18 μ m, to weld silicon/aluminum silicate and silicon/borosilicate glass samples without preheating or post-heating. This method yielded mechanical strengths of up to 85MPa and 45MPa, respectively, surpassing anodic bonding at high spatial resolutions (<20 μ m). Nordin et al. [63] investigated the influence of wavelength and pulse duration on micro-welding of single crystal silicon and glass using four types of laser beams: nanosecond pulses with wavelengths of 532 and 1064nm, and femtosecond pulses with the same wavelengths. Fig. 28(a-d) depicts their findings, indicating that femtosecond pulses with infrared wavelengths result in stable weld pool formation and higher fracture strength in glass-silicon micro-welding. For optically contacted samples (with a sample gap less than a quarter wavelength), when the laser focus traverses the interface between materials, fusion typically occurs within the focal region, as shown in Fig. 28(a-d). This is because optical contact effectively suppresses laser-induced surface ablation and the splattering of molten material, confining the thermo-fluid dynamic behavior of the materials within the solid-state boundaries. Experimental studies on silicon and glass micro-welding have elucidated the effects of wavelength and pulse duration (Fig. 28(e)). Single crystal silicon shows high absorption efficiency in the green laser wavelength range, suggesting potential for improved silicon-glass welding. However, there have been limited reports on using green picosecond (ps) lasers for micro-welding crystalline silicon to amorphous glass substrates. Zhang et al. [98] successfully achieved direct bonding between glass and silicon using 515nm picosecond lasers, resulting in a fracture strength of up to 122MPa, with crack-free and residue-free welding joints.



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Fig. 28. Ultrafast Laser Micro-Welding of Glass-Silicon (a-d) Cross-section of Si/D263 with Higher Pulse Repetition Rate $f=2$ MHz, $Q_0=2$ J, and $N=10$. Distribution of Silicon and

Oxygen is also Shown [62]; (e) Elemental Analysis of the Glass-Silicon Interface [63].

Increasing the gap between materials significantly reduces the bonding strength of non-optically contacted samples compared to optically contacted ones. To mitigate the need for optical contact during laser welding, expanding the heat-affected zone via higher pulse repetition rates, pulse widths, or laser absorption rates can be effective. This strategy ensures that the material remains in a high-temperature environment, promoting the formation of covalent bonds between samples. Furthermore, research on laser welding of single crystal silicon and glass shows that varying pulse widths and wavelengths affect the process, with picosecond pulse lasers yielding higher fracture strength in the welding.

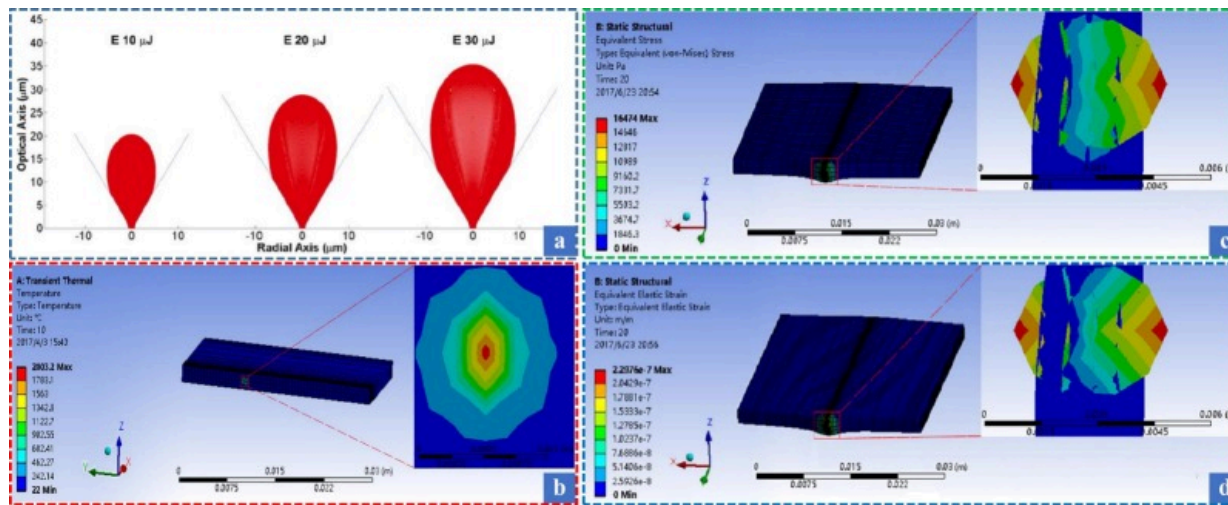
Achieving reliable and robust bonding through ultrafast laser irradiation is influenced by factors such as the central wavelength, pulse width, repetition rate, focal position, and material combination [37,60,122]. The term ‘ultrafast burst irradiation’ refers to a technique where multiple femtosecond pulses are delivered in rapid succession within a short time interval (typically sub-nanosecond). This approach enhances energy accumulation and can significantly affect the interaction mechanisms with transparent materials. These factors significantly impact the interaction between the laser and materials, determining material melting behavior, the formation of welded joints, and the resulting joint strength. For example, shorter wavelengths lead to higher absorption but can cause cracking at lower pulse energies, while longer wavelengths can achieve wider crack-free welds [123]. Laser power affects the heat-affected zone (HAZ) size, and high peak power densities can induce thermal damage, which negatively impacts weld quality [64]. Pulse width influences thermal effects and stress responses, with shorter pulses typically yielding smaller, more precise welds [23,53,[124], [125], [126],127]. High repetition rates facilitate thermal accumulation and efficient welding, especially for non-optical contact conditions [30,122,[116], [128], [129], [130], [131], [132], [133], [134]]. Focal position is crucial for effective energy deposition and bonding, with precise control needed to optimize weld strength and quality [74,38,37,98,135,136,137].

3. Numerical simulation advances in ultrafast laser micro-welding

Current research by domestic and foreign scholars on ultra-fast laser micro-welding mainly focuses on two key areas: (1) Assessing the viability of ultra-fast laser welding for glass materials; and (2) Analyzing the impact of process parameters on welding quality. However, conducting extensive experimental studies is time-consuming, labor-intensive, and expensive, often challenging to meet engineering demands. With the progress of computer technology, simulation techniques offer a promising solution to replicate the ultra-fast laser welding process of brittle materials, reducing costs and enhancing work efficiency. This discussion primarily addresses advancements in temperature field simulation, interface molecular dynamics simulation, plasma density simulation and electric field simulation concerning ultrafast laser micro-welding.

3.1. Numerical simulation of temperature fields

In their numerical simulation, Miyamoto et al. [122] explored the interaction between picosecond laser pulses (10ps pulse width) and glass materials, focusing on the temperature field during ultra-short pulse laser welding. Their findings revealed that picosecond lasers could weld glass materials with higher coefficients of thermal expansion, such as Foturan glass, particularly at repetition rates exceeding 200kHz. Kongsuwan et al. [75] investigated the influence of single-pulse energy on absorption volume microstructure using numerical simulation. They predicted tear-shaped absorption volumes, which closely aligned with experimental results. TAN et al. [138] utilized transient thermal and static structural models to simulate the temperature field, residual stress field, and strain field during femtosecond laser pulse welding of double-fused quartz substrates. Their simulations, depicted in Fig. 29(b-d), revealed that Gaussian heat sources deposited periodically at the sample's interface could induce residual stresses of up to 16.474kPa, which were insufficient to cause cracks in fused quartz within a 20-second timeframe. Additionally, they investigated the influence of an external magnetic field on the welding process, finding that it notably reduced both residual stress and strain values.



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Fig. 29. Numerical Simulation Results (a) Simulated Absorption Volume at Different Pulse Energies [75]; (b) Cross-Sectional Temperature Field along the Heat Movement Direction [138]; (c-d) Residual Stress and Strain Field Maps [138].

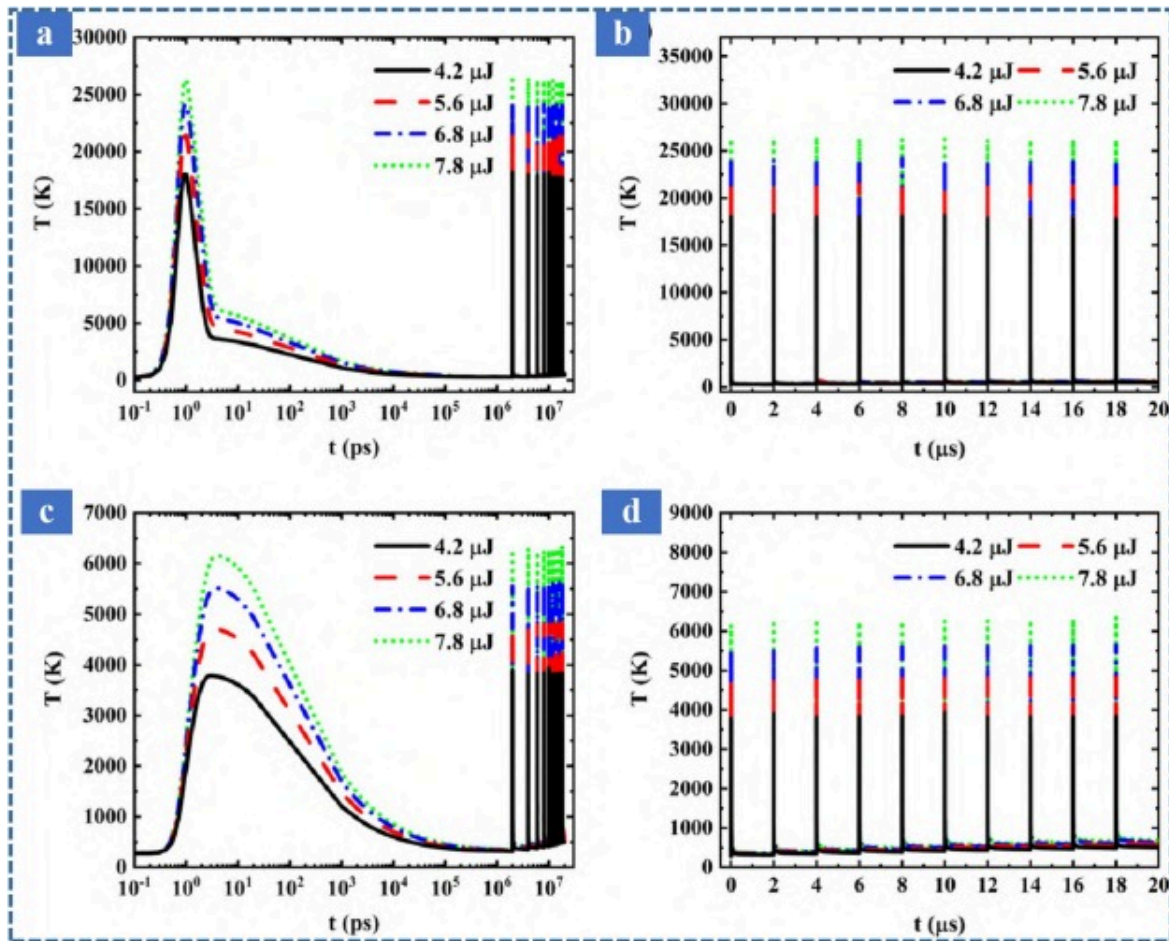
At present, the temperature field simulation for femtosecond laser welding of dissimilar materials is seldom reported, and the influence law of laser single pulse energy on weld quality needs to be studied in depth. Yuan et al. [139] employed finite element modeling to analyze the temperature field during the femtosecond laser welding process of quartz glass and 304 stainless steel, validating the weldability of the laser welding process parameters used. The heat transfer process of femtosecond laser welding of quartz glass to 304 stainless steel is described by the following two-temperature equations [139]:

$$C_e \frac{\partial T_e}{\partial t} = \nabla \cdot (-k_e \nabla T_e) - G(T_e - T_l) + Q$$

$$C_l \frac{\partial T_l}{\partial t} = \nabla \cdot (-k_l \nabla T_l) - G(T_e - T_l)$$

where the subscripts e and l represent the electron and lattice systems, respectively, and

G is the electron–phonon coupling coefficient. The laser energy is Gaussian distributed and is absorbed by the 304 stainless steel and converted to heat, so the laser is equated to a Gaussian body heat source applied to the 304 stainless steel in the simulation. During laser welding, the temporal distribution of electron temperature and lattice temperature is depicted in Fig. 30. It can be observed that upon the action of a single laser pulse, the electron temperature rises rapidly. After reaching its peak, the electron temperature quickly decreases. In contrast, the lattice temperature begins to rise only when the electron temperature approaches its peak, exhibiting significant relaxation. As the electron temperature approaches the lattice temperature, the rate of temperature change for both systems slows down. Once the electron and lattice temperatures become equal, both subsystems attain a steady-state temperature (i.e., the local temperature remains essentially unchanged) until the next pulse arrives. Even upon the arrival of the next pulse, the temperature remains elevated above room temperature, indicating heat accumulation during the femtosecond laser welding process. As Shown in Fig. 30 (a-d), the results indicate that at a lower single pulse energy of $4.2\mu\text{J}$, the maximum temperature at the interface between quartz glass and 304 stainless steel reaches 4101 K, exceeding the softening point of quartz glass (1857 K) and the melting point of 304 stainless steel (1727 K). This indicates the feasibility of the selected laser welding process parameters. With the increase in laser single pulse energy, there is an overall increase in welding temperature, and the time taken for the electron system and lattice system to reach thermal equilibrium also increases.



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Fig. 30. Temperature simulation results. (a) logarithmic horizontal time distribution of electron temperature; (b) linear abscissa time distribution of electron temperature; (c) logarithmic horizontal time distribution of lattice temperature; (d) linear abscissa time distribution of lattice temperature [139].

Encapsulation, insulation, and insulation of semiconductor core components is an indispensable process in the production of optoelectronic products. The most common

practice is to utilize anode bonding technology to weld insulating glass materials to semiconductor materials. However, the limitations of this welding process are large, generally need to be carried out at high temperatures (300 to 500 °C) assisted by an applied electric field, and the glass material is required to contain a large number of reactive ions. In view of this, in recent years, researchers have begun to try to use ultrashort pulsed laser to realize the direct welding of glass and semiconductor materials. The welding technology does not require the assistance of an external electric field, can be realized at room temperature by laser direct writing between the samples of the selected area of the weld, with high welding accuracy, small heat-affected area and so on. However, the current research in this field mainly focuses on glass and semiconductor materials with similar physical properties, and the research on the welding mechanism is still far from adequate. Zhang et al. [140] conducted a numerical simulation to investigate the temperature evolution during ultrafast laser micro-welding between fused silica glass and high-melting-point silicon carbide semiconductor. The expression for the heat conduction equation is given below:

$$\rho C_i \frac{\partial}{\partial t} T(r, z, t) = \frac{1}{r} \frac{\partial}{\partial r} \left(r K_i \frac{\partial T(r, z, t)}{\partial r} \right) + \frac{\partial}{\partial z} \left(K_i \frac{\partial T(r, z, t)}{\partial z} \right) + q(r, z, t)$$

The heat source term is expressed as:

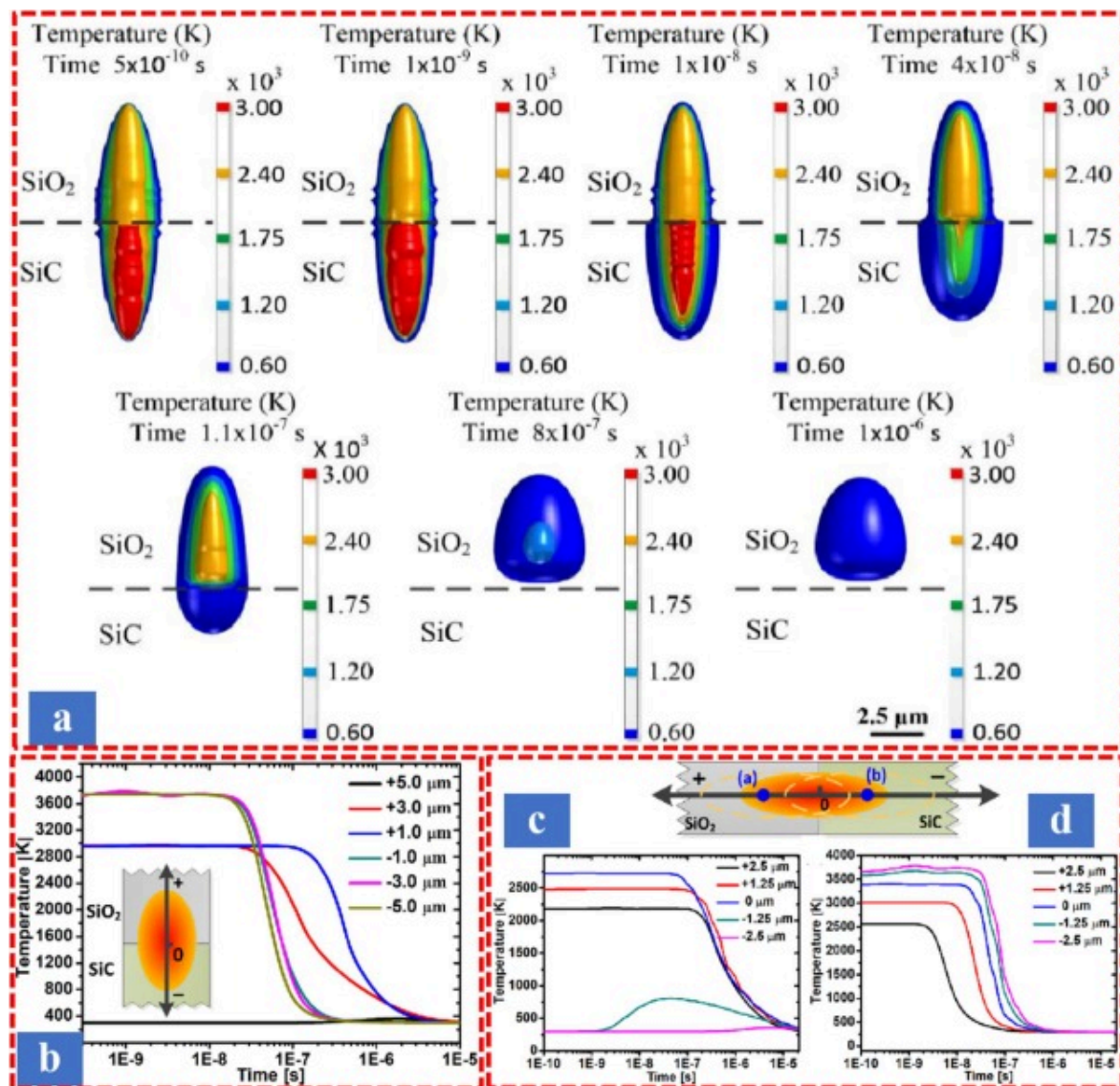
$$q(r, z, t) = I_p \alpha(z) \gamma g(t) G_p(z) G_p^2(r)$$

where I_p is the peak optical power density at the center of the laser's focal field, $\alpha(z)$ is the absorption rate of the laser energy by the material, γ is the thermal conversion efficiency of the absorbed energy, $g(t)$ is the normalized time distribution of the ultrashort pulsed laser energy, $G_p(z)$ and $G_p(r)$ is the spatial distribution of the pulsed laser in the axial and radial directions, respectively, approximated as a normalized Gaussian. The variation of specific heat capacity C_i ($z > 0$) and thermal conductivity K_i ($z < 0$) of silicon carbide semiconductor with temperature can be described by the following equation.

$$C_i = 1267 + 0.049T_i - \frac{1.227 \times 10^8}{T_i^2} + \frac{2.05 \times 10^{10}}{T_i^3}$$

$$K_i = \frac{61100}{T_i - 115}$$

Numerical simulation plays a crucial role in understanding the response of materials to ultrashort pulsed laser irradiation, providing valuable insights for the selection of parameters in laser welding processes. In a study depicted by [Fig. 31\(a\)](#), researchers conducted numerical simulations to analyze the temperature evolution at the interface of a sample under the influence of a single-pulse femtosecond laser. The results revealed that when exposed to a femtosecond laser with a pulse energy of 1 μJ, localized temperatures on both sides of the sample exceeded their respective melting (or softening) points.



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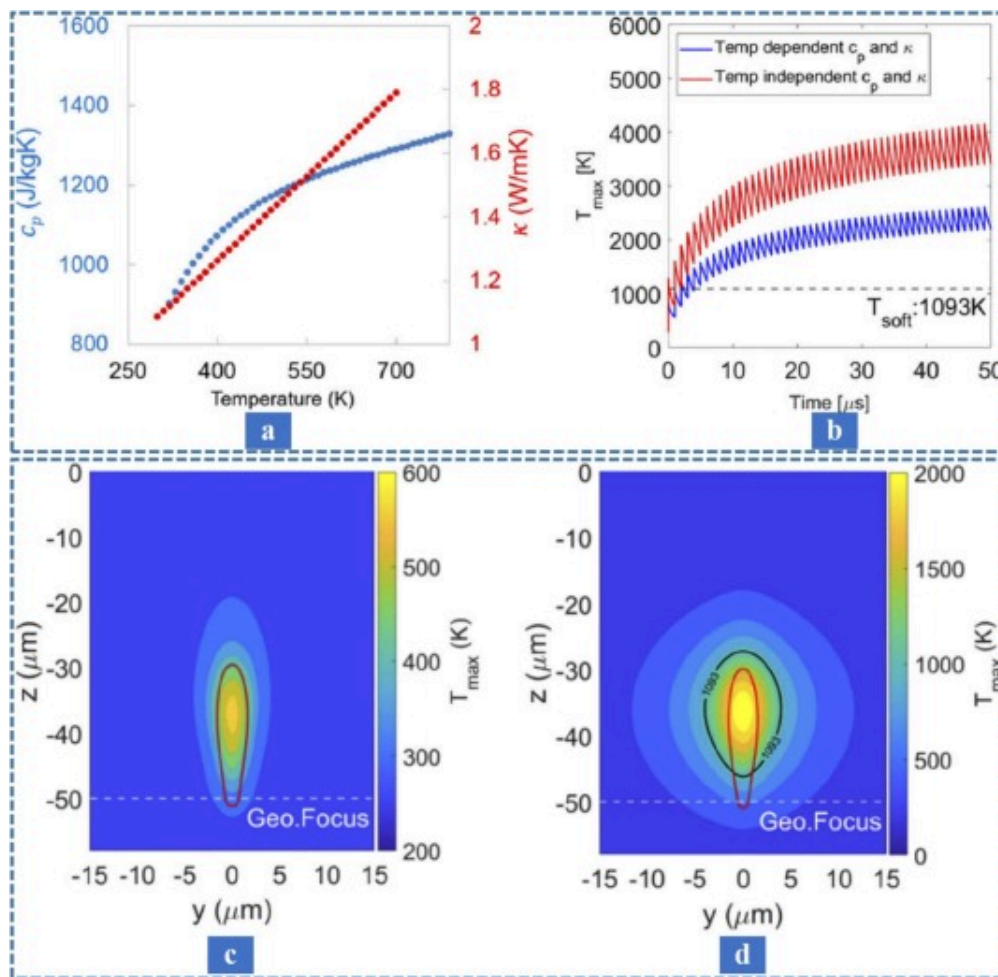
Fig. 31. Thermal simulation analysis of sample welding interface under femtosecond laser irradiation (a) Evolution of the thermal field on sample cross section after irradiated by single pulse femtosecond laser; (b) Thermal relaxation process at different position near

sample interface; (c-d) Thermal relaxation process of material at point (a) and (b) with varying focus position [140].

Furthermore, the researchers simulated the laser-induced thermal relaxation process to obtain temperature–time profiles at arbitrary positions on both sides of the interface, as shown in Fig. 31(b). Additionally, they investigated the influence of the movement of the laser focal point on the temperature field near the sample interface. Fig. 31(c) illustrates the temporal evolution of the temperature at a position located 5.0 μm away from the sample interface on the side of fused silica glass, considering the laser focus at different positions. The results demonstrate significant variations in the temperature field depending on the central position of the laser focus. Specifically, when the focus center is on the silicon carbide side and positioned more than 1.25 μm away from the interface, the temperature at the monitoring position remains below the melting point of glass, preventing its melting. Similarly, Fig. 31(d) presents the temporal evolution of the temperature at a position situated 3.0 μm away from the sample interface on the side of silicon carbide, considering different laser focus positions. Here, when the laser focus center is on the side of fused silica glass and positioned more than 1.25 μm away from the interface, the temperature at the monitoring point remains below the melting point of silicon carbide. These findings highlight the importance of precise control of the laser focus to achieve high-quality laser welding, ensuring simultaneous melting of materials on both sides of the interface. By leveraging numerical simulations, researchers can optimize laser welding parameters to achieve desired outcomes in terms of joint quality and material integrity.

It is essential to understand femtosecond (fs) laser-matter interaction, through which temperature distribution within heat affected zone and the dimension of the internal modification induced by plasma generation in solid can be evaluated. Nonlinear absorptivity is needed to predict the two aforementioned phenomena. However, in actual experimental conditions, these material-specific thermal properties significantly vary with increasing temperature during laser-matter interaction. Particularly for laser based welding by fs-pulses with high repetition rate, the heat-accumulation-induced

temperature change modifies the thermal properties of the material locally. Thus, omission of their temperature dependency will most likely over or under estimate the temperature. Adynamic pulse propagation modeling for femtosecond laser bonding of Borofloat glass is presented [141]. The temperature evolution and internal modifications are predicted by incorporating the nonlinear electron dynamics along with temperature dependent thermal properties (Fig. 32). Numerical simulations have been developed to capture both the spatial and temporal evolution of energy absorption and plasma formation during ultrafast laser welding, enabling quantitative predictions of the heat-affected zone (HAZ) and resulting joint morphology. By systematically varying the focal position relative to the interface, these models reveal how the overlap between the laser focal volume and the material boundary governs melting dynamics and HAZ dimensions. Applying this approach to Borofloat glass–glass interfaces demonstrated excellent agreement between predicted and experimental offset distances required for defect-free bonding. Such computational frameworks are readily extendable to other dielectric combinations and provide a powerful tool for optimizing focus placement in complex material systems.



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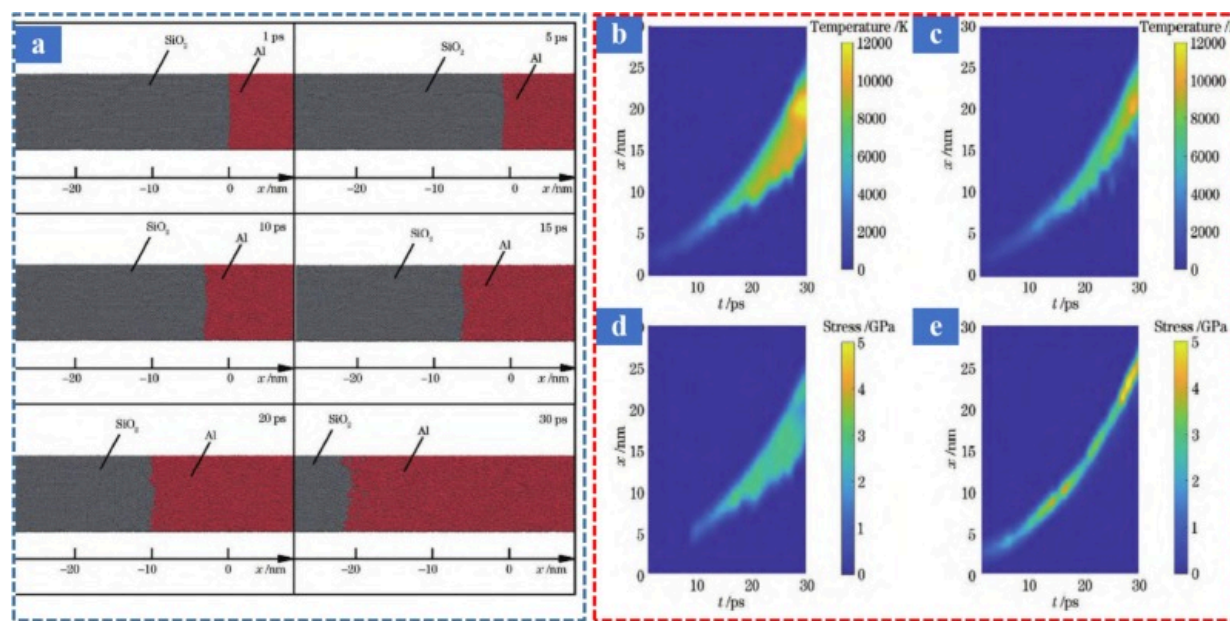
Fig. 32. The temperature evolution and internal modifications (a) Temperature dependent heat capacity (c_p) and thermal conductivity (κ) for B33; (b) Blue curve represents the time evolution of maximum temperature for 50 pulses, taking into consideration of the temperature dependence of c_p and κ ; Temperature distribution (yz-cross section) after 1 μ s for the (c) 1st pulse and (d) 50th pulse [141].

3.2. Molecular dynamics simulation of ultrafast laser micro-welding interfaces

The connection of dissimilar materials, especially metals and glass, holds great potential for applications in industries such as electronics and automotive manufacturing. Ultrafast laser welding of dissimilar materials offers advantages such as high precision, fast welding speed, small heat-affected zone, and high-quality joints, making it a topic of widespread interest in recent years. However, most of the existing research has been experimental in nature, focusing on extensive experimental and theoretical analyses of welding parameters, quality assessment, and microstructural characterization of the welded joints [[40], [41], [42], [43], [44], [60]].

On the theoretical front, the interaction mechanisms between ultrafast lasers and metal-glass interfaces are not fully elucidated. The processes associated with the interaction of ultrafast lasers with materials pose challenges for simulation using traditional methods like finite element analysis. In this context, molecular dynamics has emerged as a commonly employed approach to study phenomena such as ultrafast laser ablation and surface modification. This makes it an ideal method for simulating the microscopic-scale interaction between ultrafast lasers and dissimilar materials. Wen et al. [142] employed molecular dynamics to theoretically simulate the interaction between femtosecond lasers and the interface of aluminum and quartz glass (Fig. 33). For the simulation, a Lennard-Jones (LJ) potential function was constructed for quartz glass based on its melting point and elastic constants. The LJ potential function for the interaction between aluminum and quartz glass was established using the adhesive work between them, simplifying and accelerating the simulation process while preserving macroscopic properties. A small-scale molecular dynamics simulation of the aluminum and quartz glass interface under femtosecond laser irradiation was carried out using a coupled molecular dynamics and dual-temperature model. Following femtosecond laser irradiation, the localized instantaneous temperature in the welding zone reached up to 10000K, and the stress reached 20 GPa, leading to the diffusion and movement of aluminum atoms towards the quartz glass side. This phenomenon unveils the diffusion mechanism underlying the

bonding of the two materials, thus providing a theoretical foundation for ultrafast laser heterogenous material welding.



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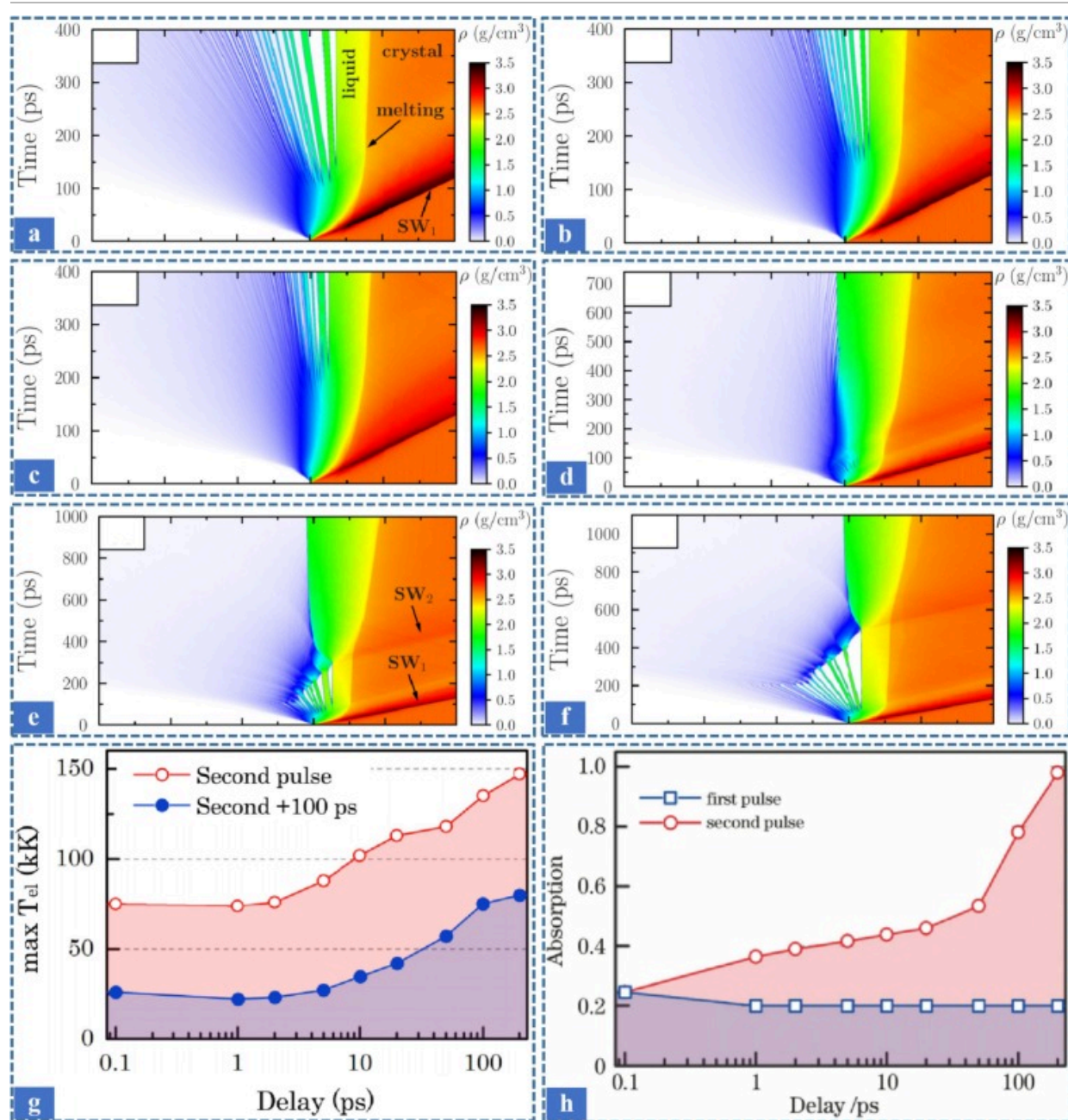
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Fig. 33. Molecular Dynamics Simulation of Aluminum-Fused Silica Interface Shot by Femtosecond Laser (a) Snapshots of the aluminum-silica glass welding process simulated by molecular dynamics. (b-e) Thermodynamic property contours in the mixed region: (b) Temperature contour plot of aluminum atoms; (c) temperature contour plot of fused silica; (d) stress contour plot of aluminum atoms; (e) stress contour plot of fused silica [142].

3.3. Plasma density simulation of ultrafast laser micro-welding processes

The interaction process between ultra-short pulse lasers and brittle materials is highly intricate, involving phenomena such as plasma ionization, heat conduction, and phase transitions. The mechanism of interaction between ultra-short pulse lasers and brittle

materials, as well as the various effects during this interaction, remain unclear. To further investigate the transmission and ionization processes induced by ultrashort, high-intensity laser pulses in glass, Zhang et al. [143] established a coupled rate-equation and heat-transfer model to simulate plasma-density evolution and temperature-field dynamics during high-speed scanning welding. Simulations were carried out for a picosecond laser (10ps pulse width, 1064nm wavelength, 4μJ pulse energy, 20mm/s scanning speed) at focal depths ranging from 5 to 30μm. Results indicate that achieving defect-free welds requires sub-micrometer precision in focal-position control to ensure simultaneous melting across the interface. Povarnitsyn et al. [144] used the MD-TTM model to study the effect of different pulse intervals on the process of sub-picosecond laser ablation of aluminum. The results show that the absorption rate of the second pulse by the newborn plasma plume increases from 25% to 98% when the delay increases from 0ps to 200ps, as shown in Fig. 34(a-h). For the pulse interval greater than 20ps, the plasma shielding effect can be observed, and the second pulse does not reach the target. These reasons caused the depth of the ablation pit to decrease monotonously.



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Fig. 34. Molecular dynamics simulation of subpicosecond double-pulse laser ablation of metals (a) Contour plots of the density of ions for 0ps delay between pulses; (b) Contour plots of the density of ions for 10ps delay between pulses; (c) Contour plots of the density of ions for 20ps delay between pulses; (d) Contour plots of the density of ions for 50ps delay between pulses; (e) Contour plots of the density of ions for 100ps delay between pulses; (f) Contour plots of the density of ions for 200ps delay between pulses; (g) The electron temperature maximum [empty (red) circles] and the electron temperature 100ps after the second pulse [filled (blue) circles]; (h) Absorption of the first (blue squares) and second (red circles) pulses by the plasma plume as a function of the delay between them [144].

3.4. Electric field simulation of ultrafast laser micro-welding processes

It should be noted that energy transport within materials under ultrafast laser irradiation follows two steps: the conduction electrons absorb the photons energy through photon-electron interactions; and energy then transfers to the lattice through electron-phonon coupling. Such two-step process will bring in a severe thermal non-equilibrium state within several picoseconds. Two-temperature model was proposed based on Fourier's law to describe the electron and lattice temperature evolution formulated as:

$$C_e \frac{\partial T_e}{\partial t} = \nabla(k_e \nabla T_e) - G(T_e - T_l) + Q_s * pulse(t)$$

$$C_l \frac{\partial T_l}{\partial t} = \nabla(k_l \nabla T_l) + G(T_e - T_l)$$

where C represents the specific heat capacity. T is the temperature of system. k is the heat conductivity. G is the electron-phonon coupling factor, and Qs is heat source which depends on the electric field around the materials. The subscripts e and l denote electron and lattice, respectively.

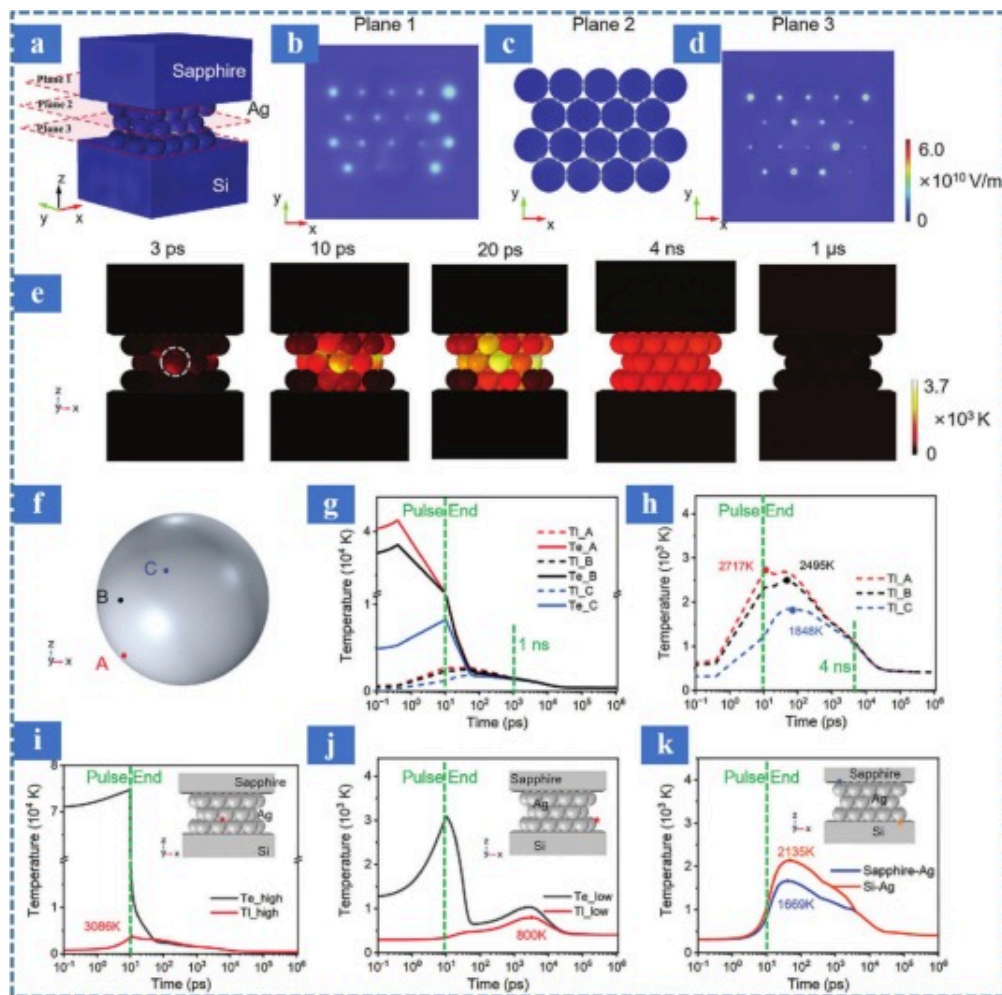
Electromagnetic induction heating in metal is primarily the result of Joule effect and determined by the electric field on the surface and inside the materials volume, which is

given by:

$$Q_s = \frac{1}{2} \epsilon_0 \omega \text{Im} \epsilon(\omega) |E|^2$$

where ϵ_0 is the permittivity in vacuum. ϵ_ω is the permittivity of material relevant to frequency ω . E is the electric field intensity, which can therefore be given by finite element method (FEM) simulation in the heterogeneous structures.

Well-ordered hexagonal periodic nanoparticle (NP) structures and monodisperse NPs were modeled using the finite element method (FEM) to simulate a closely packed layer, contrasting with denser layers of mixed NP sizes [93]. Complex internal nanostructures within this NP layer could potentially enhance plasmonic effects, although the current model may not fully capture maximum field enhancement. Simulations revealed significant plasmonic-induced localized electric field enhancement in Sapphire-Ag-Si structures across a range of Ag NP layer thicknesses under 1030nm ultrafast laser irradiation. This enhancement was particularly pronounced at a 610nm NP layer thickness, corresponding with experimental conditions, and intensified as NP layer thickness decreased (Fig. 35(a-d)). Fig. 35(e) illustrates the calculated lattice temperature distribution and evolution in the Sapphire-Ag-Si structure following single-pulse laser excitation at a fluence of 0.2J/cm². The electron and lattice temperature evolution at three points on selected Ag NP surfaces is shown in Fig. 35(f-h). At point A, significant photon energy absorption led to a high electron temperature and the highest field enhancement (“hotspot”), while peak electron temperatures at points B and C were lower. This temperature gradient facilitated electron–electron energy relaxation, achieving thermal equilibrium within 1 ns.



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Fig. 35. Electric field distribution within Sapphire-Ag-Si under ultrafast laser irradiation at laser fluence of 0.2 J/cm^2 [93].

Further FEM simulations on Sapphire-Ag-Si configurations were conducted to investigate the influence of film densification on the plasmonic effect. Results indicated that as NP density increased, the electric field intensity enhancement among Ag NPs gradually

decreased. However, the enhancement at the Ag-Si and Ag-Sapphire interfaces persisted until the relative density of the Ag NP film reached 100%. Notably, localized electric field enhancement was also observed with an extremely thin (10nm) dense Ag film inserted between Sapphire and Si. Nevertheless, this thin layer was insufficient for efficient energy confinement and localized material interactions, achieving a maximum shear strength of approximately 2.22MPa, significantly lower than the 5.4MPa strength observed with the optimized NP layer.

4. Conclusion and perspective

Ultrafast laser micro-welding offers high machining accuracy and a broad range of applications, including direct welding of transparent brittle materials like glass-metal, glass-semiconductor, and glass-glass, due to its high localization and minimal heat-affected area. The three-dimensional spatial selectivity allows for precise welding of complex-shaped transparent objects. However, the technology is still in its nascent stage, facing challenges in fundamental principles, key technologies, and process development. Despite these advantages, ultrafast laser welding is still in the early stages of basic research and key technology development. Several challenges need to be addressed in fundamental principles, key technologies, core equipment development, and process R&D before it can see widespread industrial application.

Key issues include micro-nano bubble generation, thermal residual stress, and limited operational temperature range. The process of forming heterogeneous welds using ultra-short, high-energy-density laser pulses is highly complex, primarily due to its transient nature and the mismatch of thermal properties between the two components.

In addition to the conventional welding mechanisms, ultrashort pulse laser welding can also be achieved through the self-focusing effect [20,29,60,92,93]. This approach does not require a high pulse repetition frequency but relies on sufficient peak power. Because there is no thermal

accumulation, thermal diffusion is minimal, resulting in a weld width of only a few micrometers due to the rapid cooling of the laser-affected zone.

A major challenge in ultrafast laser welding of different materials is accurately focusing the laser at the interface between them.

The physical process of ultrafast laser selective welding can be divided into two main stages: the selective deposition of ultrafast laser energy and the stress response relaxation of the material under laser-induced multi-physical fields. The energy deposition involves the coupling between photons and electrons, influenced by photon energy, light field intensity, material electronic configuration, and lattice structure. This process, described by nonlinear Maxwell equations, is central to ultrafast laser welding. During this stage, the material at the laser focus undergoes plasma evolution, thermal shock, thermal expansion, phase transitions, and interface ablation within localized multi-physical fields. Strong bonding between material interfaces primarily forms during this phase.

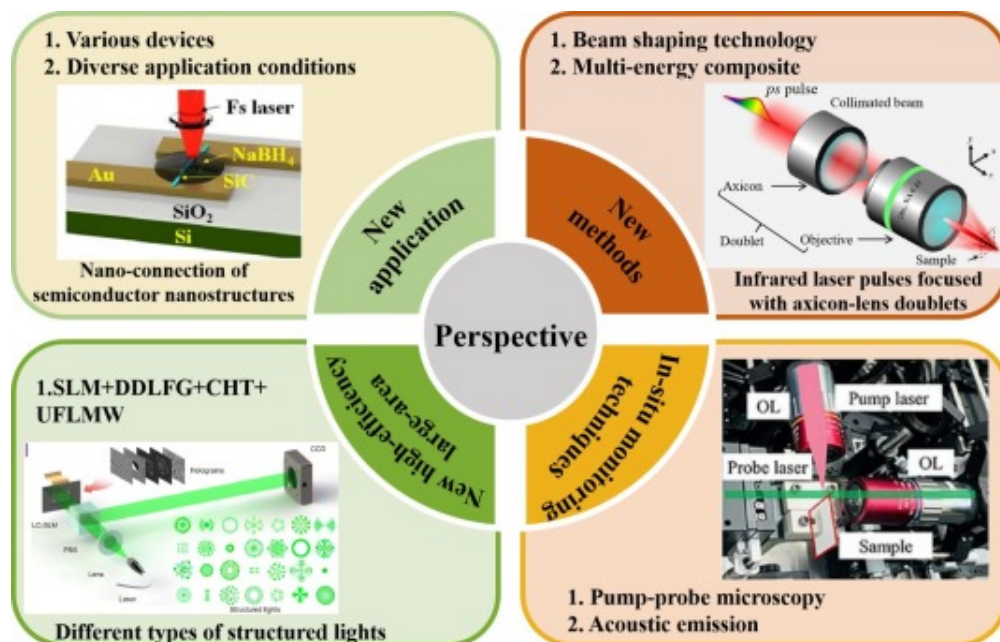
Challenges such as the elimination of residual stresses, control of interface units, and the understanding of resulting synergistic strengthening mechanisms remain unresolved. These issues significantly affect the joint quality and optical performance of specimens, limiting the advancement of brittle material-metal ultrafast laser micro-welding. Reducing the contact gap between brittle materials is particularly challenging in large-area welding, especially for complex-shaped brittle materials, where applying external force proves insufficient. Addressing the problem of ultrafast laser welding of brittle materials under conditions of large contact gaps remains a critical bottleneck in this field.

Based on research progress in ultrafast laser micro-welding (UFLMW) over the past two decades, as illustrated in [Fig. 36](#), the future research focus and development of this field is

expected to concentrate on the following directions:

- At present, ultrafast laser micro-welding of cross-scale nanomaterials, particularly two-dimensional materials, remains relatively underexplored. Future research should delve deeper into the formation of heterogeneous interfaces and the integration of heterostructures in nano-devices. Ultrafast laser micro-welding, due to its ultra-short pulse duration, high peak power, and precise energy localization, offers unique advantages in enabling reliable nano-connections between dissimilar materials. This will play a pivotal role in promoting the fabrication and application of micro-nano devices in fields such as optoelectronics, sensing, and quantum technology.
- Emerging strategies based on beam shaping technologies (e.g., axicon-lens doublets) and multi-energy composite welding (e.g., ps/fs hybrid pulses) offer enhanced control over energy deposition. These methods help optimize the spatial-temporal distribution of laser energy, improve coupling with complex materials, and reduce thermal effects, thereby expanding the application range of UFLMW to more challenging material systems and geometries.
- Although there has been considerable progress in characterizing the microstructure and composition of welded heterogeneous interfaces, the underlying welding mechanisms are still not fully understood. Advanced in-situ diagnostic methods, such as pump-probe microscopy, plasma imaging, and acoustic emission-based monitoring, can be employed to capture the dynamic evolution of the interface during the welding process. A clearer understanding of these mechanisms will provide solid theoretical support for: real-time process control and quality prediction, multi-energy composite UFLMW strategies, micro-welding of new materials, and expansion into more complex and demanding application scenarios.

- In recent years, numerous research groups worldwide have shown that applying spatial light modulation technology to ultrafast laser micro-welding has become one of the mainstream techniques to enhance processing efficiency. Traditional ultrafast laser micro-welding techniques suffer from low processing efficiency, and the use of multi-beam/multi-focus processing methods is relatively inflexible and limited in versatility. By combining spatial light modulation (SLM), dynamic digital light field generation (DDLFG), and computational holography techniques (CHT), ultrafast laser micro-welding (UFLMW) can achieve flexible and controllable light modulation and area projection exposure during the welding process. This enables high-efficiency, large-area processing with resolution comparable to that of single-focus ultrafast laser micro-welding. This method is an important approach for manufacturing high-precision, highly functionalized, highly integrated, and high-quality three-dimensional large-area welding, as well as diverse micro-nano structural device packaging. It makes it possible to industrially package integrated micro-nano photonic devices, micro-nanomechanical devices, and biomedical devices in the optoelectronic industry.
- Ultrafast laser micro-welding, as a high-precision, high-performance bonding technology, is poised to become a key enabler for next-generation micro-nano manufacturing. Continued innovation in interconnection strategies, in-situ diagnostics, beam modulation, and composite energy delivery will drive its advancement toward scalable, intelligent, and application-oriented solutions.



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Fig. 36. Perspective of ultrafast laser micro-welding in brittle optical materials [[145], [146], [147]].

Ultrafast laser micro-welding plays a crucial role in bonding specialized materials, yet challenges such as high system costs and nonlinear interaction control remain. Research into laser-material interactions and process optimization holds promise for overcoming these hurdles. Key priorities include: 1) Reducing system costs to enhance industry accessibility; 2) Improving control over nonlinear interactions to achieve precise results; 3) Exploring advanced processing techniques for a broader range of materials; and 4) Shaping and designing ultrafast lasers in the time, space, and frequency domains to modulate photon-electron-plasma interactions, thereby enhancing weld quality, strength, and speed. These efforts will drive ultrafast laser micro-welding to surpass existing

barriers, delivering significant economic and application benefits across high-end manufacturing sectors and fostering new capabilities in component fabrication.

CRediT authorship contribution statement

Changhao Ji: Writing – review & editing, Writing – original draft, Investigation, Data curation. **Cheng Yang:** Writing – original draft, Data curation. **Chengyun Wang:** Writing – original draft, Investigation. **Ziyue Yu:** Writing – review & editing, Formal analysis. **Jindou Wu:** Writing – review & editing. **Lili Yuan:** Writing – review & editing. **Yu Long:** Writing – review & editing, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Key R&D Program of Guangxi Province, China (Grant No. GKAB23026101), Transportation Operation Subsidy Project of Guangxi Key Laboratory of International Join for China-ASEAN Comprehensive Transportation in 2021 (No. 21-220-21), the National Key R&D Program of China (Grant No. 2022YFB4601601), and Guangxi Natural Science Foundation, China (Grant No. 2023GXNSFBA026287).

Data availability

Data will be made available on request.

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